

COMMON OBJECTS
OF THE
MICROSCOPE

BY THE LATE
Rev. J. G. WOOD

COMMON OBJECTS
OF THE
MICROSCOPE

BY THE LATE
Rev. J. G. WOOD

The Project Gutenberg eBook of Common objects of the microscope

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Common objects of the microscope

Author: J. G. Wood

Editor: Edward Collins Bousfield

Illustrator: Tuffen West

Release date: July 18, 2017 [eBook #55146]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/55146

Credits: Produced by Thiers Halliwell, Chris Curnow and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK COMMON
OBJECTS OF THE MICROSCOPE ***

Transcriber's notes:

Minor punctuation errors have been corrected silently (e.g. missing full stops after abbreviated words such as Fig), as have the following misspellings: Bretahing → Breathing, Pedicillaria → Pedicellaria, Pedicelaria → Pedicellaria, Chætonotus → Chætonotus, Spurganium → Sparganium, veiw → view. Unorthodox spelling and inconsistent hyphenation has not been altered. Several wrongly numbered cross-references to Plates and Figures have been corrected.

Plate VIII (and its accompanying key) was originally displayed at the beginning of the book, before the Title Page, but has been repositioned in the body of the text in correct numerical sequence.

A black underline indicates a hyperlink to a page or illustration (hyperlinks are also highlighted when the mouse pointer hovers over them). Page numbers are shown in the right margin.

COMMON OBJECTS OF THE MICROSCOPE

BY THE LATE

REV. J. G. WOOD, M.A., F.L.S., ETC.

AUTHOR OF
“COMMON OBJECTS OF THE COUNTRY” “COMMON OBJECTS OF THE SEA-
SHORE”
“MY FEATHERED FRIENDS” ETC. ETC.

WITH ILLUSTRATIONS BY TUFFEN WEST

SECOND EDITION, REVISED AND RE-WITTEN, BY

E. C. BOUSFIELD, L.R.C.P.(LOND.)

AUTHOR OF
“A GUIDE TO THE SCIENCE OF PHOTO-MICROGRAPHY”

WITH ADDITIONAL ILLUSTRATIONS BY THE REVISER

LONDON
GEORGE ROUTLEDGE AND SONS, LIMITED
BROADWAY HOUSE, LUDGATE HILL
1900

PREFACE TO THE SECOND EDITION

The task of revising and bringing up to date a work which has been the guide, philosopher, and friend of thousands of commencing microscopists has been, in the present case, one of no small difficulty. On the one hand, there was the natural desire to interfere as little as possible with the original work; and on the other, the necessity of rendering available, to some extent at least, the enormous advance in every department which has taken place in the thirty-six years which have elapsed since the work was first offered to the public. The reviser has done his best not only to fulfil these two objects, but to keep in view the original purpose of the book.

In the popular department of pond-life especially, about fifty new illustrations have been added, mostly from the reviser's own notebook sketches. The whole of the botanical part has been revised by one of our first English authorities, and, in short, no effort has been spared to make the work as accurate as its necessarily condensed form permits of. It is hoped, therefore, that it may be found not less useful than its predecessor by those for whom it is alone intended.

PREFACE TO THE FIRST EDITION

In my two previous handbooks, the “Common Objects” of the Sea-shore and Country, I could but slightly glance at the minute beings which swarm in every locality, or at the wonderful structures which are discovered by the Microscope within or upon the creatures therein described. Since that time a general demand has arisen for an elementary handbook upon the Microscope and its practical appliance to the study of nature, and in order to supply that want this little volume has been produced.

I must warn the reader that he is not to expect a work that will figure and describe every object which may be found on the sea-shore or in the fields, but merely one by which he will be enabled to guide himself in microscopical research, and avoid the loss of time and patience which is almost invariably the lot of the novice in these interesting studies. Upwards of four hundred objects have been figured, including many representatives of the animal, vegetable, and, mineral kingdoms, and among them the reader will find types sufficient for his early guidance.

Neither must he expect that any drawings can fully render the lovely structures which are revealed by the microscope. Their form can be given faithfully enough, and their colour can be indicated; but no pen, pencil, or brush, however skilfully wielded, can reproduce the soft, glowing radiance, the delicate pearly translucency, or the flashing effulgence of living and ever-changing light with which God wills to imbue even the smallest of His creatures, whose very existence has been hidden for countless ages from the inquisitive research of man, and whose wondrous beauty astonishes and delights the eye, and fills the heart with awe and adoration.

Owing to the many claims on my time, I left the selection of the objects to Mr. Tuffen West, who employed the greater part of a year in collecting specimens for the express purpose, and whose well-known fidelity and wide experience are the best guarantees that can be offered to the public. To him I

also owe many thanks for his kind revision of the proof-sheets. My thanks are also due to Messrs. G. and H. Brady, who lent many beautiful objects, and to Messrs. Baker, the well-known opticians of Holborn, who liberally placed their whole stock of slides and instruments at my disposal.

CONTENTS

CHAPTER I

PAGE

Pleasures and Uses of Microscopy—Development of the Microscope—Extemporised Apparatus	<u>1</u>
---	----------

CHAPTER II

Elementary Principles of Optics—Simple Microscopes—Compound Microscope—Accessory Apparatus—Cover-glasses—Troughs—Condensers—Dissection—Dipping-tubes—Drawing—Measurement	<u>7</u>
--	----------

CHAPTER III

Examination of Objects—Principles of Illumination—Mirror and its Action—Substage Condenser—Use of Bull's-eye—Opaque Objects—Photography of Microscopic Objects	<u>28</u>
--	-----------

CHAPTER IV

Vegetable Cells and their Structure—Stellate Tissues—Secondary Deposit—Ducts and Vessels—Wood-Cells—Stomata, or Mouths of Plants—The Camera Lucida, and Mode of Using—Spiral and Ringed Vessels—Hairs of Plants—Resins, Scents, and Oils—Bark Cells	<u>37</u>
---	-----------

CHAPTER V

Starch, its Growth and Properties—Surface Cells of Petals—Pollen and its Functions—Seeds	<u>63</u>
--	-----------

CHAPTER VI

Algæ and their Growth—Desmidiaceæ, where found
—Diatoms, their Flinty Deposit—Volvox—
Mould, Blight, and Mildew—Mosses and Ferns
—Mare's-Tail and the Spores—Common Sea-
weeds and their Growth

78

CHAPTER VII

Antennæ, their Structure and Use—Eyes, Compound
and Simple—Breathing Organs—Jaws and their
Appendages—Legs, Feet, and Suckers—
Digestive Organs—Wings, Scales, and Hairs—
Eggs of Insects—Hair, Wool, Linen, Silk, and
Cotton—Scales of Fish—Feathers—Skin and its
Structure—Epithelium—Nails, Bone, and Teeth
—Blood Corpuscles and Circulation—Elastic
Tissues—Muscle and Nerve

96

CHAPTER VIII

Pond-Life—Apparatus and Instructions for Collecting
Objects—Methods of Examination—Sponge—
Infusoria

132

CHAPTER IX

Fresh-water Worms—Planarians—Hydra—Polyzoa—
Rotifers Chætonotus—Water-Bears

144

CHAPTER X

Marine Life—Sponges—Infusoria—Foraminifera—
Radiolaria—Hydroid Zoophytes—Polyzoa—
Worms—Lingual Ribbons and Gills of Mollusca
—Star-Fishes and Sea-Urchins—Cuttle-Fish—
Corallines—Miscellaneous Objects

154

CHAPTER XI

Hints on the Preparation of Objects—Preservative
Fluids—Mounting Media—Treatment of Special
Objects

168

CHAPTER XII

Section-Cutting—Staining

179

COMMON OBJECTS OF THE MICROSCOPE



CHAPTER I

Pleasures and Uses of Microscopy—Development of the Microscope—Extemporised Apparatus.

Within the last half-century the use of the microscope, not only as an instrument of scientific research, a tool in the hands of the investigator of the finer organisation of the world of nature, nor even as an adjunct to the apparatus of the chemist or the manufacturer, but as a means of innocent and instructive recreation, has become so firmly rooted amongst us that it seems hardly necessary to advocate its claims to attention on any of these grounds.

So wonderful is the information which it affords, so indispensable is it in many, if not in all, branches of scientific research, that not only would the lover of nature be deprived of one of his most valued sources of information and enjoyment, but some sciences would be brought absolutely to a standstill if by any conceivable means the microscope were to be withdrawn from their followers.

On the other hand, from every improvement in the construction of the latter, a corresponding enlargement and enlightenment of the fields reviewed by these sciences has taken place, and the beauty and interest of the revelations made by its means has attracted an ever-increasing host of earnest and intelligent volunteers, who have rendered yeoman service to the cause of knowledge.

Moreover, so vast is the scope of the instrument, that whilst discoveries in other fields of research are few and far between, comparatively speaking, in microscopic science they are of everyday occurrence, and the number of problems calling for solution by means of the instrument in question is so vast that even the humblest worker may be of the greatest assistance.

In the following pages we propose to carry out, as far as possible with reference to the microscope, the system followed in the “Common Objects of the Seashore and of the Country,” and to treat in as simple a manner as may be of the marvellous structures which are found so profusely in our fields, woods, streams, shores, and gardens. Moreover, our observations will be restricted to an instrument of such a class as to be inexpensively purchased and easily handled, and to those pieces of supplementary apparatus which can be extemporised at small cost of money and ingenuity by the observer himself, or obtained of the opticians for a few shillings.

With the same view, the descriptions will be given in language as simple and as free from technicalities as possible, though it must be remembered that for many of the organisms and structures which we shall have to describe there are none but scientific names; and since, moreover, this little work is intended to furnish a stepping-stone between the very elements of microscopic science, and those higher developments of it which should be the aim of every worker, it would be unwise to attempt to invent commonplace appellations for the purpose of this book, and leave him to discover, when he came to consult works of reference in any particular subject, that his “simplified” knowledge had all to be unlearned, and a new vocabulary acquired. Rather will it be our purpose to use correct terms, and explain them, as far as necessary, as we introduce them.

The commencing microscopist is strongly recommended, whilst not confining his interest entirely to one branch of research or observation, to adopt some one as his particular province.

The opportunities for discovery and original work, which are afforded by all alike, will be more readily appreciated and utilised by adopting such a plan than by a general and purposeless distribution of effort. To mention one or two. The student of the fascinating field of pond-life will find new organisms awaiting description by the hundred, and of the old ones, life-histories to make out; if he be attracted rather to the vegetable inhabitants of the same realm, the diatoms will furnish him with the opportunity of studying, and perhaps solving, the enigma of their spontaneous movement,

and of watching their development. The smaller fungi, and indeed the larger ones too, will amply repay the closest and most laborious study of their habits of life and processes of development. Since the first edition of this work was published, the whole subject has been practically revolutionised, and more remains to be done than has yet been accomplished.

In short, there is scarcely an organism, even of those best known and most studied, which is so completely exhausted that persevering investigation would reveal nothing new concerning it.

There can be little doubt but that if any worker, with moderate instrumental means, but with an observant mind, were to set determinately to work at the study of the commonest weed or the most familiar insect, he, or she, would by patient labour accomplish work which would not only be of value to science, but would redound to the credit of the worker.

Something like finality appears to have been reached, at least for the present, in the development of the microscope; and whilst it is beyond the scope of this work to treat of the refined and costly apparatus which is essential to useful work in certain departments of research, the result has, on the whole, been highly favourable to the worker of moderate means and ambitions, since lenses are now accessible, at the cost of a few shillings, comparatively speaking, which could not have been purchased at all when this work first appeared. It is with such appliances that we have here to deal. The worker whose finances are restricted must be contented to extemporise for himself many pieces of apparatus, and will find pleasure and occupation in doing so. And let him remember, for his encouragement, that many such home-made appliances will fulfil their purpose quite as well as the imposing paraphernalia of glittering brass and glass which decorates the table of the wealthy amateur. It is not the man who possesses the best or most costly apparatus, but the one who best understands the use of that which he possesses, that will make the most successful microscopist. A good observer will discover, with only the aid of a pocket-magnifier, secrets of Nature which have escaped the notice of a whole army of dilettante

microscopists, in spite of the advantages which, as regards instruments, the latter may enjoy.

It is for those who desire to be of the former class that this book is written, and in the course of the following pages instances will be given in which the exercise of a small amount of ingenuity and the expenditure of a few pence will be found equivalent to the purchase of costly and complicated apparatus.

An enormous amount of valuable work was done in the earliest days of microscopy, when the best apparatus available was a single lens, composed of the bead formed by fusing the drawn-out end of a rod of glass. Inserted into a plate of metal, or a piece of card, such a primitive instrument was capable of affording a large amount of information. Similar instruments are to be purchased for a few pence at the present day, and are not without their use for purposes of immediate examination of material. A very common form is a glass marble, ground flat on one side, and mounted in a tube. The material to be examined is placed upon the flat side, and is seen magnified to an extent inversely proportional to the diameter of the sphere of glass.

CHAPTER II

Elementary Principles of Optics—Simple Microscopes—Compound Microscope—Accessory Apparatus—Cover-glasses—Troughs—Condensers—Dissection—Dipping-tubes—Drawing—Measurement.

Before proceeding to deal with the microscope itself, it may be useful to give a short summary of the optical laws upon which its working depends.

To go into the minutiae of the matter here would be out of place, but it will be found very helpful, especially in the matter of illumination, to acquire some knowledge of, and facility in applying, these elementary principles. We shall confine our remarks to convex lenses, as being the form to which all the combinations in the microscope may be ultimately reduced.

Every convex lens has one “principal” focus, and an infinite number of “conjugate” foci. The principal focus is the distance at which it brings together in one point the rays which fall upon it parallel to its axis, as shown in Fig. 1, in which A is the axis of the lens L , and the rays RR are brought together in the principal focus P . Thus a ready means of finding the focal length of any lens is to see at what distance it forms an image of the sun, or of any other distant object, upon a screen, such as a piece of smooth white cardboard. In the figure this distance will be PL . Conversely, if the source of light be at P , a parallel beam of light will be emitted from the lens.

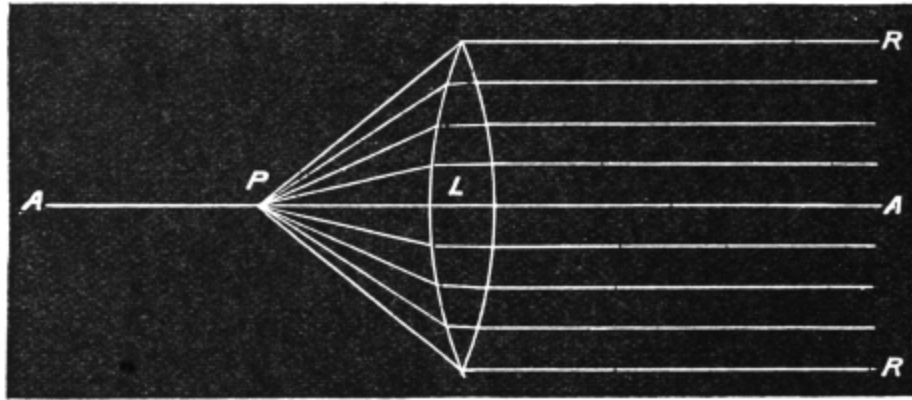


FIG. 1.

The focal length may, however, be found in another way. When an object is placed at a distance from a lens equal to twice the principal focal length of the latter, an image of the object is formed at the same distance upon the other side of the lens, inverted in position, but of the same dimensions as the original object. The object and image then occupy the equal conjugate foci of the lens, so that by causing them to assume these relative positions, and halving the distance at which either of them is from the lens, the focal length of the latter is known.

These points will be seen on reference to Fig. 2, in which L being the lens, and P the principal focus, as before, rays from the point C are brought together at the conjugate focus C' , at the same distance on the other side of L . In this case it manifestly does not matter whether the object be at one or the other of these points.

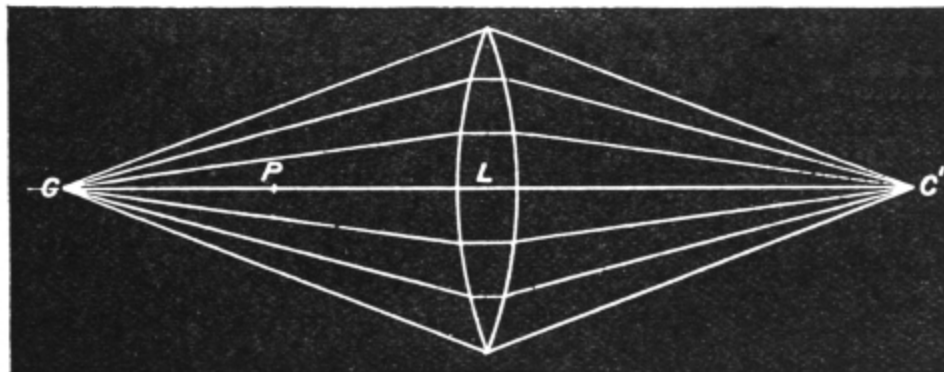


FIG. 2.

So far we have been dealing with points on the line of the axis of the lens. The facts mentioned apply equally, however, to rays entering the lens at an angle to the axis, only that in this case they diverge or converge, correspondingly, upon the other side. It is evident, from Fig. 1, that no image is formed of a point situated at the distance of the principal focus; but Fig. 3, which is really an extension of Fig. 2, shows how the rays passing along secondary axes form an inverted image of the same size as the object, when the latter is situated at twice the focal length of the lens from this last. To avoid confusion, the bounding lines only are shown, but similar lines might be drawn from each and every point of the object; and if the lines ALA' , $BL'B'$ be supposed to be balanced at L and L' respectively, they will indicate the points at which the corresponding parts of the object and image will be situated along the lines AB , $B'A'$ respectively. Moreover, rays pass from every part of the object to every part of the lens, so that we must imagine the cones LAL' , $LA'L'$ to be filled with rays diverging on one side of the lens and converging on the other.

The image so formed is a “real” image,—that is to say, it can be thrown upon a screen.

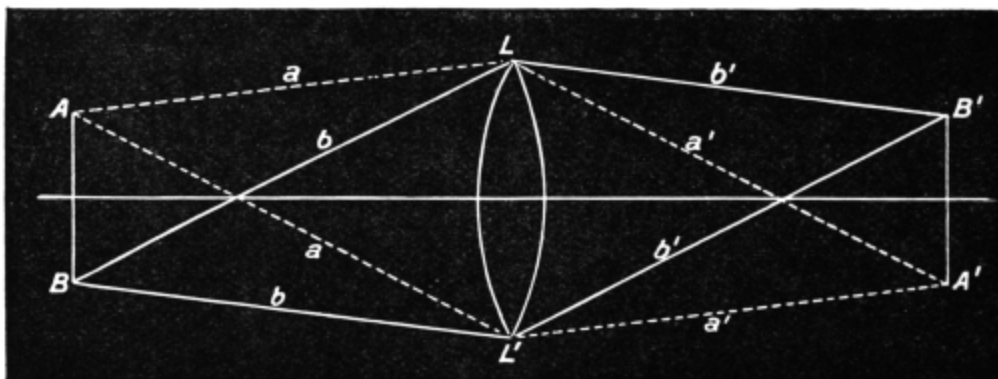


FIG. 3.

The microscopic image, on the other hand, is a virtual image, which can be viewed by the eye but cannot be thus projected, for it is formed by an object placed nearer to the lens than the principal focal length of the latter, so that the rays diverge, instead of converging, as they leave the lens, and the eye looks, as it were, back along the path in which the rays appear to travel, and so sees an enlarged image situated in the air, farther away than the object, as shown in Fig. 4. In this case, as the diagram shows, the image is upright, not inverted.

Images of the latter class are those formed by simple microscopes, of the kind described in the previous chapter. In the compound microscope the initial image, formed by the object-glass, is further magnified by another set of lenses, forming the eye-piece, by which the diverging rays of the virtual image are brought together to a focus at the eye-point; and when viewed directly, the eye sees an imaginary image, as in a simple microscope, whilst, when the rays are allowed to fall upon a screen, they form a real image of the object, larger or smaller, as the screen is farther from or nearer to the eye-point.

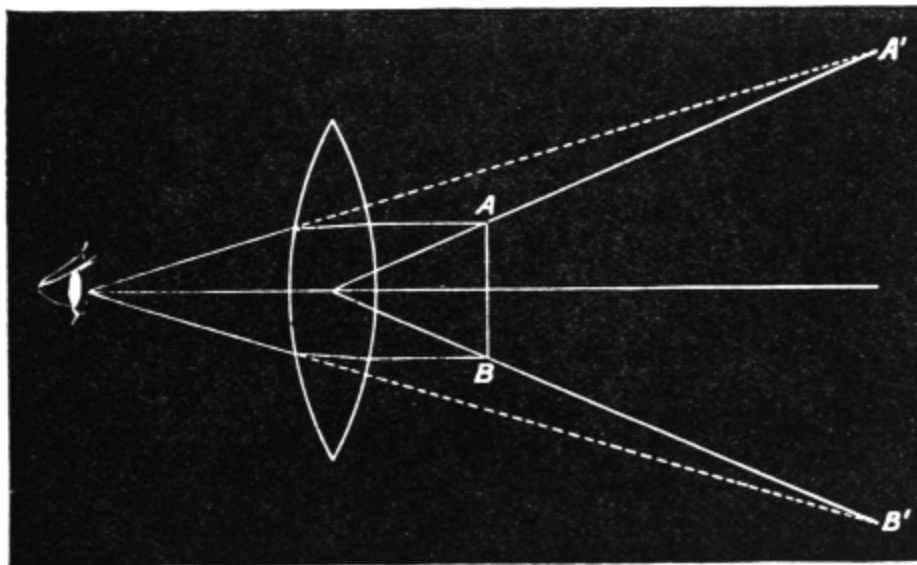


FIG. 4.

These remarks must suffice for our present purpose. Those who are sufficiently interested in the subject to desire to know more of the delicate corrections to which these broad principles are subjected in practice, that objectives may give images which are clear and free from colour, to say nothing of other faults, will do well to consult some such work as Lommel's *Optics*, in the International Science Series.

In following a work such as the present one, the simple microscope, in some form or other, will be found almost indispensable. It will be required for examining raw material, such as leaves or other parts of plants, for gatherings of life in fresh or salt water, for dissections. With such powers as those with which we shall have to deal, it will rarely happen that, for example, a bottle of water in which no life is visible will be worth the carrying-home; whilst, on the other hand, a few months' practice will render it not only possible, but easy, not only to recognise the presence, but to identify the genus, and often even the species, of the forms of life present. Moreover, these low powers, affording a general view of the object, allow the relation to each other of the details revealed by the power of the compound microscope to be much more easily grasped. A rough example may suffice to illustrate this. A penny is a sufficiently evident object to the naked eye, but it will require a sharp one to follow the details in Britannia's shield, whilst the minute scratches or the bloom upon the surface would be invisible in detail without optical aid. Conversely, however, it would be rash to conclude from an examination of a portion of the surface with the microscope alone that the portion in view was a sample of the whole surface. The more the surface is magnified, the less are the details grasped as a whole, and for this reason the observer is strongly recommended to make out all that he can of an object with a simple magnifier before resorting to the microscope.

For general purposes, the intending observer cannot do better than supply himself with a common pocket-magnifier, with one, two, or three lenses, preferably the last, as although the absolute performance is not so accurate, the very considerable range of power available by using the lenses singly, or

in various combinations, is of great advantage. Such a magnifier may be obtained from Baker for about three-and-sixpence, or, with the addition of a powerful Coddington lens (Fig. 5) in the same mount, for nine shillings more. Aplanatic lenses, such as the one shown in section in Fig. 6, with a much flatter field of vision, but of one power only each, cost about fifteen shillings, and a simple stand, which adapts them for dissecting purposes, may be obtained of the same maker for half a crown, or may easily be extemporised from a cork sliding stiffly on an iron rod set in a heavy foot, the cork carrying a loop of wire terminating in a ring which carries the lens.



FIG. 5.

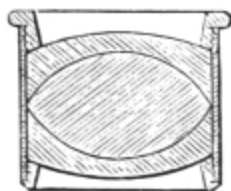


FIG. 6.

So much may suffice for the simple microscope. We pass on now to the consideration of the instrument which forms the subject of the present work, an instrument which, whilst moderate in price, is yet capable of doing a large amount of useful and valuable work in the hands of a careful owner, and of affording him a vast amount of pleasure and recreation, even if these be his only objects in the purchase, though it is difficult to understand that, an insight being once attained into the revelations effected by the instrument, without a desire being excited in any intelligent mind to pursue the subject as a study as well as a delightful relaxation. The microscope

described, and adopted as his text by the author of this work, is still made, and has shared to a very considerable extent in the general improvement of optical apparatus which has taken place during the last thirty years. It is to be obtained from Baker, 244 High Holborn, and is provided with most of the apparatus which will be found indispensable by the beginner, costing, with a case in which to keep it, the modest sum of three guineas.



FIG. 7.

If this instrument represent the limit of the purchaser's power of purse, he may very well make it answer his purpose for a considerable time. The same instrument, however, with separate objectives of good quality, together with a bull's-eye condenser (an almost indispensable adjunct), a plane mirror in addition to a concave one, and a simple but efficient form of substage condenser, may be obtained for £5, 12s. 6d., and the extra outlay

will be well repaid by the advantage in working which is gained by the possession of the additional apparatus.

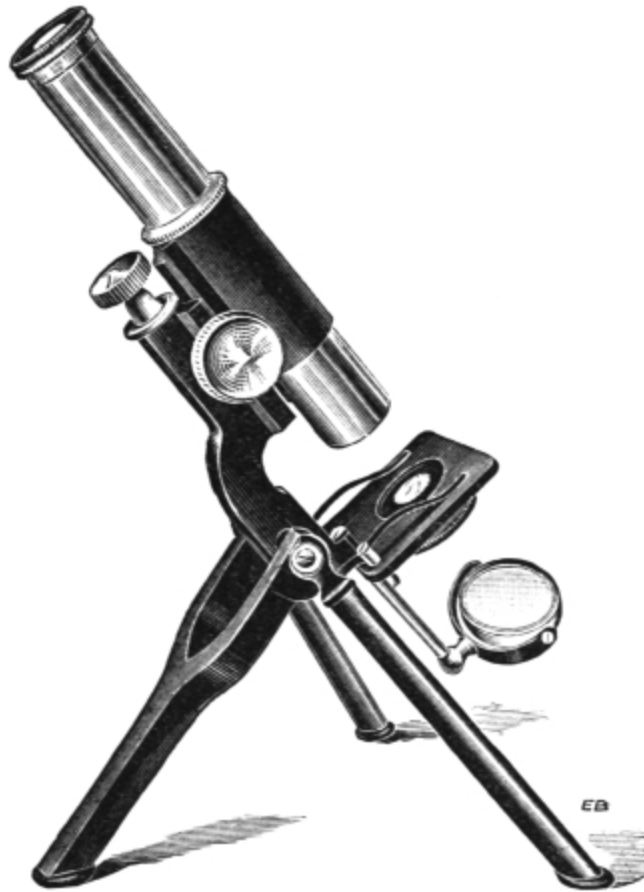


FIG. 8.

A still better stand, and one which is good enough for almost any class of work, is that shown in Fig. 8, which is known as the “Portable” microscope. In this instrument the body is made up of two tubes, so that the length may be varied at will, and this gives a very considerable range of magnification without changing the object-glass, a great convenience in practice; whilst the legs fold up for convenience of carriage, so that the whole instrument, with all necessary appliances, may be readily packed in a corner of a portmanteau for transport to the country or seaside.

The objectives supplied with the simplest form of microscope above referred to are combinations of three powers in one, and the power is varied by using one, two, or three of these in combination. This form of objective is very good, as far as it goes, though it is impossible to correct such a combination with the accuracy which is possible in manufacturing one of a fixed focal length.

Perhaps the best thing for the beginner to do would be to purchase the combination first, and, later on, to dispose of it and buy separate objectives of, say, one-inch, half-inch, and quarter-inch focal lengths. It may be explained here, that when a lens is spoken of as having a certain focal length, it is meant that the magnification obtained by its use is the same, at a distance of ten inches from the eye, as would be obtained by using a simple sphere of glass of the same focal length at the same distance. This, of course, is simply a matter of theory, for such lenses are never used actually.

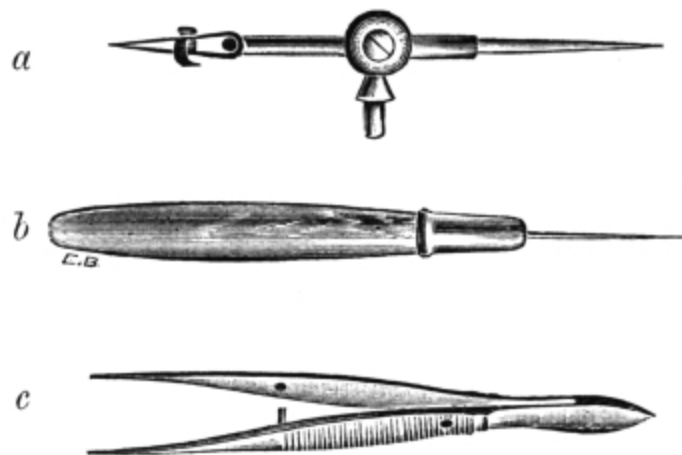


FIG. 9.

Of accessory apparatus, we may mention first the stage forceps (Fig. 9, *a*). These are made to fit into a hole upon the stage, so as to be capable of being turned about in any direction, with an object in their grasp, and for some purposes, especially such as the examination of a thin object, say the edge of a leaf, they are extremely useful.

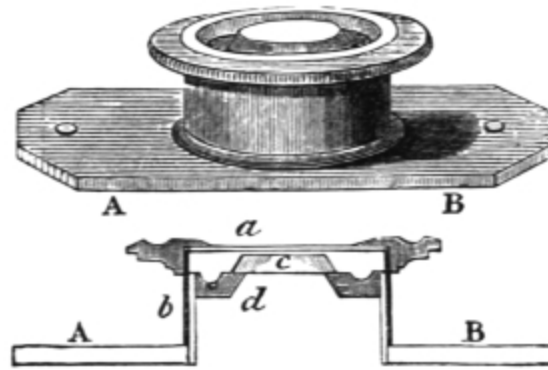


FIG. 10.

The live box, in which drops of water or portions of water-plants, or the like, may be examined, will be found of great service. By adjustment of the cap upon the cylinder, with proper attention to the thickness of the cover-glass in the cap, any required amount of pressure, from that merely sufficient to fix a restless object to an amount sufficient to crush a resistant tissue, may easily be applied, whilst the result of the operation is watched through the microscope. This proceeding is greatly facilitated if the cap of the live-box be slotted spirally, with a stud on the cylinder, so that a half-turn of the cap brings the glasses into contact. By this means the pressure may be adjusted with the greatest nicety.

In examining delicate objects, such as large infusoria, which invariably commit suicide when pressure is applied, a good plan is to restrict their movements by placing a few threads of cotton-wool, well pulled out, in the live-box with the drop of water.

A variety of instruments has been invented for the same purpose, of which Beck's parallel compressorium may be mentioned as the most efficient, though it is somewhat complicated, and consequently expensive, costing about twenty-five shillings.

A few glass slips and cover-glasses will also be required. The latter had better be those known as "No. 2," since the beginner will find it almost impossible to clean the thinner ones satisfactorily without a large percentage of fractures. The safest way is to boil the thin glass circles in

dilute nitric acid (half acid, half water) for a few minutes, wash well in several waters, first tap-water and then distilled, and finally to place the covers in methylated spirit. When required for use, the spirit may be burnt off by applying a light, the cover-glass, held in a pair of forceps, being in no way injured by the process.

In addition to the glass slides, the observer will find it advisable to be provided with a few glass troughs, of various thicknesses, in which portions of water-plants, having organisms attached to them, may be examined. Confined in the live-box, many of the organisms ordinarily found under such circumstances can rarely be induced to unfold their beauties, whilst in the comparative freedom of the trough they do so readily. The troughs may be purchased, or may be made of any desired shape or size by cutting strips of glass of a thickness corresponding to the depth desired, cementing these to a glass slide somewhat larger than the ordinary one, and cementing over the frame so formed a piece of thin glass, No. 3; the best material to use as cement being marine glue of the best quality, or, failing this, Prout's elastic glue, which is much cheaper, but also less satisfactory. The glass surface must be made, in either case, sufficiently hot to ensure thorough adhesion of the cement, as the use of any solvent entails long waiting, and considerable risk of poisoning the organisms. A useful practical hint in the use of these troughs is that the corners, at the top, should be greased slightly, otherwise the water finds its way out by capillary attraction, to the detriment of the stage of the microscope.

Of optical accessories, the bull's-eye is almost the most valuable. So much may be effected by its means alone, in practised hands, that it is well worth the while of the beginner to master its use thoroughly, and the methods of doing so will be explained in the succeeding chapter.

The substage condenser, too, even in its most simple form, is an invaluable adjunct, even though it be only a hemisphere of glass, half an inch or so in diameter, mounted in a rough sliding jacket to fit underneath the stage. Such an instrument, properly fitted, will cost about fifteen shillings, but the ingenious worker will easily extemporise one for himself.

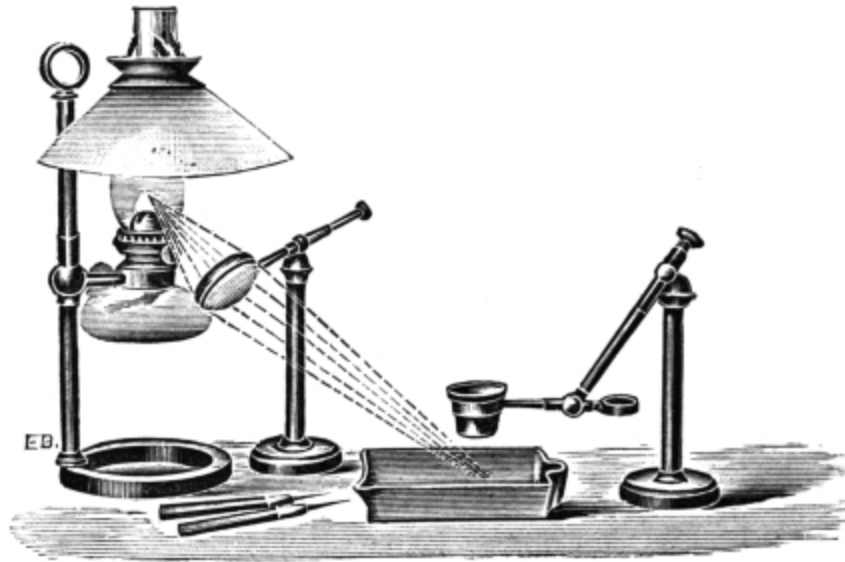


FIG. 11.

Many plants and animals require to be dissected to a certain extent before the details of their structure can be made out, and for this purpose the naked eye alone will rarely serve. The ordinary pocket magnifier, however, if mounted as described in a preceding chapter, will greatly facilitate matters, and the light may be focused upon the object by means of the bull's-eye condenser, as shown in Fig. 11. In the figure the latter is represented as used in conjunction with the lamp, but daylight is preferable if it be available, the strain upon the eyes being very much less than with artificial light. Two blocks of wood, about four inches high, will form convenient rests for the hands, a plate of glass being placed upon the blocks to support the dish, and a mirror being put in the interspace at an angle of 45° or so if required. A piece of black paper may be laid upon the mirror if reflected light alone is to be used.

As all delicate structures are dissected under fluid, a shallow dish is required. For this purpose nothing is better than one of the dishes used for developing photographic negatives. The bottom of the dish is occupied by a flat cork, to which a piece of flat lead is attached below, and the object having been pinned on to the cork in the required position, the fluid is carefully run in. This fluid will naturally vary according to the results

desired to be obtained, but it must not be plain water, which so alters all cellular structures as to practically make them unrecognisable under the microscope. Nothing could be better than a 5 per cent. solution of formalin, were it not for the somewhat irritating odour of this fluid, since it at once fixes the cells and preserves the figure. For many purposes a solution of salt, in the proportion of a saltspoonful of the latter to a pint of water, will answer well for short dissections. For more prolonged ones, a mixture of spirit-and-water, one part of the former to two of the latter, answers well, especially for vegetable structures. When the dilution is first made, the fluid becomes milky, unless pure spirit be used, but with a little trouble the Revenue authorities may be induced to give permission for the use of pure methylated spirit, which answers every purpose. The trouble then is that not less than five gallons can be purchased, an *embarras de richesses* for the average microscopist, but, after all, the spirit is extremely cheap, and does not deteriorate by keeping.

When the dissection in either of these media is completed, spirit should be gradually added to bring the strength up to 50 per cent., in which the preparation may remain for a day or two, after which it is gradually brought into pure spirit, or into water again, according to the medium in which it is to be mounted.

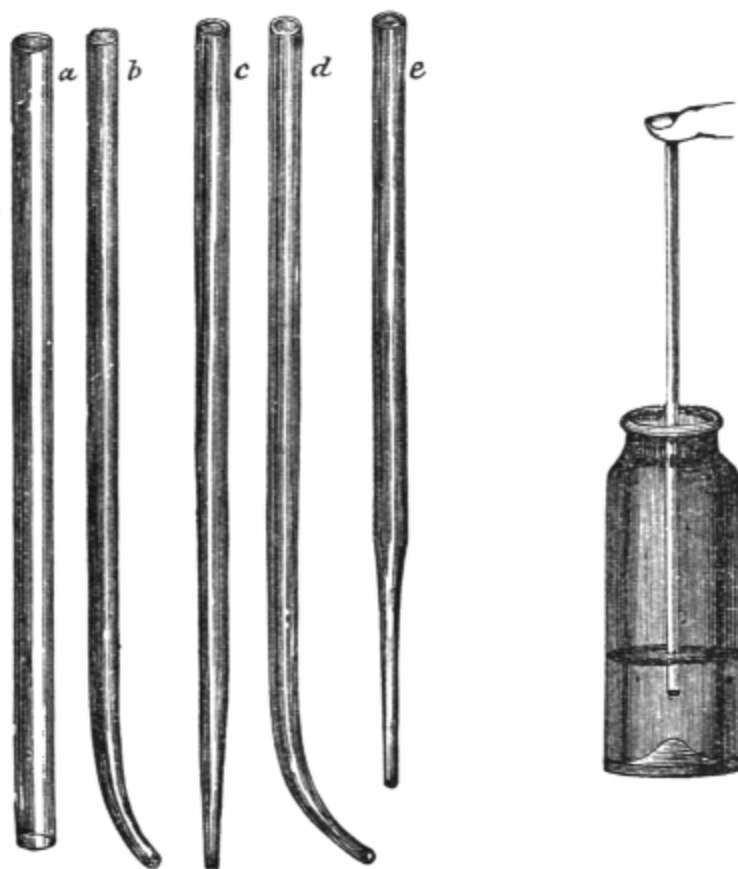


FIG. 12.

As to the tools required, they are neither numerous nor expensive. Fine-pointed but strong forceps (Fig. 9, c), curved and straight; a couple of pairs of scissors, one strong and straight, the other more delicate, and having curved blades, and a few needles of various thicknesses and curves, are the chief ones. The latter may be made by inserting ordinary needles, for three-fourths of their length, into sticks of straight-grained deal (ordinary firewood answers best), and thereafter bending them as required. A better plan, however, is to be provided with a few of the needle-holders shown in Fig. 9, b. These are very simple and inexpensive, and in them broken needles are readily replaced by others. Dipping-tubes, such as are shown in Fig. 12, will also be extremely useful for many purposes. These are very easily made by heating the centre of a piece of soft glass tubing of the required size, and, when it is quite red-hot, drawing the ends apart. The fine

tube in the centre should now be divided by scratching it with a fine triangular file, and the scratch may of course be made at such a point as to afford a tube of the required fineness. The edges should be smoothed by holding them in the flame until they just run (not melt, or the tube will close). These tubes can often be made to supply the place of a glass syringe. They may be used either for sucking up fluid or for transferring it, placing the finger over the wide end, allowing the tube to fill by displacement of air, and then re-closing it with the finger. This last method is especially useful for transferring small objects from one receptacle to another. In speaking of the dissection of objects, it should have been mentioned that the microscope itself may, under careful handling, be made to serve very well, only, as the image is reversed, it is almost impossible to work without using a prism to re-erect the image. Such a prism is shown in Fig. 13. The microscope is placed vertically, and the observer, looking straight into the prism, sees all the parts of the image in their natural positions. This appliance is extremely useful for the purpose of selecting small objects, and arranging them on slides in any desired manner. A few words may be added as to the reproduction of the images of objects.

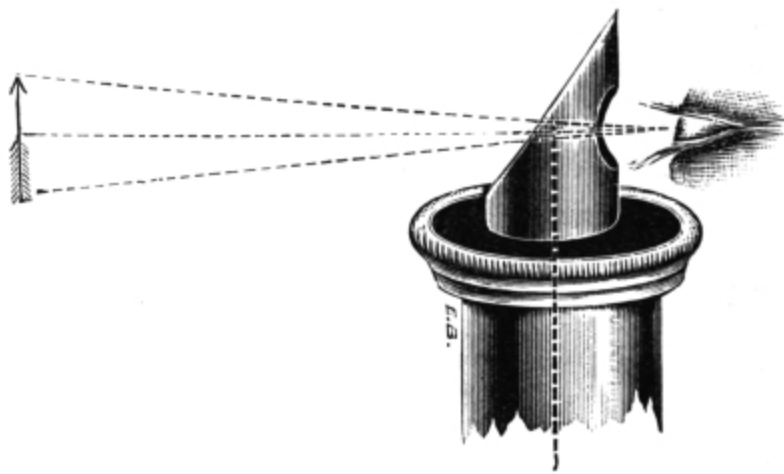


FIG. 13.

The beginner is strongly recommended to practise himself in this from the outset. Even a rough sketch is worth pages of description, especially if the magnification used be appended; and even though the worker may be devoid of artistic talent, he will find that with practice he will acquire a very considerable amount of facility in giving truthful outlines at least of the objects which he views. Various aids have been devised for the purpose of assisting in the process. The simplest and cheapest of these consists of a cork cut so as to fit round the eye-piece. Into the cork are stuck two pins, at an angle of 45° to the plane of the cork, and, the microscope being placed horizontally, a thin cover-glass is placed upon the two pins, the light being arranged and the object focused after the microscope is inclined. On looking vertically down upon the cover-glass, a bright spot of light will be seen, and as the eye is brought down into close proximity with it the spot will expand and allow the observer to see the whole of the image without looking into the microscope. If a sheet of paper be now placed upon the table at the place occupied by the image so projected, the whole of the details will be clearly seen, as will also the point of a pencil placed upon the paper in the centre of the field of view; and, after a little practice, it will be found easy to trace round the chief details of the object. Two points require attention. The first is that if the light upon the paper be stronger than that in the apparent field of the microscope, the image will not be well seen, or if the paper be too feebly lighted, it will be difficult to keep the point of the pencil in view. The light from the microscope is thrown into the eye, and the view of the image upon the paper is the effect of a mental act, the eye looking out in the direction from which the rays appear to come. The paper has therefore to be illuminated independently, and half the battle lies in the adjustment of the relative brightness of image and paper. The second point is, that it is essential to fix one particular point in the image as the starting-point of the drawing, and this being first depicted, the image and drawing of this point must be kept always coincident, or the drawing will be distorted, since the smallest movement of the eye alters the relations of the whole. The reflector must be placed at an angle of 45° , or the field will be oval instead of circular. The simple form of apparatus just described has one

drawback, inasmuch as the reflection is double, the front and back of the cover-glass both acting as reflectors. The image from the latter being much the more feeble of the two, care in illumination will do much to eliminate this difficulty; but there are various other forms in which the defect in question is got rid of. The present writer has worked with all of them, from the simple neutral tint reflector of Beale to the elaborate and costly apparatus of Zeiss, and, upon the whole, thinks that he prefers the cover-glass to them all.

A very simple plan, not so mechanical as the last-named, consists in the use of “drawing-squares,” which are delicate lines ruled upon a piece of thin glass, and dropped into the eye-piece so that the lines rest upon the diaphragm of the eye-piece, and therefore are in focus at the same time as the object. By the use of these, in combination with paper similarly ruled, a diagram of any required size can be drawn with very great facility. The squares, if compared with a micrometer, will furnish an exact standard of magnitude for each object-glass employed. The micrometer is a piece of thin glass upon which are ruled minute divisions of an inch or a millimeter. Suppose the micrometer to be placed under the microscope when the squares are in the eye-piece, and it be found that each division corresponds with one square of the latter, then, if the micrometric division be one one-hundredth of an inch, and the squares upon the paper measure one inch, it is clear that the drawing will represent the object magnified a hundred “diameters”; if two divisions of the micrometer correspond to three squares, the amplification will be a hundred and fifty diameters; if three divisions correspond to two squares, sixty-six diameters, and so on. If a draw-tube be used, it will be necessary to know the value of the squares at each inch of the length, if they are to be used for measuring magnification.

CHAPTER III

Examination of Objects—Principles of Illumination—Mirror and its Action—Substage Condenser—Use of Bull's-eye—Opaque Objects—Photography of Microscopic Objects.

So much depends upon a right method of employing the microscope, as regards both comfort and accuracy, that we propose to devote a little space to the consideration of the subject.

Let us first warn the intending observer against the use of powers higher than are required to bring out the details of the object. Mere magnification is of very little use: it increases the difficulties both of illumination and of manipulation, and, as already said, interferes with that grasp of the object which it is most desirable to obtain. Rather let the beginner lay himself out to get the very most he can out of his lowest powers, and he will find that, by so doing, he will be able far better to avail himself of the higher ones when their use is indispensable.

The essential means to this end is a mastery of the principles of illumination, which we now proceed to describe.

We suppose the microscope to be inclined at an angle of about 70° to the horizontal, with a low-power objective attached to it, a one-inch by preference. Opposite to the microscope, and about a foot away from it, is a lamp with the edge of the flame presented to the microscope, the concave mirror of which is so arranged as to receive the rays from the flame and direct them up the tube of the microscope. Upon the stage is placed a piece of ground-glass, and the mirror-arm is now to be moved up or down upon its support until the ground-glass receives the maximum of illumination, which it will do when the lamp-flame is at one conjugate focus of the mirror and the ground-glass at the other. The focus will not be an image of the flame, but a bar of light.

If an object be now placed upon the stage, instead of the ground-glass, and the objective focused upon it, it will, if the mirror be properly adjusted, be brilliantly illuminated.

It will be understood that every concave mirror has a focus, and converges the rays which fall upon it to this focus, behaving exactly like a convex lens. The principal focus of a concave mirror is its radius of curvature, and this is not difficult to determine. Place side by side a deep cardboard box and the lamp, so that the concave mirror may send the rays back, along a path only slightly inclined to that by which they reached it, to the bottom of the box. The lamp and box being equidistant from the mirror, it is evident that when the mirror forms an image of the former upon the latter equal to the flame in size, we have the equivalent of the equal conjugate foci shown in Fig. 2. Now move the box to the distance from the mirror which corresponds to the distance of the stage of the microscope from the mirror when the latter is in position upon the microscope, and then move the lamp to or fro until the mirror casts a sharp image of the flame upon the bottom of the box, which is not to be moved. The lamp distance so found will be the correct one for working with the concave mirror. The writer is led to lay special stress upon this matter, from the fact that he almost invariably finds that the mirror is arranged to be used for parallel rays, *i.e.* for daylight, and is therefore fixed far too close to the stage to be available for correct or advantageous working with the lamp, unless, indeed, the bull's-eye condenser be used, as hereinafter described, to parallelise the rays from the lamp.

Work done with the concave mirror can, however, under the most favourable conditions, only be looked upon as a *pis aller*. The advantages gained by the use of some substage condenser, even the most simple, in conjunction with the plane mirror, or even without any mirror at all, are so manifold that the beginner is strongly urged to provide himself with some form or other of it, and we now proceed to describe the way in which this should be used to produce the best effect.

To reduce the problem to its most simple elements, turn the mirror altogether out of the way, and place the microscope upon a block at such a height as shall be convenient for observation, and shall allow the rays from the lamp, placed in a line with it on the table, to shine directly into the tube of the microscope. Ascertain that this is so by removing both objective and eye-piece and looking down the tube, when the flame should be seen in the centre, edgewise. Now replace the eye-piece, and screw on to the tube the one-inch combination or objective. Place upon the stage an object, preferably a round diatom or an echinus-spine, and focus it as sharply as possible. Now place the substage condenser in its jacket, and slide it up and down until the image of the object is bisected by the image of the flame.

The centre of the object will now be brilliantly illuminated by rays travelling in the proper direction for yielding the best results. The object is situated at the common focus of the microscope and the condenser, and, whatever means of illumination be adopted, this is the result which should always be aimed at.

Satisfactory as this critical arrangement is, however, from a scientific point of view, it has its drawbacks from an artistic and æsthetic one. It is not pleasant, for most purposes, to have merely the centre of an object lighted up, and we have now to consider how the image of the edge of the flame may be so expanded as to fill the field without sacrificing more than a very small fraction of the accuracy of the arrangement just attained.

Referring to Fig. [1](#), we see that if we place the lamp at the principal focus of a lens, it will emit a bundle of parallel rays equal in diameter to the diameter of the lens. This is the key of the position. We cannot place the lamp at an infinite distance from the substage condenser, but we can supply the latter with rays approximately parallel, so that it shall bring them to a focus upon the object at very nearly its own principal focus. This we do by means of the bull's-eye condenser. Place the latter, with its flat side toward the edge of the flame, and at its principal focal distance (the method of determining which has already been described) from the latter, so that the bundle of parallel rays which issue from it may pass up to the substage

condenser. On examining the object again, it will be found that, after slight adjustments of the position of the bull's-eye have been made, the object lies in the centre of an evenly and brilliantly lighted field.

It may be necessary to place the bull's-eye a little farther from or nearer to the lamp, or to move it a little to one side or the other, but when it is at the correct distance, and on the central line between the lamp and the substage condenser, at right angles to this line, the effects will be as described. It may help in securing this result if we mention that when the bull's-eye is too far from the lamp, the image of the flame is a spindle-shaped one; whilst, when the distance between the two is too short, *i.e.* less than the principal focal length of the lens, the field is crossed by a bar or light, the ends of which are joined by a ring, whilst on either side of the bar there is a semi-circular dark space.

We have hitherto supposed the objects viewed to be transparent, but there are many, of great interest, which are opaque, and call for other means of illumination. Of these there are several. The simplest and, in many ways, the best is to use the bull's-eye condenser to bring to a focus upon the object the rays of light from some source placed above the stage of the microscope. If light can be obtained from the sun itself, no lens will be needed to concentrate it; and indeed, if this were done, there would be considerable risk of burning the object. The light from a white cloud, however, with the help of the bull's-eye, answers admirably. At night-time an artificial source of light, the more intense and the more distant the better, is required. For most cases, and with powers not higher than one inch, a good paraffin lamp, placed about two feet away from the stage, and on one side of it, so as to be about a foot above the level of the object, will give all that is needed. Such a lamp is shown in Fig. 14. Low magnifications are, as a rule, all that is called for in this method.

Lieberkuhn's condensers are useful aids, but are somewhat expensive. They are concave mirrors, which are so adjusted to the objective that the latter and the reflector come into focus together, the light being sent in from below, or from one side.

One other method of illumination must be mentioned before leaving the topic, and this is the illumination of objects upon a “dark field.” With suitable subjects, and when carefully managed, there is no method which gives more beautiful effects, and it has the great advantage of allowing the object to be brilliantly lighted, without the strain to the eyes which is involved in such lighting by the usual method of direct illumination.

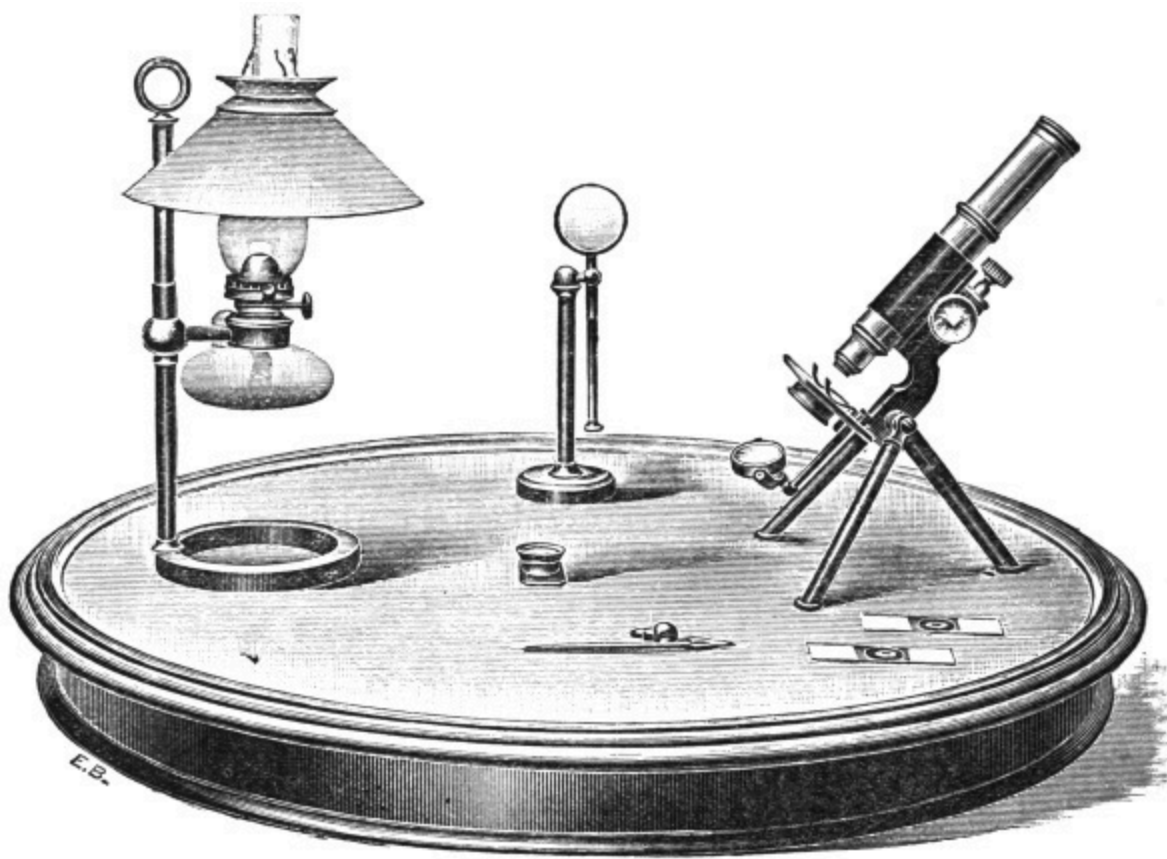


FIG. 14.

It consists essentially in allowing the light to fall upon the object from below, at such an angle that none of it can enter the objective directly. Thus the concave mirror, turned as far as possible to one side, and reflecting on to the object the rays from the lamp placed upon the opposite side, will give very fair results with low powers; this plan, however, is capable of but very limited application. Again, a disc of black paper may be stuck on to the

middle of the bull's-eye, and the latter be placed below the stage between it and the mirror. In this case everything depends upon the size of the disc, which, if too small, will not give a black ground, and if too large will cut off all light from the object.

The best and only really satisfactory plan is to arrange the illumination with the substage condenser, as previously described, and then to place below the lens of the latter a central stop of a suitable size, which can only be determined by trial. When this has been done the object will be seen brilliantly illuminated upon a field of velvety blackness. Such stops are supplied with the condenser.

We have devoted a considerable portion of space to this question, since it is, of all others, the most important to a successful, satisfactory, and reliable manipulation of the microscope; but even now, only the main points of the subject have been touched upon, and the worker will find it necessary to supplement the information given by actual experiment. A few failures, rightly considered, will afford a great amount of information, but those who desire to go thoroughly into the matter are recommended to consult the present writer's *Guide to the Science of Photomicrography*, where it is treated at much greater length, as an essential part of the subject-matter of the book.

It may be added here, that no method of reproducing the images of objects is on the whole so satisfactory as the photographic one; and whilst a lengthened reference to the topic would be out of place in a work of the character of the present one, the one just mentioned will be found to contain all that is necessary to enable the beginner to produce results which, for faithfulness and beauty, far excel any drawing, whilst they have the additional advantage that they can, if required, be exhibited to hundreds simultaneously.

CHAPTER IV

Vegetable Cells and their Structure—Stellate Tissues—Secondary Deposit—Ducts and Vessels—Wood-Cells—Stomata, or Mouths of Plants—The Camera Lucida, and Mode of Using—Spiral and Ringed Vessels—Hairs of Plants—Resins, Scents, and Oils—Bark Cells.

We will now suppose the young observer to have obtained a microscope and learned the use of its various parts, and will proceed to work with it. As with one or two exceptions, which are only given for the purpose of further illustrating some curious structure, the whole of the objects figured in this work can be obtained without any difficulty, the best plan will be for the reader to procure the plants, insects, etc., from which the objects are taken, and follow the book with the microscope at hand. It is by far the best mode of obtaining a systematic knowledge of the matter, as the quantity of objects which can be placed under a microscope is so vast that, without some guide, the tyro flounders hopelessly in the sea of unknown mysteries, and often becomes so bewildered that he gives up the study in despair of ever gaining any true knowledge of it. I would therefore recommend the reader to work out the subjects which are here mentioned, and then to launch out for himself on the voyage of discovery. I speak from experience, having myself known the difficulties under which a young and inexperienced observer has to labour in so wide a field, without any guide to help him to set about his work in a systematic manner.

The objects that can be most easily obtained are those of a vegetable nature, as even in London there is not a square, an old wall, a greenhouse, a florist's window, or even a greengrocer's shop, that will not afford an exhaustless supply of microscopic employment. Even the humble vegetables that make their daily appearance on the dinner-table are highly

interesting; and in a crumb of potato, a morsel of greens, or a fragment of carrot, the enthusiastic observer will find occupation for many hours.

Following the best examples, we will commence at the beginning, and see how the vegetable structure is built up of tiny particles, technically called “cells.”

That the various portions of every vegetable should be referred to the simple cell is a matter of some surprise to one who has had no opportunity of examining the vegetable structure, and indeed it does seem more than remarkable that the tough, coarse bark, the hard wood, the soft pith, the green leaves, the delicate flowers, the almost invisible hairs, and the pulpy fruit, should all start from the same point, and owe their origin to the simple vegetable cell. This, however, is the case; and by means of a few objects chosen from different portions of the vegetable kingdom, we shall obtain some definite idea of this curious phenomenon.

I.

FIG.

- | | |
|---|--|
| 1. Strawberry, cellular tissue | 15. Wood-cells, Elder |
| 2. Buttercup leaf, internal layer | 16. Glandular markings and resin, "Cedar" pencil |
| 3. Privet, Seed Coat, showing star-shaped cells | 17. Do. Yew |
| 4. Rush, Star-shaped cells | 18. Scalariform tissue, Stalk of Fern |
| 5. Mistletoe, cells with ringed fibre | 19. Dotted Duct, Willow |
| 6. Cells from interior of Lilac bud | 20. Do. Stalk of Wheat |
| 7. Bur-reed (<i>Sparganium</i>), square cells from leaf | 21. Wood-cell, Chrysanthemum |
| 8. Six-sided cells, from stem of Lily | 22. Do. Lime-tree |
| 9. Angular dotted cells, rind of Gourd | 23. Dotted Duct, Carrot |
| 10. Elongated ringed cells, anther of Narcissus | 24. Cone-bearing wood, Deal |
| 11. Irregular star-like tissue, pith of Bulrush | 25. Cells, outer coat, Gourd |
| 12. Six-sided cells, pith of elder | 26. Ducts, Elm |
| 13. Young cells from Wheat | 27. Cellular tissue, Stalk of Chickweed |
| 14. Do. rootlets of Wheat | 28. Holly-berry, outer coat |



I.

On Plate I. Fig. 1, may be seen three cells of a somewhat globular form, taken from the common strawberry. Any one wishing to examine these cells for himself may readily do so by cutting a very thin slice from the fruit,

putting it on a slide, covering it with a piece of thin glass (which may be cheaply bought at the optician's, together with the glass slides on which the objects are laid), and placing it under a power of two hundred diameters. Should the slice be rather too thick, it may be placed in the live-box and well squeezed, when the cells will exhibit their forms very distinctly. In their primary form the cells seem to be spherical; but as in many cases they are pressed together, and in others are formed simply by the process of subdivision, the spherical form is not very often seen. The strawberry, being a soft and pulpy fruit, permits the cells to assume a tolerably regular form, and they consequently are more or less globular.

Where the cells are of nearly equal size, and are subjected to equal pressure in every direction, they force each other into twelve-sided figures, having the appearance under the microscope of flat six-sided forms. Fig. 8, in the same Plate, taken from the stem of a lily, is a good example of this form of cell, and many others may be found in various familiar objects.

We must here pause for a moment to define a cell before we proceed further.

The cell is a close sac or bag formed of a substance called from its function "cellulose," and containing certain semi-fluid contents as long as it retains its life. In the interior of the cell may generally be found a little dark spot, termed the "núcleus," and which may be seen in Fig. 1, to which we have already referred. The object of the nucleus is rather a bone of contention among the learned, but the best authorities on this subject consider it to be the vital centre of the cells, to and from which tends the circulation of the protoplasm, and which is intimately connected with the growth and reproduction of the cell. On looking a little more closely at the nucleus, we shall find it marked with several small light spots, which are termed "nucléoli."

On the same Plate (Fig. 2) is a pretty group of cells taken from the internal layer of the buttercup leaf, and chosen because they exhibit the series of tiny and brilliant green dots to which the colour of the leaf is due. The

technical name for this substance is “chlorophyll,” or “leaf-green,” and it may always be found thus dotted in the leaves of different plants, the dots being very variable in size, number, and arrangement. A very fine object for the exhibition of this point is the leaf of *Anácharis*, the “Canadian timber-weed,” to be found in almost every brook and river. It also shows admirably the circulation of the protoplasm in the cell.

In the centre of the same Plate (Fig. [12](#)) is a group of cells from the pith of the elder-tree. This specimen is notable for the number of little “pits” which may be seen scattered across the walls of the cells, and which resemble holes when placed under the microscope. In order to test the truth of this appearance, the specimen was coloured blue by the action of iodine and dilute sulphuric acid, when it was found that the blue tint spread over the pits as well as the cell-walls, showing that the membrane is continuous over the pits.

Fig. [7](#) exhibits another form of cell, taken from the *Spargánium*, or bur-reed. These cells are tolerably equal in size, and have assumed a square shape. They are obtained from the lower part of the leaf. The reader who has any knowledge of entomology will not fail to observe the similarity in form between the six-sided and square cells of plants and the hexagonal and square facets of the compound eyes of insects and crustaceans. In a future page these will be separately described.

Sometimes the cells take most singular and unexpected shapes, several examples of which will be briefly noticed.

In certain loosely made tissues, such as are found in the rushes and similar plants, the walls of the cells grow very irregularly, so that they push out a number of arms which meet each other in every direction, and assume the peculiar form which is termed “stellate,” or star-shaped tissue. Fig. [3](#) shows a specimen of stellate tissue taken from the seed-coat of the privet, and rather deeply coloured, exhibiting clearly the beautiful manner in which the arms of the various stars meet each other. A smaller group of stellate cells

taken from the stem of a large rush, and exemplifying the peculiarities of the structure, are seen in Fig. 4.

The reader will at once see that this mode of formation leaves a vast number of interstices, and gives great strength with little expenditure of material. In water-plants, such as the reeds, this property is extremely valuable, as they must be greatly lighter than the water in which they live, and at the same time must be endowed with considerable strength in order to resist its pressure.

A less marked example of stellate tissue is given in Fig. 11, where the cells are extremely irregular, in their form, and do not coalesce throughout. This specimen is taken from the pithy part of a bulrush. There are very many other plants from which the stellate cells may be obtained, among which the orange affords very good examples, in the so-called “white” that lies under the yellow rind, a section of which may be made with a very sharp razor, and placed in the field of the microscope.

Looking toward the bottom of the Plate, and referring to Fig. 27, the reader will observe a series of nine elongated cells, placed end to end, and dotted profusely with chlorophyll. These are obtained from the stalk of the common chickweed. Another example of the elongated cell is seen in Fig. 14, which is a magnified representation of the rootlets of wheat. Here the cells will be seen set end to end, and each containing its nucleus. On the left hand of the rootlet (Fig. 13) is a group of cells taken from the lowest part of the stem of a wheat plant which had been watered with a solution of carmine, and had taken up a considerable amount of the colouring substance. Many experiments on this subject were made by the Rev. Lord S. G. Osborne, and may be seen at full length in the pages of the *Microscopical Journal*, the subject being too large to receive proper treatment in the very limited space which can here be given to it. It must be added that later researches have caused the results here described to be gravely disputed.

Fig. 9 on the same Plate exhibits two notable peculiarities—the irregularity of the cells and the copiously pitted deposit with which they are covered. The irregularity of the cells is mostly produced by the way in which the multiplication takes place, namely, by division of the original cell into two or more new ones, so that each of these takes the shape which it assumed when a component part of the parent cell. In this case the cells are necessarily very irregular, and when they are compressed from all sides they form solid figures of many sides, which, when cut through, present a flat surface marked with a variety of irregular outlines. This specimen is taken from the rind of a gourd.

The “pitted” structure which is so well shown in this figure is caused by a layer of matter which is deposited in the cell and thickens its walls, and which is perforated with a number of very minute holes called “pits.” This substance is called “secondary deposit.” That these pits do not extend through the real cell-wall has already been shown in Fig. 12.

This secondary deposit assumes various forms. In some cases it is deposited in rings round the cell, and is clearly placed there for the purpose of strengthening the general structure. Such an example may be found in the mistletoe (Fig. 5), where the secondary deposit has formed itself into clear and bold rings that evidently give considerable strength to the delicate walls which they support. Fig. 10 shows another good instance of similar structure; differing from the preceding specimen in being much longer and containing a greater number of rings. This object is taken from an anther of the narcissus. Among the many plants from which similar objects may be obtained, the yew is perhaps one of the most prolific, as ringed wood-cells are abundant in its formation, and probably aid greatly in giving to the wood the strength and elasticity which have long made it so valuable in the manufacture of bows.

Before taking leave of the cells and their remarkable forms, we will just notice one example which has been drawn in Fig. 6. This is a congeries of cells, containing their nuclei, starting originally end to end, but swelling and dividing at the top. This is a very young group of cells (a young hair, in

fact) from the inner part of a lilac bud, and is here introduced for the purpose of showing the great similarity of all vegetable cells in their earliest stages of existence.

Having now examined the principal forms of cells, we arrive at the “vessels,” a term which is applied to those long and delicate tubes which are formed of a number of cells set end to end, their walls of separation being absorbed.

In Fig. 19 the reader will find a curious example of the “pitted vessel,” so called from the multitude of little markings which cover its walls, and are arranged in a spiral order. Like the pits and rings already mentioned, the dots are composed of secondary deposit in the interior of the tube, and vary very greatly in number, function, and dimensions. This example is taken from the wood of the willow, and is remarkable for the extreme closeness with which the dots are packed together.

Immediately on the right hand of the preceding figure may be seen another example of a dotted vessel (Fig. 20), taken from a wheat stem. In this instance the cells are not nearly so long, but are wider than in the preceding example, and are marked in much the same way with a spiral series of dots. About the middle of the topmost cell is shown the short branch by which it communicates with the neighbouring vessel.

Fig. 23 exhibits a vessel taken from the common carrot, in which the secondary deposit is placed in such a manner as to resemble a net of irregular meshes wrapped tightly round the vessel. For this reason it is termed a “netted vessel.” A very curious instance of these structures is given in Fig. 26, at the bottom of the Plate, where are represented two small vessels from the wood of the elm. One of them—that on the left hand—is wholly marked with spiral deposit, the turns being complete; while, in the other instance, the spiral is comparatively imperfect, and the cell-walls are marked with pits. If the reader would like to examine these structures more attentively, he will find plenty of them in many familiar garden vegetables,

such as the common radish, which is very prolific in these interesting portions of vegetable nature.

There is another remarkable form in which this secondary deposit is sometimes arranged that is well worthy of our notice. An example of this structure is given in Fig. [18](#), taken from the stalk of the common fern or brake. It is also found in very great perfection in the vine. On inspecting the illustration, the reader will observe that the deposit is arranged in successive bars or steps, like those of a winding staircase. In allusion to the ladder-like appearance of this formation, it is called “scalariform” (Latin, *scala*, a ladder).

In the wood of the yew, to which allusion has already been made, there is a very peculiar structure, a series of pits found only in those trees that bear cones, and therefore termed the coniferous pitted structure. Fig. [16](#) is a section of a common cedar pencil, the wood, however, not being that of the true cedar, but of a species of fragrant Juniper. This specimen shows the peculiar formation which has just been mentioned.

Any piece of deal or pine will exhibit the same peculiarities in a very marked manner, as is seen in Fig. [24](#). A specimen may be readily obtained by making a very thin shaving with a sharp plane. In this example the deposit has taken a partially spiral form, and the numerous circular pits with which it is marked are only in single rows. In several other specimens of coniferous woods, such as the Araucaria, or Norfolk Island pine, there are two or three rows of pits.

A peculiarly elegant example of this spiral deposit may be seen in the wood of the common yew (Fig. [17](#)). If an exceedingly thin section of this wood be made, the very remarkable appearance will be shown which is exhibited in the illustration. The deposit has not only assumed the perfectly spiral form, but there are two complete spirals, arranged at some little distance from each other, and producing a very pretty effect when seen through a good lens.

The pointed, elongated shape of the wood-cells is very well shown in the common elder-tree (see Fig. 15). In this instance the cells are without markings, but in general they are dotted like Fig. 21, an example cut from the woody part of the chrysanthemum stalk. This affords a very good instance of the wood-cell, as its length is considerable, and both ends are perfect in shape. On the right hand of the figure is a drawing of the wood-cell found in the lime-tree (Fig. 22), remarkable for the extremely delicate spiral markings with which it is adorned. In these wood-cells the secondary deposit is so plentiful that the original membranous character of the cell-walls is entirely lost, and they become elongated and nearly solid cases, having but a very small cavity in their centre. It is to this deposit that the hardness of wood is owing, and the reader will easily see the reason why the old wood is so much harder than the young and new shoots. In order to permit the passage of the fluids which maintain the life of the part, it is needful that the cell-wall be left thin and permeable in certain places, and this object is attained either by the “pits” described on page 43, or by the intervals between the spiral deposit.

At the right-hand bottom corner of Plate I. (Fig. 28) may be seen a prettily marked object, which is of some interest. It is a slice stripped from the outer coat of the holly-berry, and is given for the purpose of illustrating the method by which plants are enabled to breathe the atmospheric air on which they depend as much as ourselves, though their respiration is slower. Among the mass of net-like cells may be seen three curious objects, bearing a rather close resemblance to split kidneys. These are the mouths, or “stómata,” as they are scientifically called.

In the centre of the mouths may be seen a dark spot, which is the aperture through which the air communicates with the passages between the cells in the interior of the structure. In the flowering plants their shape is generally rounded, though they sometimes take a squared form, and they regularly occur at the meeting of several surface cells. The two kidney-shaped cells which form the “mouth” are the “guard-cells,” so called from their function, since, by their change of form, they cause the mouth to open or shut,

according to the needs of the plant. In young plants these guard-cells are very little below the surface of the leaf or skin, but in others they are sunk quite beneath the layer of cells forming the outer coat of the tissue. There are other cases where they are slightly elevated above the surface.

Stomata are found chiefly in the green portions of plants, and are most plentiful on the under side of leaves. It is, however, worthy of notice, that when an aquatic leaf floats on the water, the mouths are only to be found on the upper surface. These curious and interesting objects are to be seen in many structures where we should hardly think of looking for them; for instance, they may be found existing on the delicate skin which envelops the kernel of the common walnut. As might be expected, their dimensions vary with the character of the leaf on which they exist, being large upon the soft and pulpy leaves, and smaller upon those of a hard and leathery consistence. The reader will find ample amusement, and will gain great practical knowledge of the subject, by taking a plant, say a tuft of groundsel, and stripping off portions of the external skin or “epidermis” from the leaf or stem, etc., so as to note the different sizes and shapes of the stomata.

On the opposite bottom corner of Plate I. Fig. 25, is an example of a stoma taken from the outer skin of a gourd, and here given for the purpose of showing the curious manner in which the cells are arranged about the mouth, no less than seven cells being placed round the single mouth, and the others arranged in a partially circular form around them.

Turning to Plate II., we find several other examples of stomata, the first of which (Fig. 1) is obtained from the under surface of the buttercup leaf, by stripping off the external skin, or “epidermis,” as it is scientifically termed. The reader will here notice the slightly waved outlines of the cell-walls, together with the abundant spots of chlorophyll with which the leaf is coloured. In this example the stomata appear open. Their closure or expansion depends chiefly on the state of the weather; and, as a general rule, they are open by day and closed at night.

A remarkably pretty example of stomata and elongated cells is to be obtained from the leaf of the common iris, and may be prepared for the microscope by simply tearing off a strip of the epidermis from the under side of the leaf, laying it on a slide, putting a little water on it, and covering it with a piece of thin glass. (See Plate II. Fig. 2.) There are a number of longitudinal bands running along the leaf where these cells and stomata appear. The latter are not placed at regular intervals, for it often happens that the whole field of the microscope will be filled with cells without a single stoma, whilst elsewhere a group of three or four may be seen clustered closely together.

Fig. 3 on the same Plate exhibits a specimen of the beautifully waved cells, without mouths, which are found on the upper surface of the ivy leaf. These are difficult to arrange from the fresh leaf, but are easily shown by steeping the leaf in water for some time, and then tearing away the cuticle. The same process may be adopted with many leaves and cuticles, and in some cases the immersion must be continued for many days, and the process of decomposition aided by a very little nitric acid in the water, or by boiling.

On the same Plate are three examples of spiral and ringed vessels, types of an endless variety of these beautiful and interesting structures. Fig. 4 is a specimen of a spiral vessel taken from the lily, and is a beautiful example of a double spire. The deposit which forms this spiral is very strong, and it is to the vast number of these vessels that the stalk owes its well-known elasticity. In many cases the spiral vessels are sufficiently strong to be visible to the naked eye, and to bear uncoiling. For example, if a leaf-stalk of geranium be broken across, and the two fragments gently drawn asunder, a great number of threads, drawn from the spiral vessels, will be seen connecting the broken ends. In this case the delicate membranous walls of the vessel are torn apart, and the stronger fibre which is coiled spirally within it unrolls itself in proportion to the force employed. In many cases these fibres are so strong that they will sustain the weight of an inch or so of the stalk.

In Fig. 5 is seen a still more bold and complex form of this curious structure; being a coil of five threads, laid closely against each other, and forming, while remaining in their natural position, an almost continuous tube. This specimen is taken from the root of the water lily, and requires some little care to exhibit its structure properly.

Every student of nature must be greatly struck with the analogies between different portions of the visible creation. These spiral structures which we have just examined are almost identical in appearance, and to some extent in their function, with the threads that are coiled within the breathing tubes of insects. This is in both cases twofold, namely, to give support and elasticity to a delicate membrane, and to preserve the tube in its proper form, despite the bending to which it may be subjected. When we come to the anatomy of the insect in a future page we shall see this structure further exemplified.

In some cases the deposit, instead of forming a spiral coil, is arranged in a series of rings, and the vessel is then termed “annulated.” A very good example of this formation is given in Fig. 6, which is a sketch of such a vessel, taken from a stalk of the common rhubarb. To see these ringed vessels properly, the simplest plan is to boil the rhubarb until it is quite soft, then to break down the pulpy mass until it is flattened, to take some of the most promising portions with the forceps, lay them on the slide and press them down with a thin glass cover. They will not be found scattered at random through the fibres, which elsewhere present only a congeries of elongated cells, but are seen grouped together in bundles, and with a little trouble may be well isolated, and the pulpy mass worked away so as to show them in their full beauty. As may be seen in the illustration, the number of the rings and their arrangement is extremely variable. A better, but somewhat more troublesome, plan is to cut longitudinal sections of the stem, as described in our concluding chapter, when not only the various forms of cells and vessels, but their relations to each other, will be well shown. The numerous crystals of oxalate of lime, which make rhubarb so injurious a food for certain persons, will also be well seen. These crystals

are called “raphides,” and are to be found in very many plants in different forms.

II.

FIG.

- | | |
|------------------------------------|---|
| 1. Cuticle, Buttercup leaf | 24. Do. Pine cone |
| 2. Do. Iris | 25. Vitta, Caraway Seed |
| 3. Do. Ivy leaf | 26. Cork |
| 4. Spiral vessel, Lily | 27. Hair, Flower of Garden Verbena |
| 5. Do. root, (rhizome) Water Lily | 28. Do. fruit of Plane |
| 6. Ringed vessel, Rhubarb | 29. Do. do. |
| 7. Chaff, after burning | 30. Do. do. |
| 8. Bifid hair, Arabis | 31. Do. Lobelia |
| 9. Hair, Marvel of Peru | 32. Do. Cabbage |
| 10. End of hair, leaf of Hollyhock | 33. Do. Dead-nettle flower |
| 11. Hair, Sowthistle leaf | 34. Do. Garden Verbena flower |
| 12. Do. Tobacco | 35. Fruit-hair, Dandelion |
| 13. Do. Southernwood | 36. Hair, Thistle leaf |
| 14. Group of hairs, Hollyhock leaf | 37. Do. Cactus |
| 15. Hair, Yellow Snapdragon | 38. Do. do. |
| 16. Do. Moneywort | 39. Do. Virginian Spider-wort |
| 17. Hair, Geum | 40. Do. Lavender |
| 18. Do. Flower of Heartsease | 41. Section, Lavender leaf, Hairs and perfume-gland |
| 19. Do. Dockleaf | 42. Section, Orange Peel |
| 20. Do. Throat of Pansy | 43. Sting of Nettle |
| 21. Do. Dead-nettle Flower | 44. Hair, Marigold flower |
| 22. Do. Groundsel | 45. Do. Ivy |
| 23. Cell, Beech-nut | |



II.

The hairs of plants form very interesting objects, and are instructive to the student, as they afford valuable indications of the mode in which plants grow. They are all appendages of and arise from the skin or epidermis; and

although their simplest form is that of a projecting and elongated cell, the variety of shapes which are assumed by these organs is inexhaustible. On Plate [II](#). are examples of some of the more striking forms, which will be briefly described.

The simple hair is well shown in Figs. 18, 19, and 32, the first being from the flower of the heartsease, the second from a dock-leaf, and the third from a cabbage. In Fig. [18](#) the hair is seen to be but a single projecting cell, consisting only of a wall and the contents. In Fig. [19](#) the hair has become more decided in shape, having assumed a somewhat dome-like form; and in Fig. [32](#) it has become considerably elongated, and may at once be recognised as a true hair.

In Fig. [8](#) is a curious example of a hair taken from the white Arabis, one of the cruciferous flowers, which is remarkable for the manner in which it divides into two branches, each spreading in opposite directions. Another example of a forked hair is seen in Fig. [13](#), but in this instance the hair is composed of a chain of cells, the three lower forming the stem of the hair, and the two upper being lengthened into the lateral branches. This hair is taken from the common southernwood.

In most cases of long hairs, the peculiar elongation is formed by a chain of cells, varying greatly in length and development. Several examples of these hairs will be seen on the same Plate.

Fig. [9](#) is a beaded hair from the Marvel of Peru, which is composed of a number of separate cells placed end to end, and connected by slender threads in a manner that strongly reminds the observer of a chain of beads strung loosely together, so as to show the thread by which they are connected with each other. Another good example is seen at Fig. [11](#), in a hair taken from the leaf of the sowthistle. In this case the beads are strung closely together, and when placed under a rather high power of the microscope have a beautifully white and pearly aspect. The leaf must be dry and quite fresh, and the hairs seen against the green of the leaf. Fig. [39](#) represents another beaded hair taken from the Virginian Spiderwort, or

Tradescantia. This hair is found upon the stamens, and is remarkable for the beautifully beaded outline, the fine colouring, and the spiral markings with which each cell is adorned.

A still further modification of these many-celled hairs is found in several plants, where the hairs are formed by a row of ordinarily shaped cells, with the exception of the topmost cell, which is suddenly elongated into a whip-like form. Fig. [22](#) represents a hair of this kind, taken from the common groundsel; and Fig. [36](#) is a still more curious instance, found upon the leaf of the thistle. The reader may have noticed the peculiar white “fluffy” appearance of the thistle leaf when it is wet after a shower of rain. This appearance is produced by the long lash-like ends of the hairs, which are bent down by the weight of the moisture, and lie almost at right angles with the thicker portions of the hair.

An interesting form of hair is seen in the “sting” of the common nettle. This may readily be examined by holding a leaf edgewise in the stage forceps, and laying it under the field of the microscope. In order to get the proper focus throughout the hair, the finger should be kept upon the screw movement, and the hair brought gradually into focus from its top to its base. The general structure of this hair is not unlike that which characterises the fang of a venomous serpent. The acrid fluid which causes the pain is situated in the enlarged base of the hair, and is forced through the long straight tubular extremity by means of the pressure exerted when the sting enters the skin. At the very extremity of the perfect sting is a slight bulb-like swelling, which serves to confine the acrid juice, and which is broken off on the least pressure. The sting is seen in Fig. [43](#).

The extremities of many hairs present very curious forms, some being long and slender, as in the examples already mentioned, while others are tipped with knobs, bulbs, clubs, or rosettes in endless variety.

Fig. [12](#) is a hair of the tobacco leaf, exhibiting the two-celled gland at the tip, containing the peculiar principle of the plant, known by the name of “nicotine.” The reader will see how easy it is to detect adulteration of

tobacco by means of the microscope. The leaves most generally used for this purpose are the dock and the cabbage, so that if a very little portion of leaf be examined the character of the hairs will at once inform the observer whether he is looking at the real article or its substitute.

Fig. [15](#) is a hair from the flower of the common yellow snapdragon, which is remarkable for the peculiar shape of the enlarged extremity, and for the spiral markings with which it is decorated. Fig. [16](#) is a curious little knobbed hair found upon the moneywort, and Fig. [17](#) is an example of a double-knobbed hair taken from the Geum. Fig. [34](#) affords a very curious instance of a glandular hair, the stem being built up of cells disposed in a very peculiar fashion, and the extremity being developed into a beautiful rosette-shaped head. This hair came from the Garden Verbena.

Curiously branched hairs are not at all uncommon, and some very good and easily obtained examples are given on Plate II.

Fig. [28](#) is one of the multitude of branched hairs that surround the well-known fruit of the plane-tree, the branches being formed by some of the cells pointing outward. These hairs do not assume precisely the same shape; for Fig. [29](#) exhibits another hair from the same locality, on which the spikes are differently arranged, and Fig. [30](#) is a sketch of another such hair, where the branches have become so numerous and so well developed that they are quite as conspicuous as the parent stem.

One of the most curious and interesting forms of hair is that which is found upon the lavender leaf, and which gives it the peculiar bloom-like appearance on the surface.

This hair is represented in Figs. 40 and 41. On Fig. [40](#) the hair is shown as it appears when looking directly upon the leaf, and in Fig. [41](#) a section of the leaf is given, showing the mode in which the hairs grow into an upright stem, and then throw out horizontal branches in every direction. Between the two upright hairs, and sheltered under their branches, may be seen a glandular appendage not unlike that which is shown in Fig. [16](#). This is the reservoir containing the perfume, and it is evidently placed under the

spreading branches for the benefit of their shelter. On looking upon the leaf by reflected light the hairs are beautifully shown, extending their arms on all sides; and the globular perfume cells may be seen scattered plentifully about, gleaming like pearls through the hair-branches under which they repose. They will be found more numerous on the under side of the leaf.

This object will serve to answer a question which the reader has probably put to himself ere this, namely, Where are the fragrant resins, scents, and oils stored? On Plate I. Fig. 16, will be seen the reply to the first question; Fig. 41 of the present Plate has answered the second question, and Fig. 42 will answer the third. This figure represents a section of the rind of an orange, the flattened cells above constituting the delicate yellow skin, and the great spherical object in the centre being the reservoir in which the fragrant essential oil is stored. The covering is so delicate that it is easily broken, so that even by handling an orange some of the scent is sure to come off on the hands, and when the peel is stripped off and bent double, the reservoirs burst in myriads, and fling their contents to a wonderful distance. This may be easily seen by squeezing a piece of orange peel opposite a lighted candle, and noting the distance over which the oil will pass before reaching the flame, and bursting into little flashes of light. Other examples are given on the same plate.

Returning to the barbed hairs, we may see in Fig. 35 a highly magnified view of the “pappus” hair of a dandelion, *i.e.* the hairs which fringe the arms of the parachute-like appendage which is attached to the seed. The whole apparatus will be seen more fully on Plate III. Figs. 44, 45, 46. This hair is composed of a double layer of elongated cells lying closely against each other, and having the ends of each cell jutting out from the original line. A simpler form of a double-celled, or more properly a “duplex” hair, will be seen in Fig. 44. This is one of the hairs from the flower of the marigold and has none of the projecting ends to the cells.

In some instances the cell-walls of the hairs become greatly hardened by secondary deposit, and the hairs are then known as spines. Two examples of these are seen in Figs. 37 and 38, the former being picked from the Indian

fig-cactus, and well known to those persons who have been foolish enough to handle the fig roughly before feeling it. The wounds which these spines will inflict are said to be very painful, and have been compared to those produced by the sting of the wasp. The latter hair is taken from the *Opuntia*. These spines must not be confounded with thorns; which latter are modified branches.

Fig. [10](#) represents the extreme tip of a hair from the hollyhock leaf, subjected to a lens of very high power.

Many hairs assume a star-like appearance, an aspect which may be produced in different ways. Sometimes a number of simple hairs start from the same base, and by radiating in different directions produce the stellate effect. An example of this kind of hair may be seen in Fig. [14](#), which is a group of hairs from the hollyhock leaf. There is another mode of producing the star-shape which may be seen in Fig. [45](#), a hair taken from the leaf of the ivy. Very fine examples may also be found upon the leaf of *Deutzia scabra*.

Hairs are often covered with curious little branches or protuberances, and present many other peculiarities of form which throw a considerable light upon certain problems in scientific microscopy.

Fig. [33](#) represents a hair of two cells taken from the flower of the well-known dead-nettle, which is remarkable for the number of knobs scattered over its surface. A similar mode of marking is seen in Fig. [31](#), a club-shaped hair covered with external projections, found in the flower of the *Lobelia*. In order to exhibit these markings well, a power of two hundred diameters is needed. Fig. [21](#) shows this dotting in another hair from the dead-nettle, where the cell is drawn out to a great length, but is still covered with these markings.

Fig. [20](#) is an example of a very curious hair taken from the throat of the pansy. This hair may readily be obtained by pulling out one of the petals, when the hairs will be seen at its base. Under the microscope it has a particularly beautiful appearance, looking just like a glass walking-stick

covered with knobs, not unlike those huge, knobby club-like sticks in which some farmers delight, where the projections have been formed by the pressure of a honeysuckle or other climbing plant.

A hair of a similar character, but even more curious, is found in the same part of the flower of the Garden Verbena (see Fig. [27](#)), and is not only beautifully translucent, but is coloured according to the tint of the flower from which it is taken. Its whole length is covered with large projections, the joints much resembling the antennæ of certain insects; and each projection is profusely spotted with little dots, formed by elevation of the outer skin or cuticle. These are of some value in determining the structure of certain appearances upon petals and other portions of the flowers, and may be compared with Figs. 33 to 35 on Plate [III](#).

Fig. [26](#) offers an example of the square cells which usually form the bark of trees. This is a transverse section of cork, and perfectly exhibits the form of bark cells. The reader is very strongly advised to cut a delicate section of the bark of various trees, a matter very easily accomplished with the aid of a sharp razor and a steady hand.

Fig. [24](#) is a transverse section through one of the scales of a pine-cone, and is here given for the purpose of showing the numerous resin-filled cells which it displays. This may be compared with Fig. [16](#) of Plate I. Fig. [25](#) is a part of one of the “vittæ,” or oil reservoirs, from the fruit of the caraway, showing the cells containing the globules of caraway oil. This is rather a curious object, because the specimen from which it was taken was boiled in nitric acid, and yet retained some of the oil globules. Immediately above it may be seen (Fig. [23](#)) a transverse section of the beechnut, showing a cell with its layers of secondary deposit.

In the cuticle of the grasses and the mare’s-tails is deposited a large amount of pure flint. So plentiful is this substance, and so equally is it distributed, that it can be separated by heat or acids from the vegetable parts of the plant, and will still preserve the form of the original cuticle, with its cell-walls, stomata, and hairs perfectly well defined.

Fig. **Z**, Plate II., represents a piece of wheat chaff, or “bran,” that has been kept at a white heat for some time, and then mounted in Canada balsam. I prepared the specimen from which the drawing was made by laying the chaff on a piece of platinum, and holding it over the spirit-lamp. A good example of the silex or flint in wheat is often given by the remains of a straw fire, where the stems may be seen still retaining their tubular form but fused together into a hard glassy mass. It is this substance that cuts the fingers of those who handle the wild grasses too roughly, the edges of the blades being serrated with flinty teeth, just like the obsidian swords of the ancient Mexicans, or the shark’s-tooth falchion of the New Zealander.

These are but short and meagre accounts of a very few objects, but space will not permit of further elucidation, and the purpose of this little work is not to exhaust the subjects of which it treats, but to incite the reader to undertake investigation on his own account, and to make his task easier than if he had done it unaided.

CHAPTER V

Starch, its Growth and Properties—Surface Cells of Petals—Pollen and its Functions—Seeds.

The white substance so dear to the laundries under the name of starch is found in a vast variety of plants, being distributed more widely than most of the products which are found in the interior of vegetable cells.

The starch grains are of very variable size even in the same plant, and their form is as variable as their size, though there is a general resemblance in those of the same plant which allows of their being fairly easily identified after a moderate amount of practice. Sometimes the grains are found loosely packed in the interior of the cells, and are then easily recognised as starch grains by their peculiar form and the delicate lines with which they are marked; but in many places they are pressed so closely together that they assume an hexagonal shape under the microscope, and bear a close resemblance to ordinary twelve-sided cells. In other plants, again, the grains never advance beyond the very minute form in which they seem to commence their existence; and in some, such as the common oat, a great number of very little granules are compacted together so as to resemble one large grain.

There are several methods of detecting starch in those cases where its presence is doubtful; and the two modes that are usually employed are polarised light and the iodide of potassium. When polarised light is employed—a subject on which we shall have something to say presently—the starch grains assume the characteristic “black-cross,” and when a plate of selenite is placed immediately beneath the slide containing the starch grains, they glow with all the colours of the rainbow. The second plan is to treat them with a very weak solution of iodine and iodide of potassium, and in this case the iodine has the effect on the starch granules of staining them blue. They are so susceptible of this reaction that when the liquid is too

strong the grains actually become black from the amount of iodine which they imbibe.

Nothing is easier than to procure starch granules in the highest perfection. Take a raw potato, and with a razor cut a very thin slice from its interior, the direction of the cut not being of the slightest importance. Put this delicate slice upon a slide, drop a little water upon it, cover it with a piece of thin glass, give it a good squeeze, and place it under a power of a hundred or a hundred and fifty diameters. Any part of the slice, provided that it be very thin, will then present the appearance shown in Plate III. Fig. 9, where an ordinary cell of potato is seen filled loosely with starch grains of different sizes. Around the edges of the slice a vast number of starch granules will be seen, which have been squeezed out of their cells by pressure, and are now floating freely in the water. As cold water has no perceptible effect upon starch, the grains are not altered in form by the moisture, and can be examined at leisure.

III.

FIG.

- | | |
|------------------------------------|--|
| 1. Laurel leaf, transverse section | 27. Do. Heath, another species |
| 2. Starch, Wheat | 28. Pollen, Furze |
| 3. Do. from Pudding | 29. Do. Tulip |
| 4. Do. Potato | 30. Petal, Pelargonium |
| 5. Outer Skin, Capsicum pod | 31. Do. Periwinkle |
| 6. Starch, Parsnip | 32. Do. Golden Balsam |
| 7. Do. Arrow Root, West Indian | 33. Do. Snapdragon |
| 8. Do. "Tousles Mois" | 34. Do. Primrose |
| 9. Do. in cell of Potato | 35. Do. Scarlet Geranium |
| 10. Do. Indian Corn | 36. Pollen, Crocus |
| 11. Do. Sago | 37. Do. Hollyhock |
| 12. Do. Tapioca | 38. Fruit, Galium, Goosegrass |
| 13. Root, Yellow Water-Lily | 39. A hook of ditto more magnified |
| 14. Starch, Rice | 40. Seed, Red Valerian |
| 15. Do. Horsebean | 41. Portion of Parachute of same, more magnified |
| 16. Do. Oat | 42. Seed, Foxglove |
| 17. Pollen, Snowdrop | 43. Do. Sunspurge |
| 18. Do. Wallflower | 44. Parachute, Dandelion seed |
| 19. Do. Willow Herb, a pollen tube | 45. Seed, Dandelion |
| 20. Do. Violet | 46. Do. Hair of Parachute |
| 21. Do. Musk Plant | 47. Do. Yellow Snapdragon |
| 22. Do. Apple | 48. Do. Mullein |
| 23. Do. Dandelion | 49. Do. Robin Hood |
| 24. Do. Sowthistle | 50. Do. Bur-reed |
| 25. Do. Lily | 51. Do. Willow Herb |
| 26. Do. Heath | 52. Do. Musk Mallow |



III.

On focusing with great care, the surface of each granule will be seen to be covered with very minute dark lines, arranged in a manner which can be readily comprehended from Fig. 4, which represents two granules of potato

starch as they appear when removed from the cell in which they took their origin. All the lines evidently refer to the little dark spots at the end of the granule, called technically the “hilum,” and represent the limits of successive layers of material deposited one after another. The lines in question are very much better seen if the substage condenser be used with a small central stop, so as to obtain partial dark-field illumination. Otherwise they are often very difficult of detection.

In the earliest stages of their growth the starch granules appear to be destitute of these markings, or at all events they are so few and so delicate as not to be visible even with the most perfect instruments, and it is not until the granules assume a comparatively large size that the external markings become distinctly perceptible.

We will now glance at the examples of starch which are given in the Plate, and which are a very few out of the many that might be figured. Fig. 2 represents the starch of wheat, the upper grain being seen in front, the one immediately below it in profile, and the two others being examples of smaller grains. Fig. 6 is a specimen of a very minute form of starch, where the granules do not seem to advance beyond their earliest stage. This specimen is obtained from the parsnip; and although the magnifying power is very great, the dimensions of the granules are exceedingly small, and except by a very practised eye they would not be recognisable as starch grains.

Fig. 3 is a good example of a starch grain of wheat, exemplifying the change that takes place by the combined effects of heat and moisture. It has already been observed that cold water exercises little, if any, perceptible influence upon starch; but it will be seen from the illustration that hot water has a very powerful effect. When subjected to the action of water at a temperature over 140° Fahr., the granule swells rapidly, and at last bursts, the contents escaping in a gelatinous mass, and the external membrane collapsing into the form which is shown in Fig. 3, which was taken out of a piece of hot pudding. A similar form of wheat starch may also be detected

in bread, accompanied, unfortunately, by several other substances not generally presumed to be component parts of the “staff of life.”

In Fig. [7](#) are represented some grains of starch from West Indian arrowroot, and Fig. [8](#) exhibits the largest kind of starch grain known, obtained from the tuber of a species of canna, supposed to be *C. edulis*, a plant similar in characteristics to the arrowroot. The popular name of this starch is “Tous les Mois,” and under that title it may be obtained from the opticians, or chemists.

Fig. [10](#) shows the starch granules from Indian corn, as they appear before they are compressed into the honeycomb-like structure which has already been mentioned. Even in that state, however, if they are treated with iodine, they exhibit the characteristics of starch in a very perfect manner. Fig. [11](#) is starch from sago, and Fig. [12](#) from tapioca, and in both these instances the several grains have been injured by the heat employed in preparing the respective substances for the market.

Fig. [13](#) exhibits the granules obtained from the root of the water-lily, and Fig. [14](#) is a good example of the manner in which the starch granules of rice are pressed together so as to alter the shape and puzzle a novice. Fig. [16](#) is the compound granule of the oat, which has already been mentioned, together with some of the simple granules separated from the mass; and Fig. [15](#) is an example of the starch grains obtained from the underground stem of the horse-bean. It is worthy of mention that the close adhesion of the rice starch into those masses is the cause of the peculiar grittiness which distinguishes rice flour to the touch.

Whilst very easily acted on by heat, starch-granules are very resistant to certain other reagents. Weak alkalies, in watery solution, readily attack them, but by treating portions of plants with caustic potash dissolved in strong spirit, the woody and other parts may be dissolved away; and after repeated washing with spirit the starch may be mounted. This, however, must never be in any glycerine medium, except that given on p. [172](#).

In Plate III. Fig. 1, may be seen a curious little drawing, which is a sketch of the laurel-leaf cut transversely, and showing the entire thickness of the leaf. Along the top may be seen the delicate layer of “varnish” with which the surface of the leaf is covered, and which serves to give to the foliage its peculiar polish. This varnish is nothing more than the translucent matter which binds all the cells together, and which is poured out very liberally upon the surface of the leaf. The lower part of this section exhibits the cells of which the leaf is built, and towards the left hand may be seen a cut end of one of the veins of the leaf, more rightly called a wood-cell.

We will now examine a few examples of surface cells.

Fig. 5 is a portion of epidermis stripped from a Capsicum pod, exhibiting the remains of the nuclei in the centre of each cell, together with the great thickening of the wall-cells and the numerous pores for the transmission of fluid. This is a very pretty specimen for the microscope, as it retains its bright red colour, and even in old and dried pods exhibits the characteristic markings.

In the centre of the Plate may be seen a wheel-like arrangement of the peculiar cells found on the petals of six different flowers, all easily obtainable, and mounted without difficulty.

Fig. 30 is the petal of a geranium (Pelargonium), a very common object on purchased slides. It is a most lovely subject for the microscope, whether it be examined with a low or a high power,—in the former instance exhibiting a most beautiful “stippling” of pink, white, and black, and in the latter showing the six-sided cells with their curious markings.

In the centre of each cell is seen a radiating arrangement of dark lines with a light spot in the middle, looking very like the mountains on a map. These lines were long thought to be hairs; but Mr. Tuffen West, in an interesting and elaborate paper on the subject, has shown their true nature. From his observations it seems that the beautiful velvety aspect of flower petals is owing to these arrangements of the surface cells, and that their rich brilliancy of colour is due to the same cause. The centre of each cell-wall is

elevated as if pushed up by a pointed instrument from the under side of the wall, and in different flowers this elevation assumes different forms. Sometimes it is merely a slight wart on the surface, sometimes it becomes a dome, while in other instances it is so developed as to resemble a hair. Indeed, Mr. West has concluded that these elevations are nothing more than rudimentary hairs.

The dark radiating lines are shown by the same authority to be formed by wrinkling of the membrane forming the walls of the elevated centre, and not to be composed of “secondary deposit,” as has generally been supposed.

Fig. 31 represents the petal of the common periwinkle, differing from that of the geranium by the straight sides of the cell-walls, which do not present the toothed appearance so conspicuous in the former flower. A number of little tooth-like projections may be seen on the interior of the cells, their bases affixed to the walls and their points tending toward the centre, and these teeth are, according to Mr. West, formed of secondary deposit.

In Fig. 32 is shown the petal of the common garden balsam, where the cells are elegantly waved on their outlines, and have plain walls. The petal of the primrose is seen in Fig. 34, and that of the yellow snapdragon in Fig. 33; in the latter instance the surface cells assume a most remarkable shape, running out into a variety of zigzag outlines that quite bewilders the eye when the object is first placed under the microscope. Fig. 35 is the petal of the common scarlet geranium.

In several instances these petals are too thick to be examined without some preparation, and glycerine will be found well adapted for that purpose. The young microscopist must, however, beware of forming his ideas from preparations of dried leaves, petals, or hairs, and should always procure them in their fresh state whenever he desires to make out their structure. Even a fading petal should not be used, and if the flowers are gathered for the occasion, their stalks should be placed in water, so as to give a series of leaves and petals as fresh as possible.

We now pass from the petal of the flower to the pollen, that coloured dust, generally yellow or white, which is found upon the stamens, and which is very plentiful in many flowers, such as the lily and the hollyhock.

This substance is found only upon the stamens or anthers of full-blown flowers (the anthers being the male organs), and is intended for the purpose of enabling the female portion of the flower to produce fertile seeds. In form the pollen grains are wonderfully diverse, affording an endless variety of beautiful shapes. In some cases the exterior is smooth and marked only with minute dots, but in many instances the outer wall of the pollen grain is covered with spikes, or decorated with stripes or belts. A few examples of the commonest forms of pollen will be found on Plate III.

Fig. 17 is the pollen of the snowdrop, which, as will be seen, is covered with dots and marked with a definite slit along its length. The dots are simply tubercles in the outer coat of the grain, and are presumed to be formed for the purpose of strengthening the membrane, otherwise too delicate, upon the same principle which gives to “corrugated” iron such strength in proportion to the amount of material. Fig. 18 is the pollen of the wall-flower, shown in two views, and having many of the same characteristics as that of the snowdrop. Fig. 19 is the pollen of the willow-herb, and is here given as an illustration of the manner in which the pollen aids in the germination of plants.

In order to understand its action, we must first examine its structure.

All pollen-grains are furnished with some means by which their contents when thoroughly ripened can be expelled. In some cases this end is accomplished by sundry little holes called pores; in others, certain tiny lids are pushed up by the contained matter; and in some, as in the present instance, the walls are thinned in certain places so as to yield to the internal pressure.

When a ripe pollen-grain falls upon the stigma of a flower, it immediately begins to swell, and seems to “sprout” like a potato in a damp cellar, sending out a slender “pollen-tube” from one or other of the apertures

already mentioned. In Fig. 19 a pollen-tube is seen issuing from one of the projections, and illustrates the process better than can be achieved by mere verbal description. The pollen-tubes insinuate themselves between the cells of the stigmas, and, continually elongating, worm their way down the “style” until they come in contact with the “ovules.” By very careful dissection of a fertilised stigma, the beautiful sight of the pollen-tubes winding along the tissues of the style may be observed under a high power of the microscope.

The pollen-tube is nothing more than the interior coat of the grain, very much developed, and filled with a substance technically named “fovilla,” composed of “protoplasm” (the semi-liquid substance which is found in the interior of cells), very minute starch grains, and some apparently oily globules.

In order to examine the structure of the pollen-grains properly, they should be examined under various circumstances—some dry, others placed in water to which a little sugar has been added, others in oil, and it will often be found useful to try the effect of different acids upon them.

Fig. 20 is the pollen of the common violet, and is easily recognisable by its peculiar shape and markings. Fig. 21 is the pollen of the musk-plant, and is notable for the curious mode in which its surface is belted with wide and deep bands, running spirally round the circumference. Fig. 22 exhibits the pollen of the apple, and Fig. 23 affords a very curious example of the raised markings upon the surface of the dandelion pollen. In Fig. 24 there are also some very wonderful markings, but they are disposed after a different fashion, forming a sort of network upon the surface, and leaving several large free spaces between the meshes. The pollen of the lily is shown in Fig. 25, and is a good example of a pollen-grain covered with the minute dottings which have already been described.

Figs. 26 and 27 show two varieties of compound pollen, found in two species of heath. These compound pollen-grains are not of unfrequent occurrence, and are accounted for in the following manner.

The pollen is formed in certain cavities within the anthers, by means of the continual subdivision of the “parent-cells” from which it is developed. In many cases the form of the grain is clearly owing to the direction in which these cells have divided, but there is no great certainty on this subject. It will be seen, therefore, that if the process of subdivision be suddenly arrested, the grains will be found adhering to each other in groups of greater or smaller size, according to the character of the species and the amount of subdivision that has taken place. The reader must, however, bear in mind that the whole subject is as yet rather obscure, and that further discovery may throw doubt on many theories which at present are accepted as established.

Fig. 28 shows the pollen of the furze, in which are seen the longitudinal slits and the numerous dots on the surface; and Fig. 29 is the curiously shaped pollen of the tulip. The two large yellow globular figures at each side of the Plate represent the pollen of two common flowers; Fig. 36 being that of the crocus, and Fig. 37 a pollen-grain of the hollyhock. As may be seen from the illustration, the latter is of considerable size, and is covered with very numerous projections. These serve to raise the grain from a level surface, over which it rolls with a surprising ease of motion, so much so indeed that if a little of this substance be placed on a slide and a piece of thin glass laid over it, the glass slips off as soon as it is in the least inclined, and forces the observer to fix it with paper or cement before he can place it on the inclined stage of the microscope. The little projections have a very curious effect under a high power, and require careful focusing to observe them properly; for the diameter of the grain is so large that the focus must be altered to suit each individual projection. Their office is, probably, to aid in fertilisation.

The seeds of plants are even easier of examination than the pollen, and in most cases require nothing but a pocket lens and a needle for making out their general structure. The smaller seeds, however, must be placed under the microscope, many of them exhibiting very curious forms. The external

coat of seeds is often of great interest, and needs to be dissected off before it can be rightly examined. The simplest plan in such a case is to boil the seed well, press it while still warm into a plate of wax, and then dissect with a pair of needles, forceps, and scissors under water. Many seeds may also be mounted in cells as dry objects, after being thoroughly dried themselves.

A few examples of the seeds of common plants are given at the bottom of Plate III.

Fig. 38 exhibits the fruit, popularly called the seed, of the common goosegrass, or Galium, which is remarkable for the array of hooklets with which it is covered. Immediately above the figure may be seen a drawing of one of the hooks much magnified, showing its sharp curve (Fig. 39). It is worthy of remark that the hook is not a simple curved hair, but a structure composed of a number of cells terminating in a hook.

Fig. 40 shows the seed, or rather the fruit, of the common red valerian, and is introduced for the purpose of showing its plumed extremity, which acts as a parachute, and causes it to be carried about by the wind until it meets with a proper resting-place. It is also notable for the series of strong longitudinal ribs which support its external structure. On Fig. 41 is shown a portion of one of the parachute hairs much more magnified.

The seed of the common dandelion, so dear to children in their play-hours, when they amuse themselves by puffing at the white plummy globes which tip the ripe dandelion flower-stalks, is a very interesting object even to their parents, on account of its beautiful structure, and the wonderful way in which it is adapted to the place which it fills. Fig. 45 represents the seed portion of one of these objects, together with a part of the parachute stem, the remainder of that appendage being shown lying across the broken stem.

The shape of the seed is not unlike that of the valerian, but it is easily distinguished from that object by the series of sharp spikes which fringe its upper end, and which serve to anchor the seed firmly as soon as it touches the ground. From this end of the seed proceeds a long slender shaft, crowned at its summit by a radiating plume of delicate hairs, each of which

is plentifully jagged on its surface, as may be seen in Fig. 46, which shows a small portion of one of these hairs greatly magnified. These jagged points are evidently intended to serve the same purpose as the spikes below, and to arrest the progress of the seed as soon as it has found a convenient spot.

Fig. 42 is the seed of the foxglove, and Fig. 43 the seed of the sunspurge, or milkwort. Fig. 47 shows the seed of the yellow snapdragon; remarkable for the membranous wing with which the seed is surrounded, and which is composed of cells with partially spiral markings. When viewed edgewise, it looks something like Saturn with his ring, or, to use a more homely but perhaps a more intelligible simile, like a marble set in the middle of a penny. Fig. 48 is a seed of mullein, covered with net-like markings on its external surface. These are probably to increase the strength of the external coat, and are generally found in the more minute seeds.

On Fig. 50 is shown a seed of the burr-reed; a structure which is remarkable for the extraordinary projection of the four outer ribs, and their powerful armature of reverted barbs. Fig. 51 shows another form of parachute seed, found in the willow-herb, where the parachute is not expanded nearly so widely as that of the valerian; neither is it set upon a long slender stem like that of the dandelion, but proceeds at once from the top of the seed, widening towards the extremity, and having a very comet-like appearance. Two more seeds only remain, Fig. 49 being the seed of Robin Hood, and the other, Fig. 52, that of the muskmallow, being given in consequence of the thick coat of hairs with which it is covered.

Many seeds can be well examined when mounted in Canada balsam.

CHAPTER VI

Algæ and their Growth—Desmidiaceæ, where found—Diatoms,
their Flinty Deposit—Volvox—Mould, Blight, and Mildew
—Mosses and Ferns—Mare's-Tail and the Spores—
Common Sea-weeds and their Growth.

On Plate [IV](#). will be seen many examples of the curious vegetables called respectively algæ and fungi, which exhibit some of the lowest forms of vegetable life, and are remarkable for their almost universal presence in all parts of this globe, and also almost all conditions of cold, heat, or climate. Many of them are well known under the popular name of sea-weeds, others are equally familiar under the titles of “mould,” “blight,” or “mildew,” while many of the minuter kinds exhibit such capability of motion, and such apparent symptoms of volition, that they have long been described as microscopic animalcules, and thought to belong to the animal rather than to the vegetable kingdoms.

Fig. [1](#) represents one of the very lowest forms of vegetable life, being known to the man of science as the *Palmella*, and to the general public as “gory dew.” It may be seen on almost any damp wall, extending in red patches of various sizes, looking just as if some blood had been dashed on the wall, and allowed to dry there. With a tolerably powerful lens this substance can be resolved into the exceedingly minute cells depicted in the figure. Generally, these cells are single, but in many instances they are double, owing to the process of subdivision by which the plant grows, if such a term may be used.

Fig. [2](#) affords an example of another very low form of vegetable, the *Palmoglæa*, that green slimy substance which is so common on damp stones. When placed under the microscope, this plant is resolvable into a multitude of green cells, each being surrounded with a kind of gelatinous substance. The mode of growth of this plant is very simple. A line appears

across one of the cells, and after a while it assumes a kind of hour-glass aspect, as if a string had been tied tightly round its middle. By degrees the cell fairly divides into two parts, and then each part becomes surrounded with its own layer of gelatine, so as to form two separate cells, placed end to end.

One of the figures, that on the right hand, represents the various processes of “conjugation,” *i.e.* the union and fusion together of two cells. Each cell throws out a little projection; these meet together, and then uniting, form a sort of isthmus connecting the two main bodies. This rapidly widens, until the two cells become fused into one large body. The whole subject of conjugation is very interesting, and is treated at great length in the *Micrographic Dictionary* of Messrs. Griffith and Henfrey, a work to which the reader is referred for further information on many of the subjects that, in this small work, can receive but a very hasty treatment.

Few persons would suppose that the slug-like object on Fig. 3, the little rounded globules with a pair of hair-like appendages, and the round disc with a dark centre, are only different forms of the same organism. Such, however, is the case, and these are three of the modifications which the *Protococcus* undergoes. This vegetable may be seen floating like green froth on the surface of rain-water.

On collecting some of this froth and putting it under the microscope, it is seen to consist of a vast number of little green bodies, moving briskly about in all directions, and guiding their course with such apparent exercise of volition that they might very readily be taken for animals. It may be noticed that the colour of the plant is sometimes red, and in that state it has been called the *Hæmatococcus*.

The “still” state of this plant is shown in the round disc. After a while the interior substance splits into two portions; these again subdivide, and the process is repeated until sixteen or thirty-two cells become developed out of the single parent-cell. These little ones then escape, and, being furnished with two long “cilia” or thread-like appendages, whirl themselves merrily

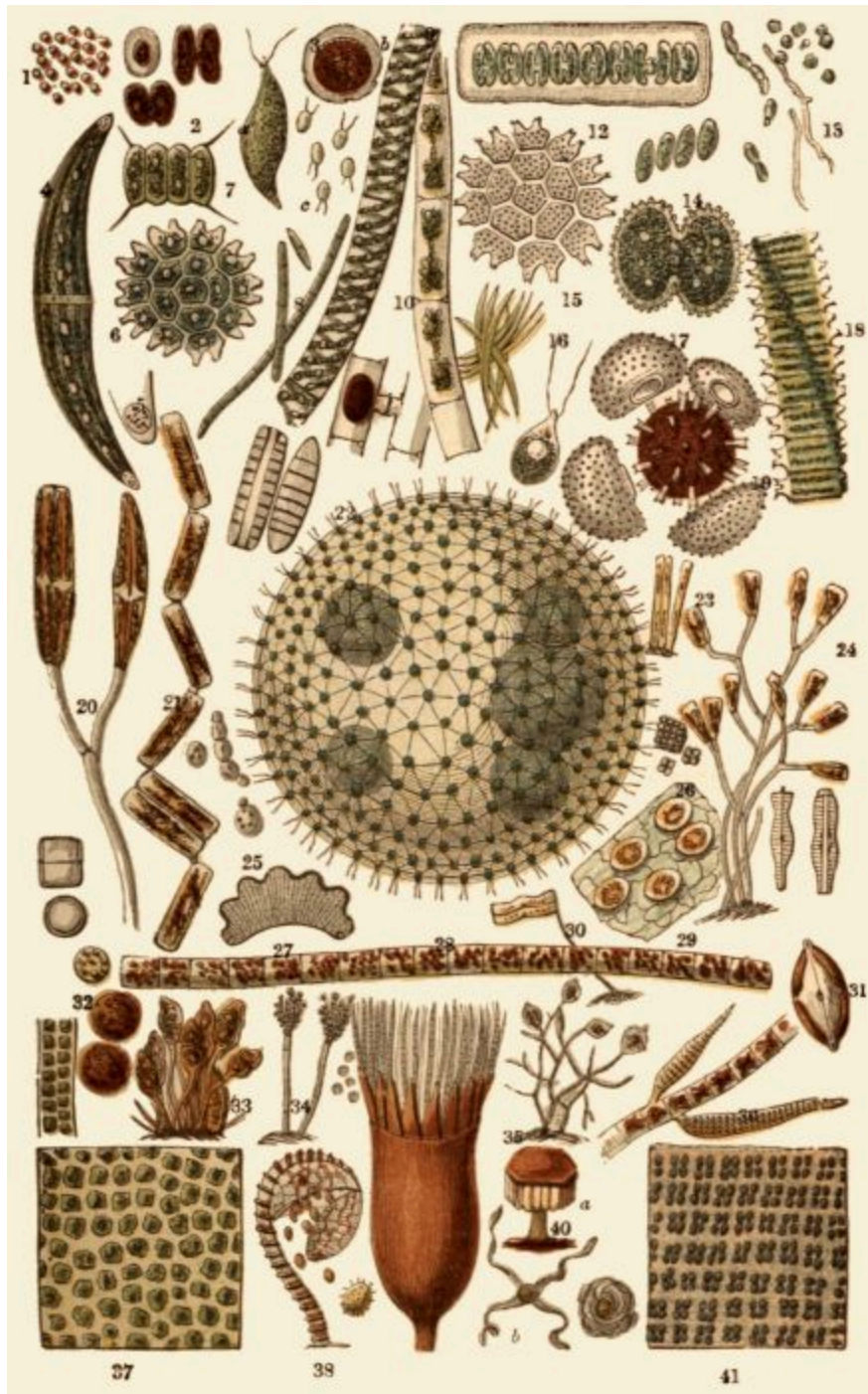
through the water. When they have spent some time in this state, growing all the while, they lose their cilia, become clothed with a strong envelope, and pass into the still stage from which they had previously emerged. This curious process is repeated in endless succession, and causes a very rapid growth of the plant. The moving bodies are technically called zoospores, or living spores, and are found in many other plants besides those of the lowest order.

IV.

FIG.

- | | |
|--|--|
| 1. Gory Dew, <i>Palmella cruenta</i> | Do. single green body, above |
| 2. <i>Palmoglæa macrococca</i> | 23. <i>Synedra</i> |
| 3. <i>Protococcus pluvialis</i> ,
<i>a</i> , in its motile,

<i>b</i> , in its fixed state,
<i>c</i> , zoospores | 24. <i>Gomphonema acuminatum</i>
Do. larger frustules, below |
| 4. <i>Closterium</i> | 25. Yeast |
| 5. Ditto, end more magnified | 26. <i>Sarcina ventriculi</i> |
| 6. <i>Pediastrum</i> | 27. <i>Eunotia diadema</i> |
| | 28. <i>Melosira varians</i>
Do. two bleached frustules |
| 7. <i>Scenedesmus</i> | 29. <i>Cocconeis pediculus</i> |
| 8. <i>Oscillatoria</i> | 30. <i>Achnanthes exilis</i> |
| 9. <i>Spirogyra</i> | 31. <i>Navicula amphisbœna</i> |
| 10. <i>Tyndaridea</i> | 32. Uredo, "Red-rust" of corn |
| 11. Do. spore | 33. <i>Puccinea</i> , Mildew of corn |
| 12. <i>Sphærozosma</i> | 34. <i>Botrytis</i> , mould on grapes
Do. Sporules, beside it |
| 13. <i>Chlorococcus</i> | 35. Do. <i>parasitica</i> , Potato blight |
| 14. <i>Scenedesmus</i> | 36. <i>Ectocarpus siliculosus</i> |
| | 37. <i>Ulva latissima</i> |
| 15. <i>Pediastrum</i> , to show cells | 38. <i>Polypodium</i>
Do. single spore, below |
| 16. <i>Ankistrodesmus</i> | 39. Moss capsule, <i>Hypnum</i> |
| 17. <i>Cosmarium</i> | 40. Mare's tail, <i>Equisetum</i> , <i>a</i> |
| 18. <i>Desmidium</i> | Do. do. <i>b</i> and <i>c</i> |
| 19. <i>Cosmarium</i> , formation of Resting Spore | 41. <i>Porphyra laciniata</i> |
| 20. <i>Cocconema lanceolatum</i> | |
| 21. <i>Diatoma vulgare</i>
Do. larger frustules, at the side | |
| 22. <i>Volvox globator</i> | |



IV.

On Fig. 13 is delineated a very minute plant, called from its colour Chlorococcus. It may be found upon tree-trunks, walls, etc., in the form of

green dust, and has recently been found to take part in forming the first stage of lichens.

A large and interesting family of the “confervoid algæ,” as these low forms of vegetable life are termed, is the Desmidiaceæ, called in more common parlance desmids. A few examples of this family are given in Plate [IV](#).

They may be found in water, always preferring the cleanest and the brightest pools, mostly congregating in masses of green film at the bottom of the water, or investing the stems of plants. Their removal is not very easy, but is best accomplished by very carefully taking up this green slippery substance in a spoon, and straining the water away through fine muslin. They may also be separated by allowing a ring, covered with muslin, to float upon the surface of the water collected in a jar, for, being great lovers of light, they assemble where it is most abundant. An opaque jar should be used. For preservation, glycerine-gelatine seems to be the best fluid. A very full and accurate description of these plants may be found in Ralfs’ *British Desmidiæ*.

Fig. [4](#) represents one of the species of Closterium, more than twenty of which are known. These beautiful objects can be obtained from the bottom of almost every clear pool, and are of some interest on account of the circulating currents that may be seen within the living plants. A high power is required to see this phenomenon clearly. The Closteria are reproduced in various ways. Mostly they divide across the centre, being joined for a while by two half-cells. Sometimes they reproduce by means of conjugation, the process being almost entirely conducted on the convex sides. Fig. [5](#) represents the end of a Closterium, much magnified in order to show the actively moving bodies contained within it.

Fig. [16](#) is a supposed desmid, called Ankistrodesmus, and presumed to be an earlier stage of Closterium.

Fig. [6](#) is a very pretty desmid called the Pediastrum, and valuable to the microscopist as exhibiting a curious mode of reproduction. The figure shows a perfect plant composed of a number of cells arranged

systematically in a star-like shape; Fig. 15 is the same species without the colouring matter, in order to show the shape of the cells. The *Pediastrum* reproduces by continual subdivision of the contents of each cell into a number of smaller cells, termed “gonidia” on account of their function on the perpetuation of the species. When a sufficient number has been formed, they burst through the envelope of the original cell, taking with them a portion of its internal layer, so as to form a vesicle, in which they move actively. In a few minutes they arrange themselves in a circle, and after a while they gradually assume the perfect form, the whole process occupying about two days. Fig. 18 exhibits an example of the genus *Desmidium*. In this genus the cells are either square or triangular in their form, having two teeth at their angles, and twisted regularly throughout their length, causing the wavy or oblique lines which distinguish them. The plants of this genus are common, and may be found almost in any water. I may as well mention that I have obtained nearly all the preceding species, together with many others, from a little pond on Blackheath.

Fig. 7 is another desmid called *Scenedesmus*, in which the cells are arranged in rows of from two to ten in number, the cell at each extremity being often furnished with a pair of bristle-like appendages. Fig. 14 is another species of the same plant, and both may be found in the water supplied for drinking in London, as well as in any pond.

A common species of desmid is seen at Fig. 12, called *Sphærozosma*, looking much like a row of stomata set chainwise together. It multiplies by self-division.

Fig. 17 is a specimen of desmid named *Cosmarium*, plentifully found in ponds on heaths and commons, and having a very pretty appearance in the microscope, with its glittering green centre and beautifully transparent envelope. The manner in which the *Cosmarium* conjugates is very remarkable, and is shown at Fig. 19.

The two conjugating cells become very deeply cleft, and by degrees separate, suffering the contents to pour out freely, and, as at present appears,

without any envelope to protect them. The mass, however, soon acquires an envelope of its own, and by degrees assumes a dark reddish-brown tint. It is now termed a sporangium, and is covered with a vast number of projections, which in this genus are forked at their tip, but in others, which also form sporangia, are simply pointed. The Closteria conjugate after a somewhat similar manner, and it is not unfrequent to find a pair in this condition, but in their case the sporangium is quite smooth on its surface.

Another very remarkable family of confervoid algæ is that which is known under the name of Oscillatoria, from the oscillating movement of the plant. They are always long and filamentous in character, and may be seen moving up and down with a curious irregularity of motion. Their growth is extremely rapid, and may be watched under a tolerably powerful lens, thus giving many valuable hints as to the mode by which these plants are reproduced. One of the commonest species is represented at Fig. 8.

Figs. 9, 10, and 11 are examples of another family, called technically the Zygnemaceæ, because they are so constantly yoked together by conjugation. They all consist of a series of cylindrical cells, set end to end, and having their green contents arranged in similar patterns. Two of the most common and typical species are here given.

Fig. 9 is the Spirogyra, so called from the spiral arrangement of the chlorophyll; and Fig. 10 is the Tyndaridea, or Zygnema, as it is called by some writers. A casual inspection will show how easy it is to distinguish the one from the other. Fig. 11 represents a portion of the Tyndaridea during the process of conjugation, showing the tube of connection between the cells and one of the spores.

We now arrive at the diatoms, so called because of their method of reproduction, in which it appears as if a cut were made right along the original cell. The commonest of these plants is the Diatoma vulgare, seen in Fig. 21 as it appears while growing. The reproduction of this plant is effected by splitting down the centre, each half increasing to the full size of

the original cell; and in almost every specimen of water taken from a pond, examples of this diatom undergoing the process of division will be distinguished. It also grows by conjugation. The diatoms are remarkable for the delicate shell or flinty matter which forms the cell skeleton, and which will retain its shape even after intense heat and the action of nitric acid. While the diatoms are alive, swimming through the water, their beautiful markings are clearly distinct, glittering as if the form were spun from crystalline glass. Just above the figure, and to the right hand, are two outlines of single cells of this diatom, the one showing the front view and the other the profile.

Fig. [20](#) is an example of a diatom—*Cocconéma lanceolátum*—furnished with a stalk. The left-hand branch sustains a “frustule” exhibiting the front view, while the other is seen sideways.

Another common diatom is shown in Fig. [23](#), and is known by the name of *Synedra*. This constitutes a very large genus, containing about seventy known species. In this genus the frustules are at first arranged upon a sort of cushion, but in course of time they mostly break away from their attachment. In some species they radiate in every direction from the cushion, like the spikes of the ancient cavalier’s mace.

Fig. [24](#) is another stalked diatom called *Gomphonéma acuminátum*, found commonly in ponds and ditches. There are nearly forty species belonging to this genus. A pair of frustules are also shown which exhibit the beautiful flinty outline without the coloured contents (technically called endochrome).

Fig. [27](#) is a side view of a beautiful diatom, called *Eunótia diadéma* from its diadem-like form. There are many species of this genus. When seen upon the upper surface, it looks at first sight like a mere row of cells with a band running along them; but by careful arrangement of the light its true form may easily be made out.

Fig. [28](#) represents a very common fresh-water diatom, named *Melosíra várians*. The plants of this genus look like a cylindrical rod composed of a

variable number of segments, mostly cylindrical, but sometimes disc-shaped or rounded. An end view of one of the frustules is seen at the left hand, still coloured with its dots of “endochrome,” and showing the cylindrical shape. Immediately above is a figure of another frustule seen under both aspects with the endochrome removed.

A rather curious species of diatom, called *Cocconeis pediculus*, is seen at Fig. 29 as it appears on the surface of common water-cress. Sometimes the frustules, which in all cases are single, are crowded very closely upon each other and almost wholly hide the substance on which they repose. Fig. 30 is another diatom of a flag-like shape, named *Achnanthes*, having a long slender filament attached to one end of the lower frustule, representing the flag-staff. There are many wonderful species of such diatoms, some running almost end to end like a bundle of sticks, and therefore called *Bacillária*; others spreading out like a number of fans, such as the genus *Licmophora*; while some assume a beautiful wheel-like aspect, of which the genus *Meridion* affords an excellent example.

A very remarkable, and not uncommon, fresh-water diatom is the *Bacillária paradóxa*. It looks, when at rest, like a broad brown ribbon of varying length. The diatoms lie across the ribbon, on edge, and slide upon each other exactly like the ladders of a fire-escape, so that the broad ribbon is converted into a fine long thread, which speedily closes up again into the original ribbon, and so *da capo*. The reason for this movement, and how it is effected, is absolutely unknown; indeed, nothing certain is known as to the way in which diatoms move, nor has ever a probable guess yet been made.

The last of the diatoms which we shall be able to mention in this work is that represented on Fig. 31. The members of this genus have the name of *Navícula*, on account of their boat-like shape and their habit of gliding through the water in a canoe-like fashion. There are many species of this genus, all of which are notable for the graceful and varied courses formed by their outlines, and the extreme delicacy of their markings. In many species the markings are so extremely minute that they can only be made

out with the highest powers of the microscope and the most careful illumination, so that they serve as test objects whereby the performance of a microscope can be judged by a practical man.

The large spherical figure in the centre of Plate [IV](#). represents an example of a family belonging to the confervoid algæ, and known by the name of *Volvox globator*. There seems to be but one species known.

This singular plant has been greatly bandied about between the vegetable and animal kingdoms, but seems now to be satisfactorily settled among the vegetables. In the summer it may be found in pools of water, sufficiently large to be visible to the naked eye, like a little green speck proceeding slowly through the water. When a moderate power is used, it appears as shown in the figure, and always contains within its body a number of smaller individuals, which after a while burst through the envelope of the parent and start upon an independent existence. On a closer examination, a further generation may be discovered even within the bodies of the children. The whole surface is profusely covered with little green bodies, each being furnished with a pair of movable cilia, by means of which the whole organism is moved through the water. These bodies are analogous to the zoospores already mentioned, and are connected with each other by a network of filaments. Reproduction also takes place by conjugation as in other algæ. A more magnified representation of one of the green bodies is shown immediately above the larger figure. The volvox is apt to die soon when confined in a bottle.

Fig. [25](#) is the common yeast-plant, consisting simply of a chain of cells, which increase by budding, and only form spores when they have exhausted the nutriment in the fluid in which they live. Fig. [26](#) is a curious object, whose scientific name is *Sárcina ventrículi*. It is found in the human stomach. Similar forms are often to be found in the air; for instance, a piece of cocoa-nut will exhibit this, and many other kinds of Bacteria and moulds, after a few days' exposure to the air, preferably in a dark cupboard.

We now come upon a few of the blights and mildews. A very interesting series of forms is first to be alluded to. Upon the bramble-leaf may often be found spots, at first red, then orange, then reddish black. These are known as *Æcidium berberidis*. Fig. 32 shows the “red-rust” of wheat, the *Urédo*; and Fig. 33 is the mildew of corn, known as *Puccinia*. The interest lies in the fact that these three forms are successive stages in the life-history of the same plant. Another species of *Urédo*, together with a *Phragmídium*, once thought to be another kind of fungus, is seen on a rose-leaf on Plate V. Fig. 1. On Fig. 10, however, of the same Plate, the *Phragmídium* may be seen proceeding from *Urédo*, thus proving them to be but two states of the same plant. There is room for any amount of observation and work in connection with the life-histories of many of these fungi.

Another species of *Puccinia*, found on the thistle, is shown on Plate V. Fig. 7. Fig. 34 is the mould found upon decaying grapes, and called therefrom, or from the clustered spores, *Botrýtis*. Some of the detached spores are seen by its side. Fig. 35 is another species of the same genus, termed *Botrýtis parasítica*, and is the cause of the well-known “potato-disease.”

The mosses and ferns afford an endless variety of interesting objects to the microscopist; but as their numbers are so vast, and the details of their structure so elaborate, they can only be casually noticed in the present work. Fig. 38 represents a spore-case of the *Polypodium*, one of the ferns, as it appears while in the act of bursting and scattering the contents around. One of the spores is seen more magnified below. The spore-cases of many ferns may be seen bursting under the microscope, and have a very curious appearance, writhing and twisting like worms, and then suddenly filling the field with a cloud of spores. Fig. 9, Plate V., is a piece of the brown, chaff-like, scaly structure found at the base of the stalk of male fern cells, showing the manner in which a flat membrane is formed. Fig. 39 is a capsule of the *Hypnum*, one of the mosses, showing the beautiful double fringe with which its edge is crowned. Fig. 2, Plate V., is the capsule of

another moss, *Polytrichum*, to show the toothed rim; on the right hand is one of the teeth much more magnified.

Fig. 3, Plate V., is the capsule of the *Jungermannia*, one of the liverworts, showing the “elaters” bursting out on every side, and scattering the spores. Fig. 4 is a single elater much magnified, showing it to be a spirally coiled filament, that, by sudden expansion, shoots out the spores just as a child’s toy-gun discharges the arrow. Fig. 5 is a part of the leaf of the *Sphagnum* moss, common in fresh water, showing the curious spiral arrangement of secondary fibre which is found in the cells, as well as the circular pores which are found in each cell at a certain stage of growth. Just below, and to the left hand, is a single cell greatly magnified, in order to show these peculiarities more strongly. Fig. 8 is part of a leaf of *Jungermannia*, showing the dotted cells.

Fig. 6, Plate V., is a part of a rootlet of moss, showing how it is formed of cells elongated and joined end to end.

On the common mare’s-tail, or *Equisetum*, may be seen a very remarkable arrangement for scattering the spores. On the last joint of the stem is a process called a fruit-spike, being a pointed head around which are set a number of little bodies just like garden-tables, with their tops outward. One of these bodies is seen in Fig. 40. From the top of the table depend a number of tiny pouches, which are called sporangia; these lie closely against each other, and contain the spores. At the proper moment these pouches burst from the inside, and fling out the spores, which then look like round balls with irregular surfaces, as shown in Fig. 40, c. This irregularity is caused by four elastic filaments, knobbed at the end, which are originally coiled tightly round the body of the spore, but by rapidly untwisting themselves cause the spore to leap about, and so aid in the distribution. A spore with uncoiled filaments is seen at Fig. 40, b. By breathing on them they may be made to repeat this process at will.

Fig. 36 is a common little sea-weed, called *Ectocarpus siliculósus*, that is found parasitically adhering to large plants, and is figured in order to show

the manner in which the extremities of the branches are developed into sporangia. Fig. 37 is a piece of the common green laver, *Ulva latíssima*, showing the green masses that are ultimately converted into zoospores, and by their extraordinary fertility cause the plant to grow with such rapid luxuriance wherever the conditions are favourable. Every possessor of a marine aquarium knows how rapidly the glass sides become covered with growing masses of this plant. The smaller figure above is a section of the same plant, showing that it is composed of a double plate of cellular tissue.

Fig. 41 is a piece of purple laver or “sloke,” *Porphýra laciniáta*, to show the manner in which the cells are arranged in groups of four, technically named “tetraspores.” This plant has only one layer of cells.

On Plate V. may be seen a number of curious details of the higher algæ.

Fig. 11 is the *Sphacelária*, so called from the curious capsule cells found at the end of the branches, and termed *sphacelæ*. This portion of the plant is shown more magnified in Fig. 12. Another sea-weed is represented in Fig. 13, in order to show the manner in which the fruit is arranged; and a portion of the same plant is given on a larger scale at Fig. 14.

A very pretty little sea-weed called *Cerámium* is shown at Fig. 15; and a portion showing the fruit much more magnified is drawn at Fig. 22. Fig. 23 is a little alga called *Myrionéma*, growing parasitically on the preceding plant.

Fig. 16 is a section of a capsule belonging to the *Hálydris siliquósa*, showing the manner in which the fruit is arranged; and Fig. 17 shows one of the spores more magnified.

Fig. 18 shows the *Polysiphónia parasítica*, a rather common species of a very extensive genus of sea-weeds, containing nearly three hundred species. Fig. 19 is a portion of the stem of the same plant, cut across in order to show the curious mode in which it is built up of a number of longitudinal cells, surrounding a central cell of large dimensions, so that a section of this plant has the aspect of a rosette when placed under the microscope. A

capsule or “ceramídium” of the same plant is shown at Fig. [20](#), for the purpose of exhibiting the pear-shaped spores, and the mode of their escape from the parent-cell previous to their own development into fresh plants. The same plant has another form of reproduction, shown in Fig. [21](#), where the “tetraspores” are seen imbedded in the substance of the branches. There is yet a third mode of reproduction by means of “antheridia,” or elongated white tufts at the extremities of the branches. The cells produced by these tufts fertilise the rudimentary capsules, and so fulfil the function of the pollen in flowering plants.

Fig. [25](#) is the *Cladóphora*, a green alga, figured to illustrate its mode of growth; and Fig. [26](#) represents one of the red sea-weeds, *Ptilóta élegans*, beautifully feathered, and with a small portion shown also on a larger scale, in order to show its structure more fully. A good contrast to this species is seen on Fig. [27](#), and the mode in which the long, slender, filamentary fronds are built up of many-sided cells is seen just to the left hand of the upper frond. Fig. [24](#) is a portion of the lovely *Delesséria sanguínea*, given in order to show the formation of the cells, as also the arrangement by which the indistinct nervures are formed.

FIG.

- | | |
|----------------------------------|---------------------------------------|
| 1. Rose Leaf, with fungus | 15. Ceramium |
| 2. Moss capsule, Polytrichum | 16. Capsule, Halidrys |
| 3. Jungermannia, capsule | 17. Spore of do. |
| 4. Do. an elater more magnified | 18. Polysiphonia parasitica |
| 5. Leaf of Moss, Sphagnum | 19. Do. stem, more magnified |
| 6. Rootlet, Moss | 20. Do. Capsule, tetraspores escaping |
| 7. Puccinia, from Thistle | 21. Do. fruit, another form |
| 8. Jungermannia, leaf | 22. Ceramium, fruit |
| 9. Scale from stalk of male fern | 23. Myrionema, parasitic Seaweed |
| 10. Uredo | 24. Delesseria sanguinea, Frond |
| 11. Sphacelaria filicina | 25. Cladophora |
| 12. Do. top, more magnified | 26. Ptilota elegans |
| 13. Seaweed, showing fruit | 27. Enteromorpha clathrata |
| 14. Do. fruit, more magnified | 28. Nitophyllum laceratum |



V.

The figure on the bottom left-hand corner of Plate V. is a portion of the pretty *Nitophyllum lacerátum*, a plant belonging to the same family as the

preceding one. The specimen here represented has a gathering of spores upon the frond, in which state the frond is said to be “in fruit.”

Fig. [27](#) represents a portion of the common sea-grass (*Enteromorpha*), so common on rocks and stones between the range of high and low water. On the left hand of the figure, and near the top, is a small piece of the same plant much more magnified, in order to show the form of its cells.



CHAPTER VII

Antennæ, their Structure and Use—Eyes, Compound and Simple—Breathing Organs—Jaws and their Appendages—Legs, Feet, and Suckers—Digestive Organs—Wings, Scales, and Hairs—Eggs of Insects—Hair, Wool, Linen, Silk, and Cotton—Scales of Fish—Feathers—Skin and its Structure—Epithelium—Nails, Bone, and Teeth—Blood Corpuscles and Circulation—Elastic Tissues—Muscle and Nerve.

We now take leave of the vegetables for a time, and turn our attention to the animal kingdom.

On Plate [VI](#), may be seen many beautiful examples of animal structures, most of them being taken from the insect tribes. We will begin with the antennæ, or horns, as they are popularly termed, of the insect.

The forms of these organs are as varied as those of the insects to which they belong, and they are so well defined that a single antenna will, in almost every instance, enable a good entomologist to designate the genus to which the insect belonged. The functions of the antennæ are not satisfactorily ascertained. They are certainly often used as organs of speech, as may be seen when two ants meet each other, cross their antennæ, and then start off simultaneously to some task which is too much for a single ant. This pretty scene may be witnessed on any fine day in a wood, and a very animated series of conversations may readily be elicited by laying a stick across their paths, or putting a dead mouse or large insect in their way.

I once saw a very curious scene of this kind take place at an ant's nest near Hastings. A great daddy long-legs had, unfortunately for itself, settled on the nest, and was immediately "pinned" by an ant or two at each leg, so effectually that all its struggles availed nothing. Help was, however, needed, and away ran four or five ants in different directions, intercepting every

comrade they met, and by a touch of the antennæ sending them off in the proper direction. A large number of the wise insects soon crowded round the poor victim, whose fate was rapidly sealed. Every ant took its proper place, just like a gang of labourers under the orders of their foreman; and by dint of pushing and pulling, the long-legged insect was dragged to one of the entrances of the nest, and speedily disappeared.

Many of the ichneumon-flies may also be seen quivering their antennæ with eager zeal, and evidently using them as feelers, to ascertain the presence of the insect in which they intend to lay their eggs; and many other similar instances will be familiar to anyone who has been in the habit of watching insects and their ways.

It is, however, most likely that the antennæ serve other purposes than that which has just been mentioned, and many entomologists are of opinion that they serve as organs of hearing.

Fig. [15](#), Plate VI., represents a part of one of the joints belonging to the antennæ of the common house-fly; it is seen to be covered with a multitude of little depressions, some being small, and others very much larger. A section of the same antenna, but on a larger scale, is shown by Fig. [16](#), in order to exhibit the real form of these depressions. Nerves have been traced to these curious cavities, which evidently serve some very useful purpose, some authors thinking them to belong to the sense of smell, and others to that of hearing. Perhaps they may be the avenues of some sensation not possessed by the human race, and of which we are therefore ignorant. Fig. [17](#) represents a section of the antennæ of an ichneumon-fly, to show the structure of these organs of sense.

We will now glance cursorily at the forms of antennæ which are depicted in the Plate.

Fig. [1](#) is the antenna of the common cricket, which consists of a vast number of little joints, each a trifle smaller than the preceding one, the whole forming a long, thread-like organ. Fig. [2](#) is taken from the

grasshopper, and shows that the joints are larger in the middle than at either end.

Figs. [3](#) and [5](#) are from two minute species of cocktailed beetles (*Staphylínidæ*), which swarm throughout the summer months, and even in the winter may be found in profusion under stones and moss. The insect from which Fig. [5](#) was taken is so small that it is almost invisible to the naked eye, and was captured on the wing by waving a sheet of gummed paper under the shade of a tree. These are the tiresome little insects that so often get into the eye in the summer, and cause such pain and inconvenience until they are removed.

Fig. [4](#) shows the antenna of the tortoise beetle (*Cássida*), so common on many leaves, and remarkable for its likeness to the reptile from which it derives its popular name. Fig. [3](#) is from one of the weevils, and shows the extremely long basal joint of the antennæ of these beetles, as well as the clubbed extremity. Fig. [7](#) is the beautifully notched antenna of the cardinal beetle (*Pyrochróa*), and Fig. [11](#) is the fan-like one of the common cockchafer. This specimen is taken from a male insect, and the reader will find his trouble repaid on mounting one of these antennæ as a permanent object.

Fig. [12](#) is an antenna from one of the common ground beetles (*Cárabus*) looking like a string of elongated pears, from the form of the joints. The reader will see that in beetles he is sure to find eleven joints in the antennæ.

Fig. [10](#) is the entire antenna of a fly (*Syrphus*), one of those pretty flies which may be seen hovering over one spot for a minute, and then darting off like lightning to hang over another. The large joint is the one on which are found those curious depressions that have already been mentioned. Fig. [8](#) is one of the antennæ of a tortoise-shell butterfly (*Vanessa*), showing the slender, knobbed form which butterfly antennæ assume; and Figs. [13](#) and [14](#) are specimens of moths' antennæ, showing how they always terminate in a point. Fig. [13](#) is the beautiful feathery antenna of the ermine moth (*Spilosóma*); and Fig. [14](#) is the toothed one of the tiger moth (*Arctia*

caja). In all these feathered and toothed antennæ of moths, the male insects have them much more developed than the female, probably for the purpose of enabling them to detect the presence of their mates, a property which some possess in wonderful perfection. The male oak-egger moth, for example, can be obtained in any number by putting a female into a box with a perforated lid, placing the box in a room, and opening the window. In the course of the evening seven or eight males are seen to make their appearance, and they are so anxious to get at their intended mate that they will suffer themselves to be taken by hand.

Fig. 9 is an antenna of the male gnat, a most beautiful object, remarkable for the delicate transparency of the joints, and the exquisitely fine feathering with which they are adorned.

We now arrive at the eyes of the insects, all of which are very beautiful, and many singularly full of interest.

In the centre of Plate VI. may be seen the front view of the head of a bee, showing both kinds of eyes, three simple eyes arranged triangularly in the centre, and two large masses, compound eyes, at the sides.

The simple eyes, termed “ocelli,” are from one to three in number, and usually arranged in a triangular form between the two compound eyes. Externally they look merely like shining rounded projections, and can be seen to great advantage in the dragon-flies. The compound eyes may be considered as aggregations of simple eyes, set closely together, and each assuming a more or less perfect six-sided form. Their number varies very greatly; in some insects, such as the common fly, there are about four thousand of these simple eyes in one compound one, in the ant only fifty, in the dragon-fly about twelve thousand, and in one of the beetles more than twenty-five thousand.

Fig. 18 shows a portion of the compound eye of the *Atalanta* butterfly, and Fig. 20 the same organ of the death’s-head moth. A number of the protecting hairs may be seen still adhering to the eye of the butterfly. Fig. 22 is a remarkably good specimen of the eye of a fly (*Heliophilus*),

showing the facets, nearly square, the tubes to which they are attached, and portions of the optic nerves. Fig. 23 is part of the compound eye of a lobster, showing the facets quite square. All these drawings were taken by the camera lucida from my own preparations, so that I can answer for their authenticity.

On Plate VIII. Figs. 6 and 12, the reader will find two more examples of eyes, these being taken from the spiders. Fig. 6 is an example of the eight eyes of the well-known zebra spider, so common on our garden walls and similar situations, hunting incessantly after flies and other prey, and capturing them by a sudden pounce. The eyes are like the ocelli of insects, and are simple in their construction. The number, arrangement, and situation of the eyes is extremely varied in spiders, and serves as one of the readiest modes of distinguishing the species. Fig. 12, Plate VIII., represents one of the curious eyes of the common harvest spider, perched on a prominence or “watch-tower” (as it has been aptly named), for the purpose of enabling the creature to take a more comprehensive view of surrounding objects.

Returning to Plate VI., in Fig. 21 we see a curiously branched appearance, something like the hollow root of a tree, and covered with delicate spiral markings. This is part of the breathing apparatus of the silkworm, extracted and prepared by myself for the purpose of showing the manner in which the tubes branch off from the “spiracle” or external breathing-hole, a row of which may be seen along the sides of insects, together with the beautiful spiral filament which is wound round each tube for the purpose of strengthening it. One of these spiracles may be seen in the neck of the gnat (Fig. 27). Another spiracle, more enlarged, may be seen on Plate VII. Fig. 34, taken from the wireworm, *i.e.* the larva of the skipjack beetle (*Eláter*), to show the apparatus for excluding dust and admitting air. The object of the spiral coil is very evident, for as these breathing-tubes extend throughout the whole body and limbs, they would fail to perform their office when the limbs were bent, unless for some especial provision. This is

achieved by the winding of a very strong but slender filament between the membranes of which the tube is composed, so that it always remains open for the passage of air throughout all the bends to which it may be subjected. Flexible tubes for gas and similar purposes are made after the same fashion, spiral metal wire being coiled within the india-rubber pipe. A little piece of this thread is seen unwound at the end of a small branch towards the top, and this thread is so strong that it retains its elasticity when pulled away from the tube, and springs back into its spiral form. I have succeeded in unwinding a considerable length of this filament from the breathing-tube of a humble bee.

Fig. 28 represents the two curious tubercles upon the hinder quarters of the common green-blight, or *Aphis*, so very common on our garden plants, as well as on many trees and other vegetables. From the tips of these tubercles exudes a sweet colourless fluid, which, after it has fallen upon the leaves, is popularly known by the name of honey-dew. Ants are very fond of this substance, and are in the habit of haunting the trees upon which the aphides live, for the purpose of sucking the honey-dew as it exudes from their bodies. A drop of this liquid may be seen on the extremity of the lower tubercle.

The head of the same insect may be seen in Fig. 24, where the reader may observe the bright scarlet eye, and the long beak with which the *aphis* punctures the leaves and sucks the sap. Fig. 29 is the head of the sheep-tick, exhibiting the organ by which it pierces the skin of the creature on which it lives. Fig. 25 is the head of another curious parasite found upon the tortoise, and remarkable for the powerful hooked apparatus which projects in front of the head.

Turning to Plate VII. Fig. 4, we find the head of a ground beetle (*Cárabus*), valuable as exhibiting the whole of the organs of the head and mouth.

Immediately above the compound eyes are seen the roots of the antennæ, those organs themselves being cut away. Above there are two pairs of similarly constructed organs termed the “maxillary palpi,” because they

belong to the lesser jaws or maxillæ, seen just within the pair of great curved jaws called the mandibles, which are extended in so threatening a manner. The “labial palpi,” so called because they belong to the “labium,” or under lip, are seen just within the others; the tongue is seen between the maxillæ, and the chin or “mentum” forms a defence for the base of the maxillæ and the palpi. A careful examination of a beetle’s mouth with the aid of a pocket lens is very instructive as well as interesting.

Fig. 1 on the same Plate shows the jaws of the hive bee, where the same organs are seen modified into many curious shapes. In the centre may be seen the tongue, elongated into a flexible and hair-covered instrument, used for licking the honey from the interior of flowers. At each side of the tongue are the labial palpi, having their outermost joints very small, and the others extremely large, the latter acting as a kind of sheath for the tongue. Outside the labial palpi are the maxillæ, separated in the specimen, but capable of being laid closely upon each other, and outside all are the mandibles.

VI.

FIG.

- | | |
|---------------------------------|---------------------------------|
| 1. Antenna, Cricket | 16. Do. do. section |
| 2. Do. Grasshopper | 17. Do. Ichneumon |
| 3. Do. Staphylinus | 18. Eye of Butterfly, Atalanta |
| 4. Do. Cassida | 19. Eyes, Bee |
| 5. Do. Staphylinus | 20. Eye, Death’s Head Moth |
| 6. Do. Weevil | 21. Breathing-tube, Silkworm |
| 7. Do. Pyrochroa | 22. Eye, Heliophilus |
| 8. Do. Butterfly, Tortoiseshell | 23. Do. Lobster |
| 9. Do. Gnat, male | 24. Do. Aphis of Geranium |
| 10. Do. Syrphus | 25. Head, Parasite of Tortoise |
| 11. Do. Cockchafer, male | 26. Hind leg, Aphis of Geranium |
| 12. Do. Ground Beetle | 27. Head, Gnat |
| 13. Do. Ermine Moth | 28. “Paps” of Aphis |
| 14. Do. Tiger Moth | 29. Head, Sheep-tick |
| 15. Do. Blowfly | 30. Foot, Tipula |



VI.

The curiously elongated head of the scorpion-fly (*Panorpa*), seen at Fig. 7, affords another example of the remarkable manner in which these organs are developed in different insects. Another elongated head, belonging to the

daddy long-legs, is seen in Plate VI. Fig. 27, and well shows the compound eyes, the antennæ, and the palpi. Fig. 2 represents the coiled tongue of the *Atalanta* butterfly; it is composed of the maxillæ, very greatly developed, and appearing as if each had originally been flat, and then rolled up so as to make about three-fourths of a tube. A number of projections are seen towards the tip, and one of these little bodies is shown on a larger scale at Fig. 3. These curious organs have probably some connection with the sense of taste. Along the edges of the semi-tubes are arranged a number of very tiny hooks, by means of which the insect can unite the edges at will.

Fig. 11, in the centre of the Plate, shows one of the most curious examples of insect structure, the proboscis or trunk of the common bluebottle-fly. The maxillary palpi covered with bristles are seen projecting at each side, and upon the centre are three lancet-like appendages, two small and one large, which are used for perforating various substances on which the insect feeds. The great double disc at the end is composed of the lower lip greatly developed, and is filled with a most complex arrangement of sucking-tubes, in order to enable it to fulfil its proper functions. The numerous tubes which radiate towards the circumference are strengthened by a vast number of partial rings of strong filamentary substance, like that which we have already seen in the breathing-tube of the silkworm. Some of these partial rings are seen on Fig. 12, a little above. The mode in which the horny matter composing the rings is arranged upon the tubes is most wonderful, and requires a tolerably high power to show it. The fine hairs upon the proboscis itself afford most admirable practice for the young microscopist. They should, when properly lighted and focused, be quite black and sharp. Any errors of manipulation will cause them to be “fuzzy.”

Fig. 5 shows the tongue of the common cricket, a most elegantly formed organ, having a number of radiating bands covered with zigzag lines, due to the triangular plates of strengthening substance with which they are furnished, instead of the rings. A portion more highly magnified is shown at Fig. 6, exhibiting the manner in which the branches are arranged.

The legs of insects now claim our attention.

Fig. 9, Plate VII., shows the “pro-leg” of a caterpillar. The pro-legs are situated on the hinder parts of the caterpillar, and, being set in pairs, take a wonderfully firm hold of a branch or twig by pressure toward each other. Around the pro-legs are arranged a series of sharp hooks, set with their points inwards, for greater power in holding. Fig. 10 represents one of the hooks more magnified.

Fig. 15 is the lower portion of the many-jointed legs of the long-legged spider (*Phalángium*), the whole structure looking very like the antenna of the cricket. Fig. 17 is the leg of the glow-worm, showing the single claw with which it is armed. Fig. 26 shows the foot of the flea, furnished with two simple claws. Fig. 16 is the foot of the *Trombídium*, a genus of parasitic creatures, to which the well-known harvest-bug belongs. Fig. 26, Plate VI., shows the leg of the green *Aphis* of the geranium, exhibiting the double claw, and the pad or cushion, which probably serves the same purpose as the pads found upon the feet of many other insects. Fig. 8 is the lower portion of the leg of the ant, showing the two claws and the curious pad in the centre, by means of which the insect is able to walk upon slippery surfaces. The *Típula* has a foot also furnished with a single pad (see Plate VI. Fig. 30). This organ is seen under a very high power to be covered with long hair-like appendages, each having a little disc at the end, and probably secreting some glutinous fluid which will enable the creature to hold on to perpendicular and smooth surfaces. Many of my readers will doubtless have noticed the common fly, towards the end of autumn, walking stiffly upon the walls, and evidently detaching each foot with great difficulty, age and infirmity having made the insect unable to lift its feet with the requisite force.

Fig. 21 is the foot of one of the ichneumon-flies (*Ophíon*), the hairy fringe being apparently for the purpose of enabling it to hold firmly to the caterpillar in which it is depositing its eggs, and which wriggles so violently under the infliction that it would soon throw its tormentor had not some special means been provided for the purpose of enabling the latter to keep

its hold. Fig. [20](#) is a beautiful example of a padded foot, taken from the little red parasitic creature so plentifully found upon the dor or dung beetle (*Geotrúpes*), and of which the afflicted insect is said to rid itself by lying on its back near an ant's nest, and waiting until the ants carry off its tormentors.

Fig. [18](#) is the foot of the common yellow dung-fly (plentiful in pasture lands), having two claws and two pads; and Fig. [19](#) shows the three pads and two claws found in the foot of the hornet-fly (*Ásilus*).

Few microscopic objects call forth such general and deserved admiration as the fore-foot of the male water-beetle (*Dytiscus*), when properly prepared and mounted, for which see Fig. [13](#).

On examining this preparation under the microscope, it is seen that three of the joints are greatly expanded, and that the whole of their under surface is covered profusely with certain wonderful projections, which are known to act as suckers. One of them is exceedingly large, and occupies a very considerable space, its hairs radiating like the rays of the heraldic sun. Another is also large, but scarcely half the diameter of the former, and the remainder are small, and mounted on the extremities of delicate foot-stalks, looking something like wide-mouthed trumpets. In the specimen from which the drawing was taken the smaller suckers are well shown, as they protrude from the margin of the foot.

One of the larger suckers is seen more magnified on Fig. [14](#).

Plate VIII. Fig. [1](#), exemplifies the manner in which the muscles of insects do their work, being well attached in the limbs to the central tendon, and pulling "with a will" in one direction, thus giving very great strength. This leg is taken from the water boatman (*Notonecta*), and has been mounted in Canada balsam.

On Plate VII. Fig. [29](#), may be seen a curiously formed creature. This is the larva of the tortoise beetle (*Cássida*), the skin having been flattened and mounted in Canada balsam. The spiracles are visible along the sides, and at

the end is seen a dark fork-like structure. This is one of the peculiarities of this creature, and is employed for the purpose of carrying the refuse of its food, which is always piled upon its back, and retained in its place by the forked spines, aided probably by the numerous smaller spines that project from the side.

Fig. 33 shows part of the stomach and gastric teeth of the grasshopper. This structure may be seen to perfection in the “gizzard,” as it is called, of the great green locust of England (*Acrida viridíssima*). The organ looks like a sudden swelling of the œsophagus, and when slit longitudinally under water, the teeth may be seen in rows set side by side, and evidently having a great grinding power. The common house cricket has a similar organ of remarkable beauty. Just above (Fig. 27) is the corresponding structure in the hive bee, three of the teeth being shown separately at Fig. 28.

We now cast a rapid glance at the wings of insects.

They have no analogy, except in their use, with the wings of birds, as they are not modifications of existing limbs, but entirely separate organs. They consist of two membranes united at their edges, and traversed and supported by sundry hollow branches or “nervures,” which admit air, and serve as useful guides to entomologists for separating the insects into their genera. Indeed, the general character of the wings has long been employed as the means of dividing the insect race into their different orders, as may be seen in any work on entomology. The typical number of wings is four, but it often happens that two are almost wholly absent, or that the uppermost pair are thickened into a shelly kind of substance which renders them useless for flight; while in many insects, such as the ground beetles and others, the upper wings become hardened into firm coverings for the body, and the lower pair are shrivelled and useless.

Fig. 22 shows two of the wings of a humble bee, together with their nervures, and the peculiar system by which the upper and lower pair are united together at the will of the insect. At the upper edge of the lower

wing, and nearly at its extremity, may be seen a row of very tiny hooks, shown on a larger scale at Fig. 25. These hooklets hitch into the strengthened membrane of the upper wing, which is seen immediately above them, and so conjoin the two together. The curious wing-hooks of the *Aphis* may be seen on Fig. 24, very highly magnified.

Fig. 31 is the wing of the midge (*Psychóda*), that odd little insect which is seen hopping and popping about on the windows of outhouses and similar localities, and is so hard to catch. The whole wing is plentifully covered with elongated scales, and is a most lovely object under any power of the microscope. These scales run along the nervures and edges of the wings, and part of a nervure is shown more highly magnified at Fig. 32.

At Fig. 23 is shown the wing of one of the hemipterous insects, common along the banks of ditches and in shady lanes, and known by the name of *Cíxius*. It is remarkable for the numerous spots which stud the nervures, one being always found at each forking, and the others being very irregularly disposed.

Fig. 30 is one of the balancers or “haltéres” of the house-fly. These organs are found in all the two-winged insects, and are evidently modifications of the second pair of wings. They are covered with little vesicles, and protected at their base by scales. Some writers suppose that the sense of smell resides in these organs. Whatever other purpose they may serve, they clearly aid in the flight, as, if the insect be deprived of one or both of the balancers, it has the greatest difficulty in steering itself through the air.

The wings of insects are mostly covered with hairs or scales, several examples of which are given in Plate VIII. Fig. 4 shows one of the scales of the *Adippe* or fritillary butterfly, exhibiting the double membrane—part of which has been torn away—and the beautiful lines of dots with which it is marked. The structure of the scales is further shown by a torn specimen of tiger moth scale seen on Fig. 16. On many scales these dots assume a “watered” aspect when the focus or illumination changes, an example of which may be seen in Fig. 15, a scale of the peacock butterfly.

Fig. [11](#) is one of the ordinary scales of the azure blue butterfly, and Fig. [10](#) shows one of the curious “battledore” scales of the same insect, with its rows of distinct dottings. Fig. [14](#) is one of the prettily tufted scales of the orange-tip butterfly, and Fig. [8](#) is the splendid branched scale of the death’s-head moth. Fig. [19](#) shows a scale of the sugar-runner (*Lepisma saccharina*), a little silvery creature with glistening skin, and long bristles at the head and tail, that is found running about cupboards, window-sills, and similar places. It is not easy to catch with the fingers, as it slips through them like oil; but by holding a cover-glass in a pair of forceps, and pressing it upon one of the little creatures, a number of the scales may be caused to adhere to it, and these should be mounted dry for examination. The gnats also possess very pretty scales, with the ribs projecting beyond the membrane.

VII.

FIG.

- | | |
|---|---|
| 1. Tongue, Hive Bee | 18. Do. Dung fly |
| 2. Do. Tortoiseshell Butterfly | 19. Do. Asilus |
| 3. Do. do. one of the barrel-shaped bodies | 20. Do. Acarus of Dor-beetle |
| 4. Head, Violet Ground Beetle (Carabus) | 21. Claws and Pad, Ophion |
| 5. Tongue, Cricket | 22. Wings, Humble Bee |
| 6. Do. do. | 23. Do. |
| 7. Head, Scorpion Fly (Panorpa) | 24. Wing hooks, hind wing of Aphis |
| 8. Leg, Ant | 25. Wing hooks, Humble Bee |
| 9. Proleg, Caterpillar | 26. Foot, Flea |
| 10. Do. do. single hook | 27. Stomach and gastric teeth, Bee |
| 11. Proboscis, Fly | 28. Three teeth of do. |
| 12. Do. do. "modified trachea" | 29. Cast skin, Larva of Tortoise Beetle (Cassida) |
| 13. Part of Foreleg of Water Beetle (Acilius) | 30. Balancer, Blow fly |
| 14. Do. large sucker | 31. Wing, Midge (Psychoda) |
| 15. Leg, long-legged Spider (Phalangium) | 32. Do. do. part of a nervure with scales |
| 16. Do. Harvest-bug (Trombidium) | 33. Stomach and gastric teeth, Grasshopper |
| 17. Do. Glow-worm | 34. Spiracle, Wire-worm |



VII.

Fig. 21 is a scale from the common spring-tail (*Podúra plúmbea*), a little creature which is found plentifully in cellars and other damp places, skipping about with great activity. Some flour scattered on a piece of paper

is a sure trap for these little beings. Fig. [3](#) is one of the scales taken from the back of the celebrated diamond beetle, showing the cause of the magnificent gem-like aspect of that insect. We have in England many beetles of the same family—the weevils—which, although much smaller, are quite as splendid when exhibited under a microscope by reflected light. The wing-case or “elytron” of a little green weevil, very common in the hedges, may be seen on Plate XII. Fig. [10](#).

The reader will observe that all these scales are furnished with little root-like appendages, by means of which they are affixed to the insect. Fig. [13](#) shows a portion of the wing of the azure blue butterfly, from which nearly all the scales have been removed, for the purpose of exhibiting the pits or depressions in which they had formerly been fastened, and one or two of the scales are left still adherent to their places. The scales are arranged in equal rows like the slates of a housetop, as may be seen on Fig. [18](#), which represents part of the same wing, to show the scales overlapping each other, and the elegant form which they take near the edges of the wing, so as to form a delicate fringe. The long hair-like down which covers the legs and bodies of the moths and butterflies (which are called *Lepidóptera*, or scale-winged insects, in consequence of this peculiarity), is seen under the microscope to be composed of scales very much elongated, as is shown in Fig. [17](#), a portion taken from the leg of a tiger moth.

The eggs of insects are all very beautiful, and three of the most curious forms are given on Plate VIII.

Fig. [2](#) is the empty egg of the gad-fly, as it appears when fastened to a hair of the horse. Fig. [5](#) represents the pretty ribbed egg of the common tortoiseshell butterfly; and Fig. [7](#) is the very beautiful egg of the very horrid bed-bug, worthy of notice on account of the curious lid with which its extremity is closed, by means of which the young larva creeps out as soon as it is hatched.

The feathers of birds, and the fur of animals, will furnish many examples of the eggs of parasites, some of which are of extreme beauty. The feather or hair may be mounted in a cell without disturbing the eggs, which should, however, be heated sufficiently to kill the embryo if present.

Fig. [9](#) shows the penetrating portions of the sting of the wasp. The two barbed stings, which seem to be the minute prototypes of the many-barbed spears of the South Sea islanders, are seen lying one at each side of their sheath, and a single barb is drawn a little to the left on a very much larger scale. It is by reason of these barbs that the sting is always left adhering to the wound, and is generally drawn wholly out of the insect, causing its death in a short while.

The sting is only found in female insects, and is supposed to be analogous to the “ovipositor” of other insects, *i.e.* the instrument by which the eggs are deposited in their places. Fig. [20](#) shows the curious egg-placing apparatus of one of the saw-flies. The backs of these “saws” work in grooves, and they work alternately, so that the fly takes but a very short time in cutting a slit in the young bark of a tender shoot, and laying her eggs in the slit. When she has completed one of these channels, she sets to work upon another, and in the early spring the young branches of the gooseberry bushes may be seen plentifully covered with these grooves and the eggs. When hatched, black caterpillar-like grubs from the eggs issue, and devastate the bushes sadly, turning in process of time into blackish flies, which are seen hovering in numbers over the gooseberries, and may be killed by thousands.

The scales and hairs of other animals deserve great attention. Fig. [23](#) is a single hair of the human beard, as it often appears when tied in a knot—by Queen Mab and her fairies, according to Mercutio. Fig. [22](#) is a portion of the same hair as it appears when splitting at its extremity. The structure of the hair is not, however, so well seen in this object as in that represented on Fig. [24](#), which is a beautiful example of white human hair that once adorned

the head of the victor of Waterloo. It formed one of a tiny lock given to me by a friend, and is so admirable an example of human hair, that I forthwith mounted it for the microscope. In this hair the cells may be seen extending down its centre, and the peculiar roughened surface produced by the flattened cells which are arranged around its circumference are also seen. By steeping in caustic potash, these scales can be separated, but generally they lie along the hair in such a manner that if the hair be drawn through the fingers from base to point, their projecting ends permit it to pass freely; whilst if it be drawn in the reverse direction, they cause it to feel very harsh to the touch.

In the sheep's wool (Fig. 30) this structure is much more developed, and gives to the fibres the "felting" power that causes them to interlace so firmly with each other, and enables cloth—when really made of wool—to be cut without unravelling. Fig. 37 is the smooth hair of the badger; and Fig. 34 is the curious hair of the red deer, which looks as if it had been covered with a delicate net.

Fig. 28 is the soft, grey, wool-like hair of the rat; and Fig. 29 is one of the larger hairs that protrude so plentifully, and form the glistening brown coat of that animal. Fig. 38 is the curiously knobbed hair of the long-eared bat, the knobs being formed of protuberant scales that can easily be scraped off. Fig. 31 shows a hair of the common mole; and Fig. 32 is one of the long hairs of the rabbit. Fig. 27 is a flat hair of the dormouse, slightly twisted, the difference in the breadth showing where the twist has taken place. The hair of the mouse is beautifully ribbed, so as to look like a ladder. Fig. 26 is one of the very long hairs that so thickly clothe the tiger moth caterpillar; and Fig. 25 is a beautifully branched hair taken from the common humble bee.

All hairs should be examined by polarised light, with a plate of selenite, when most gorgeous colour effects may be obtained.

The four fibres mostly used in the manufacture of apparel are: wool, Fig. 30, which has already been described; linen, Fig. 39; cotton, Fig. 40; and silk, Fig. 41. The structure of each is very well marked and easily made

out with the microscope; so that an adulterated article can readily be detected by a practised eye. Cotton is the most common adulteration of silk and linen fabrics, and may at once be detected by its flat twisted fibre. Silk is always composed of two parallel threads, each proceeding from one of the spinnerets of the caterpillar, and it may be here remarked that if these threads are not quite parallel the silk is of bad quality. Silken fibre is always covered, when new, with a kind of varnish, usually of a bright orange colour, which gives the undressed “floss” silk its peculiar hue, but which is soluble and easily washed away in the course of manufacture.

Figs. [35](#) and 36 are the small and large hairs of that magnificent creature, the sea mouse (*Aphrodíte aculeáta*), whose covering, although it lies in the mud, glows with every hue of the rainbow, and in a brilliant light is almost painfully dazzling to the eye.

VIII.

FIG.

- | | |
|-----------------------------------|----------------------------|
| 1. Boat-fly, leg | 23. Do. Human Beard |
| 2. Gadfly, empty egg | 24. Do. do. aged |
| 3. Diamond Beetle, scale | 25. Do. Humble Bee |
| 4. Scale, Fritillary, Adippe | 26. Do. Tiger Moth, Larva |
| 5. Egg, Tortoiseshell Butterfly | 27. Do. Dormouse |
| 6. Head and Eyes, Zebra Spider | 28. Do. Rat |
| 7. Eyes, Bed-Bug | 29. Do. do. long hair |
| 8. Scale, Death's-Head Moth | 30. Do. Sheep |
| 9. Sting, Wasp | 31. Do. Mole |
| 10. Scale, battledore, Azure blue | 32. Do. Rabbit |
| 11. Do. ordinary scale | 33. Scale, Greenbone Pike |
| 12. Eye, Harvest Spider | 34. Hair, Red Deer |
| 13. Wing Membrane, Azure Blue | 35. Do. fine, Sea Mouse |
| 14. Scale, Anthocera cardaminis | 36. Do. do. large |
| 15. Do. Peacock Butterfly | 37. Do. do. Badger |
| 16. Do. Tiger Moth | 38. Do. do. long-eared Bat |
| 17. Do. Thigh of Tiger Moth | 39. Fibre, Linen |
| 18. Wing and Scales, Azure Blue | 40. Do. Cotton |
| 19. Scale, Lepisma | 41. Do. Silk |
| 20. Saws, Sawfly | 42. Scale, Perch |
| 21. Scale, Podura | 43. Do. do. |
| 22. Hair, Black Human | |



VIII.

The scales of some of the fishes are shown on Plate [VIII.](#), in order to exhibit their mode of growth by successive layers. The scales are always enveloped in membranous sacs, and in some cases, as in the eel, they do not project

beyond the surface, and require some little observation to detect them. A scale of an eel is shown on Plate XI. Fig. 14, and is a magnificent object under polarised light. Fig. 33 is a scale of the greenbone pike; and Figs. 42 and 43 are scales of the perch, showing the roots by which they are held in their places. The roach, dace, bleak, and many other similar fish have a beautiful silvery substance on the under surface of the scales, which was greatly used in the manufacture of artificial pearls, glass beads being thinly coated in the interior with the glittering substance, and then filled in with wax. A piece of sole-skin, when preserved in Canada balsam and placed under the microscope, is a very beautiful object.

More examples of hairs, and other processes from the skin, together with the structure of the skin itself, of bone, of blood, and the mode in which it circulates, are given on Plate X.

In all important points of their structure the feathers of birds are similar to the hairs of animals, and are developed in a similar manner. They are all composed of a quill portion, in which the pith is contained, and of a shaft, which carries the vane, together with its barbs. The form of each of these portions varies much, even in different parts of the same bird, and the same feather has almost always two kinds of barbs; one close and firm, and the other loose, floating, and downy. If a small feather be plucked from the breast or back of a sparrow or any other small bird, the upper part of the feather is seen to be close and firm, while the lower is loose and downy, the upper part being evidently intended to lie closely on the body and keep out the wet, while the lower portion affords a soft and warm protection to the skin.

Fig. 12, Plate X., shows the feather of a peacock, wherein the barbs are very slightly fringed and lie quite loosely side by side. Fig. 18 is part of the same structure, in a duck's feather, wherein are seen the curious hooks which enable each vane to take a firm hold of its neighbour, the whole feather being thus rendered firm, compact, and capable of repelling water. The reader will not fail to notice the remarkable analogy between these hooks and those which connect the wings of the bee.

Fig. [17](#) is a part of the shaft of a young feather taken from the canary, given for the purpose of showing the form of the cells of which the pith is composed. Fig. [20](#) is part of the down from a sparrow's feather, showing its peculiar structure; and Fig. [21](#) is a portion of one of the long drooping feathers of the cock's tail.

Fig. [13](#) exhibits a transverse section of one of the large hairs or spines from the hedgehog, and shows the disposition of the firm, horn-like exterior, and the arrangement of the cells. Sections of various kinds of hair are interesting objects, and are easily made by tying a bundle of them together, soaking them in gum, hardening in spirit, and then cutting thin slices with a razor. A little glycerine will dissolve the gum, and the sections of hair will be well shown. Unless some such precaution be taken, the elasticity of the hair will cause the tiny sections to fly in all directions, and there will be no hope of recovering them.

Several examples of the skin are also given. Fig. [27](#) is a section through the skin of the human finger, including the whole of one of the little ridges which are seen upon the extremity of every finger, and half of two others. The cuticle, epidermis, or scarf-skin, as it is indifferently termed, is formed of cells or scales, much flattened and horny in the upper layers, rounder and plumper below. The true skin, or "cutis," is fibrous in structure, and lies immediately beneath, the two together constituting the skin, properly so called. Beneath lies a layer of tissue filled with fatty globules, and containing the glands by which the perspiration is secreted.

One of the tubes or channels by which these glands are enabled to pour their contents to the outside of the body, and, if they be kept perfectly clean, to disperse them into the air, is seen running up the centre of the figure, and terminating in a cup-shaped orifice on the surface of the cuticle. On the palm of the hand very nearly three thousand of these ducts lie within the compass of a square inch, and more than a thousand in every square inch of the arm and other portions of the body, so that the multitude of these valuable organs may be well estimated, together with the absolute necessity for keeping the skin perfectly clean in order to enjoy full health.

Fig. [1](#) shows a specimen of epidermis taken from the skin of a frog, exhibiting the flattened cells which constitute that structure, and the oval or slightly elongated nuclei, of which each cell has one. In Fig. [32](#), a portion of a bat's wing, the arrangement of the pigment is remarkably pretty. Immediately above, at Fig. [31](#), is some of the pigment taken from the back of the human eye-ball. The shape of the pigment cells is well shown. Similar specimens may easily be obtained from the back of a sheep's eye which has been hardened in spirit, or from that of a boiled fish. Fig. [33](#) shows the pigment in the shell of the prawn.

On various parts of animal structures, such as the lining of internal cavities, the interior of the mouth, and other similar portions of the body, the cells are developed into a special form, which is called "Epithélium," and which corresponds to the epidermis of the exterior surface of the body. The cells which form this substance are of different shapes, according to their locality. On the tongue, for example (for which see Fig. [11](#)), they are flattened, and exhibit their nucleus, in which the nucléolus may be discovered with a little care. Cells of this kind are rounded, as in the case just mentioned, or angular, and in either case they are termed squamous (*i.e.*, scaly) epithelium. Sometimes they are like a number of cylinders, cones, or pyramids, ranged closely together, and are then called cylindrical epithelium. Sometimes the free ends of cylindrical epithelium are furnished with a number of vibrating filaments or cilia, and in this case the structure is called "ciliated" epithelium. Cylindrical epithelium may be found in the ducts of the glands which open into the intestines, as well as in the glands that secrete tears; and ciliated epithelium is seen largely in the windpipe, the interior of the nose, etc. A specimen taken from the nose is seen at Fig. [15](#). A beautiful example of ciliated epithelium is to be found in the gills of the mussel. A portion of one of the yellowish bands which lie along the edge of the shell on the opening side is carefully removed with sharp scissors, and examined in the shell-liquor, being protected from pressure by placing a piece of paper beneath each end of the cover-glass. Such a preparation is

shown in Plate IX. Fig. 39, but no drawing can give an idea of its wonderful beauty and interest. The cilia will continue to move for a long time after removal from the shell.

Bone in its various stages is figured on Plate X.

Fig. 9 is a good example of human bone, and is a thin transverse section taken from the thigh. When cut across, bone exhibits a whitish structure filled with little dottings that become more numerous towards the centre, and are almost invisible towards the circumference. In the centre of the bone there is a cavity, which contains marrow in the mammalia and air in the birds. When placed under a microscope, bone presents the appearance shown in the illustration.

The large aperture in the centre is one of innumerable tubes that run along the bone, and serve to allow a passage to the vessels which convey blood from one part of the bone to another. They are technically called Haversian canals, and if a longitudinal section be made they will be found running tolerably parallel, and communicating freely with each other. Around each Haversian canal may be seen a number of little black spots with lines radiating in all directions, and looking something like flattened insects. These are termed bone-cells or “lacunæ,” and the little black lines are called “canaliculi.” In the living state they contain cells which are concerned in the growth of the bone, and these may be made evident by softening fresh bone with acid, cutting sections of it, and staining. When viewed by transmitted light the lacunæ and canaliculi are black; but when seen by dark-field illumination the Haversian canals become black, and the lacunæ are white.

As these canaliculi exist equally in every direction, it is impossible to make a section of bone without cutting myriads of them across; and when a high power is employed they look like little dots scattered over the surface. A very pretty object can be made of the bone taken from a young animal which has been fed with madder, as the colour gets into the bone and settles

chiefly round the Haversian canal. A young pig is a very good subject, so is a rabbit.

Fig. [16](#) is a similar section cut from the leg-bone of an ostrich.

The development of bone is beautifully shown in Fig. [30](#), a delicate slice taken from a pig's rib. Above may be seen the gristle or cartilage, with the numerous rows of cells; below is the formed bone, with one of the Haversian canals and its contents; while between the two may be seen the cartilage-cells gathering together and arranging themselves into form. The cartilage-cells are well shown in Fig. [28](#), which is a portion of the cup which had contained the eye of a haddock.

The horn-like substances at the end of our fingers, which we call the nails, are composed of innumerable flattened cells. These cells are generally so fused together as to be quite indistinguishable even with a microscope, but can be rendered visible by soaking a section of nail in liquor potassæ, which causes the cells to swell up and resume to a degree their original rounded form.

It is worthy of remark that the animal form is built up of cells, as is the case with the vegetables, although the cells are not so variable in shape. They generally may be found to contain well-marked nuclei, two or more of the latter being often found within a single cell, and in many cases the tiny nucleoli are also visible. Good examples of these cells may be obtained from the yolk of an egg, and by careful management they may be traced throughout every part of the animal form.

The teeth have many of the constituents of bone, and in some of their parts are made after precisely the same fashion. When cut, the teeth are seen to consist of a hard substance, called enamel, which coats their upper surfaces, of dentine, or ivory, within the enamel, and of "cement," which surrounds the fangs. In Fig. [26](#), Plate X., which is a longitudinal section of the human "eye" tooth, is seen the ivory occupying the greater part of the tooth, coated by the enamel at the top and the cement at the bottom. In the centre of each tooth there is a cavity, which is plentifully filled with a pulpy substance by

which the tooth is nourished, and which conveys the nerves which endow it with sensation. A traverse section of the same tooth is seen in Fig. 25.

The enamel is made of little elongated prisms, all pointing to the centre of the tooth. When viewed transversely, their ends are of a somewhat hexagonal shape, something like an irregular honeycomb. The dentine is composed of a substance pierced with myriads of minute tubes. They require a rather high power—say 300 diameters—to show them properly. The cement is found at the root of the fangs, and is best shown in the tooth of an aged individual, when it assumes very clearly the character of bone.

Sections may be made by sawing a slice in the required direction, polishing one side, and cementing it with old Canada balsam to a slide. It may then be filed down to nearly the required thinness, finished by carefully rubbing with a hone, and polished with buff leather. Canada balsam may then be dropped upon it, and a glass cover pressed firmly down.

Sections of young bone form magnificent objects for the polariser.

Fig. 29 is a section cut from one of the palate teeth of the ray (*Myliobates*).

A rather important element in the structure of animals is the “elastic ligament,” which is found in the back of the neck and other parts of the body, especially about the spine. It is made of a vast number of fibres of variable shape and length, branching and communicating, arranged generally in bundles, and remarkable for containing very few vessels, and no nerves at all. At Fig. 14 may be seen an example of elastic ligament, popularly called “paxwax,” taken from the neck of a sheep.

The white fibrous tissue by which all the parts of the body are bound together is seen at Fig. 10; and at Fig. 11 is a beautiful example of the “ultimate fibres” of the crystalline lens of a sturgeon’s eye.

The muscles of animals are of two kinds, the one termed the striped, and the other the unstriped. Of these, the latter belongs to organs which work independently of will, such as the stomach, etc., while the former belongs to those portions of the body which are subject to voluntary motion, such as

the arm and the leg. The unstriped muscle is very simple, consisting merely of long spindle-shaped cells, but the striped or voluntary muscle is of more complex construction. Every voluntary muscle consists of myriads of tiny fibres, bound together in little bundles, enveloped in a kind of sheath. Fig. 24 is an example of this muscular fibre, taken from beef. When soaked in spirit, it often splits into a number of discs, the edges of which are marked by the transverse lines.

A fibre of nerve is drawn at Fig. 23, and is given for the purpose of showing the manner in which the nerve is contained in and protected by its sheath, just like a telegraph-wire in its coverings. Just above is a transverse section of the same fibre, showing the same arrangement from another point of view, and also illustrating the curious phenomenon, that when nerve-fibres are treated with carmine the centre takes up the colouring matter, while the sheath remains white as before. The best way of studying nerves is to decapitate a frog, and cut off a piece of one of the nerves, which, like fine silk threads, come out between the joints of the spine inside the abdomen. By careful teasing out it is easy to obtain preparations showing all the above points, and, in addition, the folding-in of the internal sheath which correspond to the insulators of a telegraph-line.

The blood of animals is analogous in its office to the sap of plants, but differs greatly from it under the microscope. In sap there seem to be no microscopic characters, except that when a branch is cut, as in the vine, the flowing sap may contain certain substances formed in the wounded cells, such as chlorophyll, starch, and raphides; but the blood is known to be an exceedingly complex substance both in a microscopic and a chemical point of view. When a little fresh blood is placed under the microscope, it is seen to consist of a colourless fluid filled with numerous little bodies, commonly called "blood-globules," varying very greatly in size and shape, according to the animal from which they were taken. Those of the reptiles are very large, as may be seen at Fig. 4, Plate X., which represents a blood corpuscle of the Proteus. In this curious reptile the globules are so large that they may be distinguished during its life by means of a common pocket lens.

In the vertebrated animals these corpuscles are red, and give to the blood its peculiar tint. They are accompanied by certain colourless corpuscles, spherical in form, which are sometimes, as in man, larger than the red globules, and in others, as in the siren and the newt, considerably smaller. The general view of the red corpuscles has sufficient character to enable the practised observer to name the class of animal from which it was taken, and in some cases they are so distinctive that even the genus can be ascertained with tolerable certainty. In point of size, the reptiles have the largest and the mammalia the smallest, those of the Proteus and the musk-deer being perhaps the most decidedly opposed to each other in this respect.

IX.

FIG.

- | | |
|--|---|
| 1. <i>Amœba diffluens</i> | 21. Head of Snake-headed Zoophyte |
| 2. <i>Arcella</i> | 22. <i>Bugula avicularia</i> |
| 3. Sun animalcule | 23. <i>Polyzoön</i> , <i>Eretea</i> |
| 4. <i>Miliolina</i> | 24. Do. <i>Notamia</i> |
| 5. <i>Paramœcium</i> | 25. Zoea, Young of Crab |
| 6. <i>Chilodon</i> subdividing | 26. <i>Hydra tuba</i> |
| 7. <i>Melicerta ringens</i> | 27. Medusa, cast off from above |
| 8. Spicula of Sponge, <i>Grantia</i> | 28. Naked-eyed Medusa, <i>Thaumantias</i> |
| 9. <i>Noctiluca miliaris</i> | 29. Compound Eye, Medusa |
| 10. <i>Rotifer vulgaris</i> | 30. Larva, Snake Star |
| 11. Do. jaws | 31. Water Flea |
| 12. Sponge animalcule | 32. <i>Serpula</i> , Pushing Pole |
| 13. <i>Sertularia operculata</i> | 33. <i>Comatula</i> , early stage of Starfish |
| 14. Sponge, <i>Grantia</i> | 34. Carbonate of Lime, artificial |
| 15. <i>Sertularia operculata</i> , with ovicells | 35. Sea Urchin, transverse section of spine |
| 16. <i>Actinia</i> , showing weapons | 36. <i>Serpula</i> , bundle of spears |
| 17. Do. base of weapon more magnified | 37. Sun-star, part of skin |
| 18. Sponge granule, ciliated | 38. Oyster shell in different stages |
| 19. <i>Anguinaria anguina</i> | 39. Cilia on mussel |
| 20. Spicules of sponge from Oyster Shell | |



IX.

In shape, those of the mammalia are circular discs, mostly with a concave centre, though the camel has oval ones; those of the birds are more or less oval and convex; those of the reptiles are decidedly oval, very thin, and

have the nucleus projecting; and those of the fishes are oval and mostly convex. During the process of coagulation the blood corpuscles run together into a series of rows, just as if a heap of pence had been piled on each other and then pushed down, so that each penny overlaps its next neighbour.

These objects are illustrated by six examples on Plate X. Fig. 2 is human blood, showing one of the white corpuscles; Fig. 3 is the blood of the pigeon; Fig. 4, of the *Proteus anguinus*; Fig. 5, of the tortoise; Fig. 6, of the frog, showing the projecting nucleus; and Fig. 7, of the roach. The blood possesses many curious properties, which cannot be described in these few and simple pages.

In the centre of Plate X. is a large circular figure representing the membrane of a frog's foot as seen through the microscope, and exhibiting the circulation of the blood. The mode of arranging the foot so as to exhibit the object without hurting the frog is simple enough.

Take an oblong slip of wood,—my own was made in five minutes out of the top of a cigar-box,—bore a hole about an inch in diameter near one end, and cut a number of little slits all round the edge of the wooden slip. Then get a small linen bag, put the frog into it, and dip him into water to keep him comfortable. When he is wanted, pull one of his hind feet out of the bag, draw the neck tight enough to prevent him from pulling his foot back again, but not sufficiently tight to stop the circulation. Have a tape fastened to the end of the bag, and tie it down to the wooden slide. Then fasten a thread to each of his toes, bring the foot well over the centre of the hole, stretch the toes well apart, and keep them in their places by hitching the threads into the notches on the edge of the wooden strip. Perhaps an easier plan is to secure the threads by drops of sealing-wax when in the desired position. Push a glass slide carefully between the foot and the wood, so as to let the membrane rest upon the glass, and be careful to keep it well wetted. If the frog kick, as he will most likely do, pass a thin tape over the middle of the leg, and tie him gently down to the slide.

Bring the glass into focus, and the foot will present the appearance so well depicted in the engraving. The veins and arteries are seen spreading over the whole of the membrane, the larger arteries being often accompanied by a nerve, as seen in the illustration. Through all these channels the blood continually pours with a rather irregular motion, caused most probably by the peculiar position of the reptile. It is a most wonderful sight, of which the observer is never tired, and which seems almost more interesting every time that it is beheld.

The corpuscles go pushing and jostling one another in the oddest fashion, just like a British crowd entering an exhibition, each one seeming to be elbowing its way to the best place. To see them turning the corners is very amusing, for they always seem as if they never could get round the smaller vessels, and yet invariably accomplish the task with perfect ease, turning about and steering themselves as if possessed of volition, and insinuating their ends when they could not pass crosswise.

By putting various substances, such as spirit or salt, upon the foot, the rapidity of the circulation at the spot can be greatly increased or reduced at will, or even stopped altogether for a while, and the phenomenon of inflammation and its gradual natural cure be beautifully illustrated. The numerous black spots upon the surface are pigment-cells.

The tails of young fish also afford excellent objects under the microscope, as the circulation can be seen nearly as well as in the frog's foot. The gills of tadpoles can also be arranged upon the stage with a little care, and the same organs in the young of the common newt will also exhibit the circulation in a favourable manner. The frog, however, is perhaps the best, as it can be arranged on the "frog-plate" without difficulty, and the creature may be kept for months by placing it in a cool, damp spot, and feeding it with flies, little slugs, and similar creatures.

CHAPTER VIII

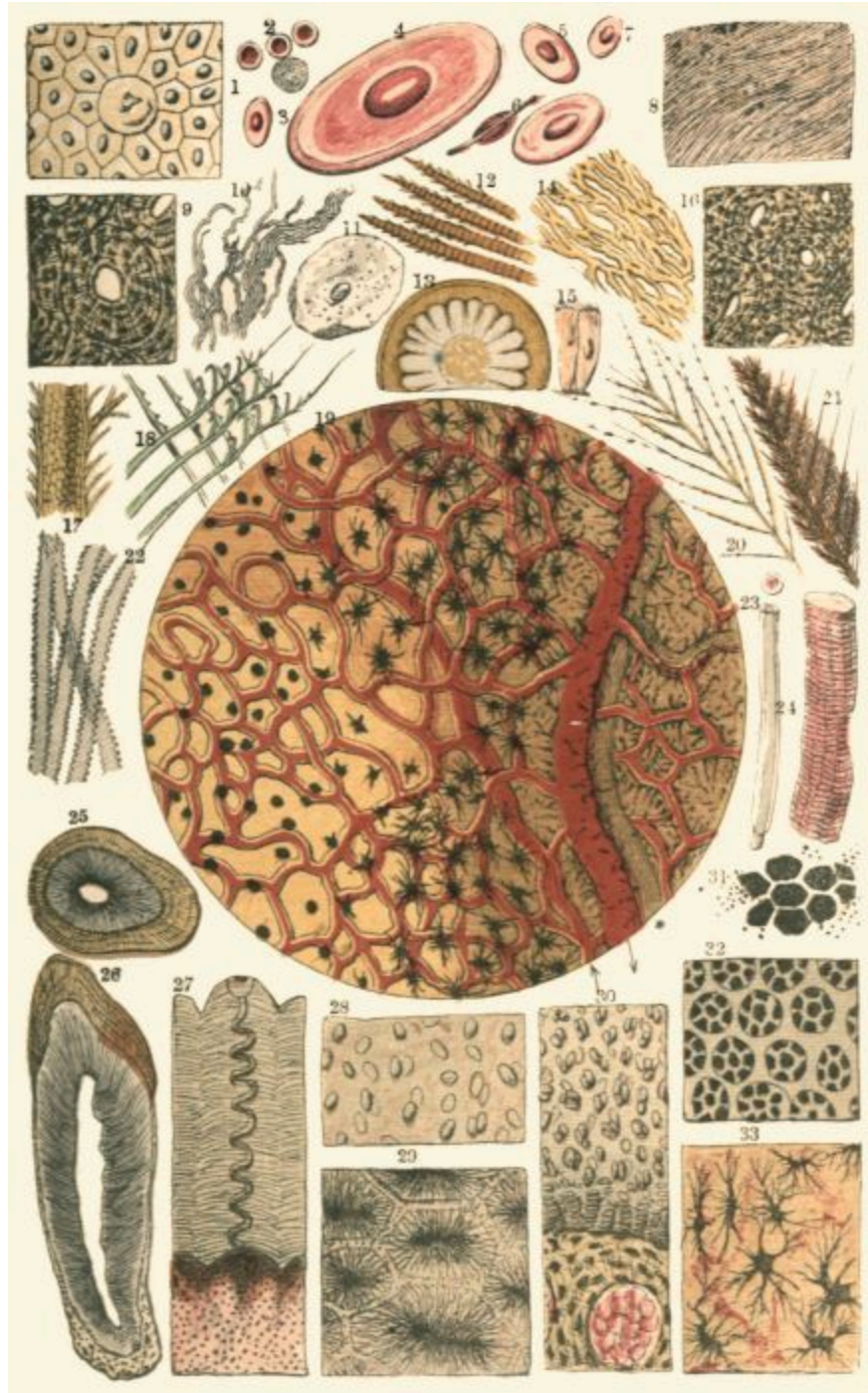
Pond-Life—Apparatus and Instructions for Collecting Objects—Methods of Examination—Sponge—Infusoria.

Of all departments of microscopic research the most fascinating and the most popular is that which deals with what is known by the generic name of “pond-life.” The minute forms of the animal creation included in this term are of such exquisite beauty, and allow the processes of their life-history to be followed with such facility, from the cradle (when they have one) to the grave (which is very often the body of another, larger, organism), that there is none which has attracted more observers. Indeed, the first application of the microscope, by Leeuwenhoek, early in the seventeenth century, was to the observation of these forms of life.

X.

FIG.

- | | |
|---|---------------------------------------|
| 1. Skin, Frog | 18. Do. Wild Duck |
| 2. Blood, Human | 19. Circulation of blood, Frog's foot |
| 3. Do. Pigeon | 20. Feather, Sparrow |
| 4. Do. Proteus | 21. Do. Cock's tail |
| 5. Do. Tortoise | 22. Fibre, crystalline lens of fish |
| 6. Do. Frog | 23. Nerve |
| 7. Do. Fish | 24. Muscle, Meat |
| 8. Human nail | 25. Tooth, transverse section |
| 9. Bone, Human | 26. Do. Longitudinal section |
| 10. White fibrous tissue | 27. Sweat duct |
| 11. Epithelial cells from tongue | 28. Eye of Haddock |
| 12. Feather, Peacock | 29. Myliobates, palate |
| 13. Spine, Hedgehog, transverse section | 30. Gristle, Pig |
| 14. Pax-wax | 31. Pigment, Human eye |
| 15. Epithelial cells from nose | 32. Do. Wing of Bat |
| 16. Bone, Ostrich | 33. Do. Shell of Prawn |
| 17. Feather, Shaft of Canary's | |



X.

A few words may be said, in the first place, as to the outfit. A very useful part of it is a walking-stick, to which can be attached either a net for capturing the larger forms of life, or a hook for collecting the weeds, to

which many forms of great interest and beauty are attached (Fig. 15). The stick is telescopic, and can also have attached to it a bottle, which, put into the water at any desired spot,—say, amongst a clump of weeds, or near the bottom, upside down, and then suddenly reversed,—will bring away samples of the inhabitants of the neighbourhood. When these are sparsely distributed through the water, the latter may be concentrated by the use of a bottle round the neck of which is firmly tied a coarse calico bag, funnel-shaped, and supported by a wire ring, somewhat as shown in the illustration. Muslin is, however, too coarse for many organisms. This net is immersed in the water so that the ring is just above the surface, and one bottleful after another poured through. The water strains off, the organisms are left behind. The immersion is necessary to reduce the pressure to which delicate organisms would otherwise be subjected. When the bottle is full, or sufficiently concentrated as to its contents, the latter are poured into one of the ordinary collecting-bottles, of which half a dozen at least should always be taken.

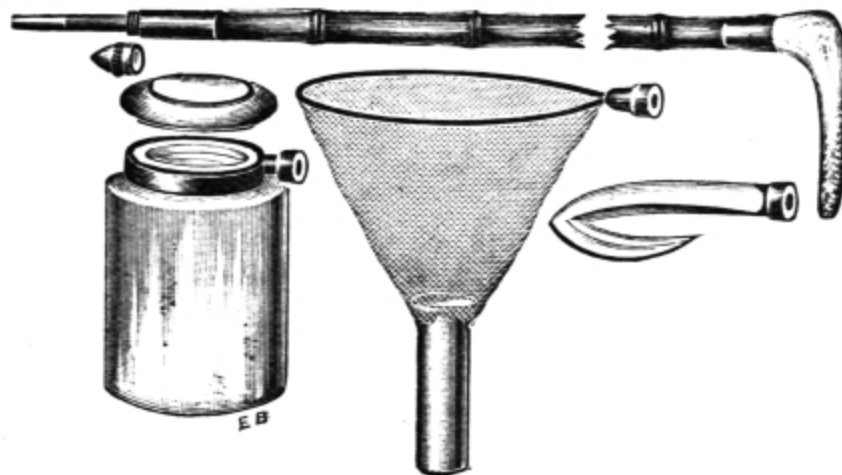


FIG. 15.

On reaching home, and as often as possible on the way, the corks should be removed, as these organisms soon use up the air in the water.

For examination a glass trough of considerable size, say three inches in length, half an inch in depth, and two inches in height, should be half filled with the water, and examined with the pocket magnifier. With a little practice it will be found easy to take up not only the larger organisms, but even very minute ones, with one of the dipping-tubes with a long tapering point already referred to. The organism, when “spotted,” is followed by eye and tube, the finger being held over the mouth of the latter, and at the critical moment the finger is removed, and the organism swept into the tube by the in-rushing water. Now wipe off the excess with a clean handkerchief, “spot” the organism in the tube again, and carefully absorb the superfluous water with a piece of blotting paper; and finally, gently but sharply blow the remainder on to the plate of the live-box, put on the cover, and examine with a one-inch power. If, as often happens, the organism sticks to the side of the tube, a little more water must be drawn in, and the process repeated. The use of the cotton-wool trap spoken of previously will often be very helpful in the examination of actively moving organisms.

In the case of weeds, a small portion should be placed in the trough and carefully examined from end to end, first with the pocket lens and then with the one-inch power. Let us now consider the objects most likely to be met with.

A piece of stick may be coated with a white layer, feeling rough to the touch, and full of small holes. The chances are that this will be a piece of fresh-water sponge, *Spongilla fluviatilis*, and by dark-field illumination particles may be seen to enter at some orifices and be ejected at others. With a very high power and a very thin section, properly prepared, these holes will be seen to be the mouths of channels which are lined by the most delicate organisms possible, each having a minute body crowned with a tiny crystal cup, in the middle of which is a long cilium, or flagellum, as it is here called (Plate XIII. Fig. [1](#)). The currents are produced by the combined action of these flagella. In point of fact, the sponge is a colony of minute animals working harmoniously for the common good. If the specimen be found in winter the sponge will be full of tiny balls, the “gemmules” of the

next season's growth. The roughness is due to the flinty spicules, which are at once the scaffolding and the protection of the sponge, and by boiling the sponge in a mixture of nitric acid and water (half and half) these spicules will be set free, and may be washed, allowed to settle, washed again, dried, and mounted in balsam. The gemmules are coated by very beautiful spicules, consisting of two wheels connected by a rod. These may be treated in the same way. The life-history of the common sponge is as yet but imperfectly known.

Perhaps the lowest form of life is the *Amœba*, shown in Plate IX. Fig. 1, a mere lump of jelly, which flows along, and when it comes into contact with any likely subject for digestion flows round it, encloses it, absorbs what it can from it, and leaves it behind. A near relative of the *Amœba* is the *Arcella* (Fig. 2), which is simply an *Amœba* with a shell. Being unable to swim, these organisms are naturally to be most often found at the bottom of the collecting bottle, and it is always advisable to take up a portion of the débris with a dipping tube, which is then held upright on a slide with the finger upon it until the dirt settles on to the slide, when it is removed, a cover-glass put upon the dirt, and a quarter-inch power used for examination. Many forms will be discovered in this way which would otherwise escape observation.

Another curious organism, of great size (comparatively) and extreme beauty, is the sun animalcule (*Actinophrys*), which has a round body and long tentacles (Fig. 3), to which free-swimming organisms adhere, and by the combined action of the neighbouring ones are drawn to the body and received into it; one cannot say swallowed.

Fig. 6, Plate IX., shows the curious arithmetical process whereby the Infusoria multiply by division, a groove appearing at one point, rapidly deepening, and finally separating the animal completely into two. The species is the *Chilodon*, a flattened creature, ciliated all over, having a set of teeth arranged in the form of a tube, and at its fore-part a kind of membranous lip. A similar phenomenon, in an earlier stage, is shown in Fig. 26, Plate XIII., the organism in this case being *Euplotes*.

It has been said that sponges are colonies of extremely minute organisms, each furnished with a membranous collar or funnel, the whole looking like an exquisite wine-glass without a foot. These organisms are not always grouped in colonies, however. Many are free-growing, and may be found attached to the stems of water-plants, but they are extremely minute, and will hardly be noticed until the microscopist has acquired considerable experience, nor even then—with such an instrument as we have postulated—will he see more than a tiny pear, with a straight line, the margin of the cup, on each side of its summit. The flagellum will be quite invisible.

Some similar organisms may, nevertheless, be found which, though still minute, are within the range of a properly managed quarter-inch objective. Such an one, of extreme beauty, is the *Dinobryon* shown in Plate XIII. Fig. 3. Each “zoöid,” as the separate animals are called, among the Infusoria, or each generation of zoöids, stands upon its parent and has two flagella. When alarmed, the zoöid sinks to the bottom of its cell, and withdraws its flagella. In Fig. 2 (*Eugléna*) we have a similar zoöid, but of far greater size, and free-swimming. It is a very common object, and possesses a red eye-speck close to the “contractile vesicle.” All Infusoria have the latter, some a great number, as in Fig. 9. The vesicle contracts at regular intervals, and is then simply blotted out, but reforms in the same place, so that it is probably the heart or the urinary bladder of these minute animals.

The lovely rosette shown in Fig. 4 is the *Synura*, a spherical colony of zoöids, each of which has two flagella, and is in addition clothed with rows of cilia. A beautiful sight it is to watch these colonies rolling through the field of view. Not uncommon, especially in brackish water, is the *Peridinium* (Fig. 5), with its plate armour, long flagellum, and girdle of cilia. A gigantic species of the same family is common in sea-water, and will be easily recognised by its body, not much larger than that of *Peridinium*, being furnished with three long arms, curiously bent. It is called *Ceratium*, and is sometimes present in such abundance as to thicken the water, near the surface of which it swims.

We now come to a class of Infusoria which is characterised by the possession of a complete covering of cilia, arranged in rows all over the body. The number of these is enormous; we can only glance at a few types, by mastering which the observer will, at all events, know whereabouts he is. The first we will take is the *Coleps* (Fig. 6), a very common kind, whose body is marked by a series of geometrical lines, so that the organism looks very much like an elongated geographical globe. These markings are on the tunic, which is of a brownish colour. Very different is the *Trachelocerca* (Fig. 7), with its long flexible neck, which is in constant movement from side to side as the creature swims along. As seen in the figure, the neck is clear and the head has a fringe of longer cilia.

The *Trachelius* (Fig. 8) is perhaps the largest of all the Infusoria, being readily visible to even an inexperienced eye. Its body is richly furnished with contractile vesicles, and the protoplasm is curiously reticulated. We may here remark that the *Trachelius* is especially prompt in doing what most of these organisms do when put under pressure in a live-box, namely, in performing a kind of *harakiri*. The outline first becomes irregular, then the body rapidly swells and finally comes to pieces, the fragments dancing mockingly away under the influence of their still-moving cilia. The remedy is to use the cotton-wool trap and the lightest possible pressure.

A very elegant organism is shown in the bottom right-hand corner of the Plate (Fig. 25). It is the *Loxophyllum*, and has a strongly marked contractile vesicle.

Another large form is *Amphileptus* (Fig. 9), already referred to as having a large number of contractile vesicles arranged in a regular row; and more massive still is *Bursaria* (Fig. 10), a very curious organism, very much like a purse indeed, and possessing a wonderful arrangement of cilia inside the funnel. These are arranged like a ladder, a series of rows of short stiff cilia, which move at short intervals in unison, and tend to sweep down into the cavity any small particles of food. This arrangement is here described for the first time, and appears to be quite unlike anything else among the Infusoria. Not unlike *Bursaria*, but having no ladder, and being furnished

with a delicate membranous pouch in front of the slit of the purse, is *Condylostoma*, which we shrewdly suspect to be the young form of *Bursaria*. This is a point which requires elucidation.

One of the most beautiful of all these forms is shown in Fig. 11, *Folliculina*, a type of a large group characterised by the possession of a transparent case, of extremely elegant form, within which the animal retreats on the slightest alarm.

Fearless and independent, as becomes its size, is the trumpet-shaped *Stentor* (Fig. 12), which may easily be seen when present, as it is in almost every good gathering of water-weed. The particular form drawn (*S. Mülleri*) does not make a case, but many members of the genus do, and it is very common to see a stem almost covered with them. Such a sight, once seen under dark-field illumination, will never be forgotten. The method of multiplication of the *Stentors* (by division) is extremely easy to watch, and very instructive.

A curious organism is *Trichodina* (Fig. 13), which, though a free-swimmer, is always parasitic upon the body of some higher animal. We have found it sometimes upon *Hydra*, and always in hundreds upon the stickleback. The next group of Infusoria is distinguished by the body's being only ciliated at particular points, usually round the mouth, or what acts as such. The first form is *Vorticella* (Fig. 14), a beautiful vase-like creature upon a stem. Down the stem runs a muscular fibre, and on the least shock the fibre contracts and draws the stem into a beautiful spiral, whilst the cilia are drawn in, and the zoöid assumes the appearance of a ball at the end of a watch-spring. An exquisite sight is a colony of *Vorticellæ*, for these actions are always going on, as, for example, when one member of the family touches another, which is quite sufficient to provoke the contraction.

XI.

POLARIZED LIGHT.

FIG.

- | | |
|--------------------------------|---|
| 1. Carbonate of Lime | 16. Chlorate of Potash, Crystals |
| 2. Starfish | 17. Cellularia reptans |
| 3. Thistle down | 18. Star-shaped hair, Stalk of Yellow Water-Lily |
| 4. Starch, Wheat | 19. Teeth, Palate of Whelk |
| 5. Do. Potato | 20. Zoophyte, Bowerbankia |
| 6. Prawn-shell | 21. Raphides, <i>i.e.</i> crystalline formations in vegetable cells, Bulb of Hyacinth |
| 7. Starch, "Tous les mois" | 22. Do. Rhubarb |
| 8. Bone, cancellous | 23. Sulphate of Magnesia, Crystals |
| 9. Gun-cotton | 24. Bone, Skate |
| 10. Cow's hair | 25. Cherrystone, transverse section |
| 11. Hoof, donkey, longitudinal | 26. Sugar, Crystals in honey |
| 12. Do. transverse | 27. Tendon, Ox |
| 13. Nitre, Crystals | 28. Calcareous plates. Tooth of Echinus |
| 14. Scale, Eel | |
| 15. Wing, Water-Boatman | |



XI.

Many compound tree-like forms of *Vorticella* are known, one of which, *Carchesium* (Fig. 15), may serve as a type of all. In the case of this organism, the colony contracts in sections on a moderate shock; in the

second, *Zoothamnium*, as a whole; whilst in *Epistylis* the stalks are rigid, and the individuals contract singly. When the shock is violent, the appearance presented by the two former is that shown in Fig. 16. In all three cases the colonies are usually so large that they are visible as trees to the naked eye, and some members of the group are extremely common. Moreover, they are often parasitic, as, for example, upon Cyclops, which is frequently loaded with them.

Another compound form is *Ophrydium*, a colony of which (not unusually large) is shown of the natural size in Fig. 18, with a single zoöid, magnified, by the side of it, in Fig. 19.

Lastly, we have an exquisite group of organisms related to Vorticella, but possessing a transparent envelope, the forms of which are most varied, but always graceful. *Vaginicola* (Fig. 17) is a good example of this, and *Cothurnia* (Fig. 20) still more so. Many of these organisms, too, are furnished with a plate, attached either to the head or to the body, which plate, when they withdraw into their cases, closes the latter perfectly, as in the case of the exquisite *Pyxicola* (Fig. 21).

A very interesting but singularly obtrusive organism is the *Stylonychia* (Figs. 22, 23). How often has it happened to us to have an interesting object nicely in the field of view, and then to have it knocked out of sight by the blundering incursion of this burly fellow, who runs so rapidly by means of his “styles” that he gives nothing time to get out of the way. He is of interest to us, however, as the representative of a class in which the body is not ciliated, or very partially and slightly so, usually round the mouth. We have frequently found *Stylonychia*, in company with Vorticella and *Paramœcium* (Plate IX. Fig. 6), in the water in which flowers have been standing for a few days; sometimes the numbers are so great as to make the water quite milky.

One more form must conclude this short sketch of the great Infusorial family. It is the *Acineta* (Fig. 24), which, attached by its foot-stalk, and devoid of cilia, patiently waits, with outspread arms, to receive and embrace

smaller members of the family as they dance merrily about. Alas! its embrace is as fatal as that of the image of the Virgin which bore beneath its robe spikes and daggers, for the victim struggles vainly to escape, and the nourishment from its body is rapidly absorbed.

And here we take our leave of a group which, simple as is the construction of the animals which it includes (for every one, great and small alike, is composed of a single cell), is yet full of beauty and interest. He who wishes to pursue the matter further will find in Saville Kent's *Manual of the Infusoria* a perfect mine of information, to which we gladly acknowledge our indebtedness, both now and in time past.

CHAPTER IX

Fresh-water Worms—Planarians—Hydra—Polyzoa—Rotifers—
Chætonotus—Water-Bears.

The fresh-water worms form a large and well-defined group, and a few words regarding them may be useful.

They are very common, and very difficult to find information about, most of the work relating to them having been done in Germany. At the same time, they are so highly organised and so transparent that the process of their life-history may be easily followed.

One large group has the peculiarity of multiplying by division, the last joints or segments being devoted to the formation of the new individual. At one time of the year the ordinary sexual process of reproduction takes the place of this method, and each worm is then surrounded by a belt such as may be seen in the common earthworm under similar conditions. Further information on this subject is greatly needed.

The type is the common *Naiis*, which has a body of thirty segments or more, two eye-specks on the head, and a double row of bristles along the back; whilst below, each segment carries strong hooked bristles, nearly buried in the body, by means of which the worm crawls. Inside the mouth is a large proboscis, which can be protruded, and this leads into a stomach which is merely an enlargement of the intestine which succeeds it. The circulation of the blood (which is colourless) can be easily watched. It begins at the tail with a contraction of the dorsal vessel, passes up to the head, and then down below the intestine to the tail again. The intestine is ciliated inside, and it is by a current of water carried into the intestine by these cilia that the blood is aërated.

In the next genus, *Dero*, this is clearly seen, for the tail (Plate XIV. Fig. [1](#)) is opened out into a wide shield, from which rise four, six, or even eight

finger-like processes. These parts are all ciliated, and contain a network of blood-vessels. The worm lives in a case which it builds in the mud, and the way to find it is to put some of the mud into a glass beaker with water, and allow it to stand. If there be members of this family in it, their tails will be seen protruding above the water. Pour out the mud sharply, fill up with water, and allow the dirt to subside, and the worms may then be made to leave their cases by pressure by a camel hair pencil on the lower end of the tube, and may be caught with the dipping tube and placed in the live-box. They have no eyes, otherwise the general outline of the body closely resembles that of Naïs.

Slavína (Fig. 2) has a row of touch-organs, like pimples, round each segment, and is a dirty looking creature, with an inordinately long first pair of bristles, but this reaches its acme in *Pristina* (Fig. 3) (sometimes, though wrongly, called *Stylaria*) *parasita*, which has three long sets of bristles upon the back, and keeps these in constant wing-like motion. The true *Stylaria* has a long trunk, set right in the head, and tubular (Fig. 6); it grows to a considerable length, and when in the stage of fission it is very funny to see the two proboscides waving about, one on the middle, as well as the original one at the head. There is also a form with a shorter proboscis of the same kind.

Bohemilla has a tremendous array of saw-like bristles upon the back, whilst *Chætogaster* has none at all in this position, and few below. *Æolosoma* has merely tufts of hair instead of bristles, and swims freely. It is easily recognised by the red, yellow, or green pigment spots in its skin, and by the ciliated head. Rarest of all the family is the one which connects it with the ordinary *Tubifex*, the red worm which lives in masses in the mud of brooks and ponds, the waving tails protruding above the water, and being instantly withdrawn when a foot is stamped upon the bank. Their Naid cousin is *Naidium*, and has red blood, but multiplies by fission, which *Tubifex* does not.

Another group of worms is the *Planarians*, small leech-like worms, black, white, or brown, which are rarely absent from a gathering. The would-be

investigator will find in them an abundant field for work, as they are but very imperfectly known or studied.

The great enemy of all these worms is the *Hydra*, a good idea of which may be formed from Plate IX. Fig. 13. There are three species, all of which are fairly common. They capture their prey in exactly the same way as sea-anemones and the marine hydroid forms, so numerous and varied.

Nor must we omit to notice the exquisitely beautiful Polyzoa, such as *Lophopus* (Plate XIV. Fig. 4), with its ciliated tentacles and transparent social home; *Fredericella* (Fig. 5), with its graceful stems, and their still more graceful inhabitants; and the wonderful *Cristatella*, whose colonies form bodies which crawl over the stems of water plants. But for grace, beauty, and variety, the Rotifers assuredly outshine all their fellow inhabitants of our ponds and streams.

We can only take a few types, and of all these the most common is the common Rotifer (Plate IX. Fig. 10). It is there shown in the act of swimming, but it can withdraw its “wheels” and creep like a leech, protruding its foot as it does so. It is distinguished by the two eye-spots on the proboscis from *Philodina*, in which they are on the breast, and *Callidina*, which has none. When at ease in its mind, the animal protrudes its wheels, and by their action draws in particles of food, these passing down to the incessantly moving jaws, which act like a mill and crush the food before it passes on to be digested. The movement of the jaws may even be seen in the young Rotifer whilst still in the egg within the body of the parent, and as the egg reaches its full development other eggs again are visible within it, so that we may have three generations in one individual. The males of most of the Rotifera are unknown. Those that are known are very lowly organised, having only the ciliary wreath and the reproductive organs, and are only found at certain seasons of the year. For the remainder of the time parthenogenesis is the rule, just as among the Aphides. We select a few individuals for illustration as types. Those who wish to pursue this study further must be referred to the monumental work of Hudson and Gosse.

The common Rotifer, already referred to, may be taken as the type of the Bdelloida, or leech-like class, so called from their mode of “looping” themselves along. The group is a comparatively small one in comparison with the next, the Ploïma, or free-swimmers. We can only select from the vast variety a few species, first of those classed as illoricated, from their being without a *loríca*, or case, and then of the loricated, which possess it. A very large and common form is *Hydátina* (Plate XIV. Fig. 7), which lives by choice in the reddish pools of water found often by the roadside. It shows the whole organisation of the class magnificently; the ciliary wreath on the head, with the striped muscles which draw the latter back, the powerful jaws, the digestive canal with its crop and intestine, the ovary with the developing eggs, the water-vascular system with the curious vibratile tags, and finally, the cloaca, which receives the waste of the body and expels it at intervals.

Another form, also common, especially in clear water, is *Synchæta* (Fig. 8), very much like a kite or peg-top in shape, which has the power of attaching itself by a glutinous thread, and spinning round at a tremendous rate. Then there is the gigantic *Asplanchna* (Fig. 9), which has no opening below, so that the waste must be discharged by the mouth; and curious *Sacculus*, which gorges itself with chlorophyll until it looks like a green bag with a string round it, but clear and sparkling. Of the *Notommatae* there is a whole host, but we can only mention the beautiful *N. Aurita* (Fig. 10), with an eye of a beautiful violet colour, composed of several spherules massed together, and two curious ear-like processes on the head, from which it takes its name. Some of the Ploïma have powers of leaping which must be noticed. The *Triarthra* (Fig. 11) has three arms, or what we may call such, which it can stretch out suddenly and leap to a considerable distance, whilst in *Polyarthra* the arms become a whole cluster of broad saw-like bristles.

We pass on to note a few species of the mail-clad or loricated Rotifers, chief among which the great *Euchlanis* (Fig. 12), a noble-looking fellow, calls for our attention, his great size rendering him easily visible to the naked eye. It is difficult to avoid using the masculine gender, but, of course, all those

figured and described are of the gentler sex. *Salpina*, too (Fig. 14), with its box-like lorica, armed with spines at each of the upper angles, and having three below, is quite easily recognised, and very common. *Brachionus* (Fig. 13) has a shield-shaped case, well furnished with spines, symmetrically arranged at the top, and an opening below for the flexible wrinkled tail, like the trunk of an elephant. *Pterodina* (Fig. 15) has a similar tail, but a round case, and the head is much more like that of the common Rotifer when extended. *Anuraea* (Fig. 16), on the other hand, has no tail, and its case is shaped like a butcher's tray, with a handle at each corner. *Dinocharis* (Fig. 17) has a roof-like case, with long spines on the root of the tail, and a forked stiff foot. *Noteus* (Fig. 18) is much like *Pterodina*, except in its foot, which more nearly resembles that of *Dinocharis*.

The list might be indefinitely extended, but sufficient has probably been said to enable the tyro to find his bearings in this large, beautiful, and interesting class.

We pass on to notice in conclusion two or three of the fixed forms, of which a beautiful example is the *Melicerta ringens* (Plate IX. Fig. 7), whose building operations have a never-ending charm. Particles of débris are accumulated in a curious little cavity in the chin, in which they are whirled round, and mixed with a secretion which binds them together, and when a brick is made the head is bent down and the brick applied to the desired spot with mathematical regularity. By supplying fine particles of innocuous colouring matters, the *Melicerta* may be made to build a variegated case. The most remarkable specimen known is the one figured in Hudson and Gosse's work, which was found by the present writer in a specimen of water from which he had already obtained five-and-twenty species of various kinds of Rotifer; the water was collected by an inexperienced person, and there was only a pint of it. It had, moreover, been kept for three weeks, and the moral of that is, to preserve one's gatherings, and keep an aquarium into which they may be poured when done with for the moment. New forms will often develop with startling rapidity, their eggs having been present in the original gathering. The young form of *Melicerta*, shown in

Plate XIV. Fig. [20](#), is strangely unlike its mother, and much more nearly resembles its father.

Another group of extreme beauty is the Flosculariæ (Fig. [19](#)), several species of which are very common. They will be easily known by their appearance, which resembles a shaving brush when closed; whilst, when opening, the shaving brush resembles a cloud of delicate shimmering threads, which at last stand out straight, radiating all round the head of the creature, and forming the trap by means of which it catches its prey. Finally, there is the lovely *Stephanoceros* (not, unfortunately, very common), with its five symmetrically placed and gracefully curved arms, perhaps the most lovely of all Rotifers, with its exquisitely transparent body, sparkling with masses of green and golden brown. He who finds this has a treasure indeed, and will be encouraged to prosecute his studies in this “Fairyland of Microscopy.”

Two irregular forms call for a word of remark. The first is *Chætonotus* (Plate XIII. Fig. [27](#)), which stands on the borderland of the Infusoria and the Rotifers, neglected as a rule by the students of both; and the second the *Tardigrada* (Plate XIV. Fig. [21](#)), or water-bears, which have feet like those of the red wriggling larva of *Chironomus*, whose silky tubes are common enough on submerged walls and on the stems of plants, these feet consisting of a mass of radially arranged hooklets, which can be protruded or withdrawn at will; whilst the head of the water-bear is far more like that of a louse, pointed and hard, and suited for burrowing about, as the animal does, among the rubbish at the bottom of the bottle. Both the genera just referred to will repay careful study, as little is known of their life-history or development.

A few words must be devoted, in conclusion, to the Entomostraca, those shrimp-like animals which, like their marine relatives, act as scavengers to the community. Fig. [22](#) is a portrait of *Cypris*, a not very handsome form, but one very commonly found. Its shell is opaque, so that the internal organs are difficult to observe. Far different in this respect is the beautiful *Daphnia*, the water-flea *par excellence*, whose carapace is of crystalline

clearness, so that every movement of every one of the internal organs may be followed with the greatest facility. There are many species of the genus, and some of them are very common, so that the opportunity of examining these lovely objects is easily obtained. Plate XIV. Fig. [23](#), shows the most common of all the class under notice, the *Cyclops*, so named from the fact that, like the fabled giants of classical literature, it has a single eye in the middle of its forehead. It is often loaded with Infusoria, especially Vorticella and Epistylis, already described, to such an extent that its movements are greatly hampered.

XII.

FIG.

- | | |
|-----------------------------------|-------------------------------------|
| 1. Tubercle, Sun-star | 14. Do. Truncatulina |
| 2. Zoophyte, Gemellaria | 15. Do. Polymorphina |
| 3. Cuttle bone | 16. Do. Miliolina |
| 4. Plate of ditto from above | 17. Gold dust, with quartz |
| 5. Zoophyte, Antennularia | 18. Foraminifer, Lagenella vulgaris |
| 6. Pedicellaria, skin of Starfish | 19. Pouches, Skin of Rat's tail |
| 7. Shell, Foraminifer | 20. Foraminifer, Biloculina ringens |
| 8. Snake-star, disc from below | 21. Ore, Copper |
| 9. Pedicellaria, Echinus | 22. Zoophyte, Membranipora pilosa |
| 10. Wing-case, Weevil | 23. Human skin, injected |
| 11. Coralline | 24. Coal, Longitudinal section |
| 12. Spine, Echinus | 25. Do. Transverse section |
| 13. Foraminifer, Polystomella | 26. Lung, Frog |



XII.

We have not space to figure more of these creatures, but other forms will be found not inferior in interest to those mentioned. The most curious of all are those which earn a dishonest and lazy living by attaching themselves to the

bodies of other and larger animals, chiefly fish. One of the largest is the *Argulus*, the bane of aquarium keepers, which is of considerable size, and attacks gold-fish, and in fact almost any fish to which it can obtain access.

The gills of the stickleback will furnish examples of the curious *Ergasilus*, which consists chiefly of an enormous pair of hooks and two long egg-bags, the latter, in varying form, being carried by many of the Entomostraca.

Upon the fins of the same fish will be found the remarkable *Gyrodactylus*, a worm-like animal which attaches itself by a large umbrella-like foot, in the centre of which are two huge claws. The head is split down the middle for some distance. We may mention, in concluding our notice of the external and involuntary guests of the unlucky stickleback, that its skin is usually frequented by hosts of the Trichodina described in the last chapter. Of the internal parasites, want of space forbids us to speak.

CHAPTER X

Marine Life—Sponges—Infusoria—Foraminifera—Radiolaria—
Hydroid Zoophytes—Polyzoa—Worms—Lingual Ribbons
and Gills of Mollusca—Star-Fishes and Sea-Urchins—
Cuttle-Fish—Corallines—Miscellaneous Objects.

Great as is the range of objects presented to the student of fresh-water life, the latter field is limited indeed as compared with that afforded by the sea. The Infusoria and Rotifers furnished by the latter are, indeed, much fewer in number and variety, but the vast host of sponges, polyzoa, hydroids, crustacea, molluscs, ascidians, and worms, to say nothing of the wealth of vegetable life, renders the sea the happy hunting-ground of the microscopist.

Whether it be along the edge of the water, as the tide retreats, especially after a gale; or in the rock-pools; or, perhaps best of all, upon those portions of the shore left uncovered only by the lowest spring-tides, the harvest is simply inexhaustible. Stones turned up will exhibit a world in miniature. Encrusted with green or pink sponges, or with gelatinous masses of ascidians, fringed at its edges with hydroids, coated above with polyzoa, a single one will often supply more work than could be got through in a week of steady application.

A description of the fresh-water sponge already given may serve very well to indicate the general outlines of the organisation of the marine ones too. The spicules of the latter are, however, not always flinty; very often, as in the case of *Grantia* (Plate IX. Figs. 8 and 14), they are calcareous, a point which can be settled by the application of a little nitric acid and water. If lime be present there will be strong effervescence, and the separation of the spicules can only be effected by gently warming a portion of the sponge in caustic potash solution, pouring the resulting mass into water, and allowing the spicules to settle. The washing and settling must be repeated several

times, and a portion of the deposit may then be taken up with a dipping-tube, spread upon a slide and dried, and then covered in balsam solution. The forms are endless, and the same sponge will often supply three or four, or even more. Among them may be seen accurate likenesses of pins, needles, marlin-spikes, cucumbers, grappling-hooks, fish-hooks, porters'-hooks, calthrops, knife-rests, fish-spears, barbed arrows, spiked globes, war-clubs, boomerangs, life-preservers, and many other indescribable forms. The flinty forms must be prepared by boiling, as described in speaking of the mounting of diatoms in Chapter XI., except that, of course, only one settlement is required after thorough washing.

Every one who has been by or on the sea on a fine summer night must have noticed the bright flashes of light that appear whenever its surface is disturbed; the wake of a boat, for example, leaving a luminous track as far as the eye can reach. This phosphorescence is caused by many animals resident in the sea, but chiefly by the little creature represented at Fig. 9, the *Noctilúca*, myriads of which may be found in a pail of water dipped at random from the glowing waves. A tooth of this creature more magnified is shown immediately above.

A large group of microscopic organisms is known to zoologists under the name of Foraminifera, on account of the numerous holes in their beautiful shells, most of which are composed of carbonate of lime, though some are horny and others are composed of aggregations of minute grains of sand, the forms in one class often closely imitating those in another. It is of the shells of these minute animals that the “white cliffs of old England” are very largely composed, and those who desire to understand the part which these tiny creatures have played, and are playing, in geology, will do well to study Huxley’s fascinating essay on “A Piece of Chalk.”

XIII.

FIG.

- | | |
|---------------------------------|---|
| 1. <i>Grantia compressa</i> | 14. <i>Vorticella nebulifera</i> |
| 2. <i>Euglena viridis</i> | 15. <i>Zoothamnium arbuscula</i> |
| 3. <i>Dinobryon sertularia</i> | 16. Do. do. contracted |
| 4. <i>Synura uvella</i> | 17. <i>Vaginicola crystallina</i> |
| 5. <i>Peridinium tabulatum</i> | 18. <i>Ophrydium versatile</i> (colony) |
| 6. <i>Coleps hirtus</i> | 19. Do. do. (single zoöid) |
| 7. <i>Trachelocerca viridis</i> | 20. <i>Cothurnia imberbis</i> |
| 8. <i>Trachelius ovum</i> | 21. <i>Pyxicola affinis</i> |
| 9. <i>Amphileptus gigas</i> | 22–23. <i>Stylonychia mytilus</i> |
| 10. <i>Bursaria Mülleri</i> | 24. <i>Acineta grandis</i> |
| 11. <i>Folliculina elegans</i> | 25. <i>Loxophyllum meleagris</i> |
| 12. <i>Stentor polymorphus</i> | 26. <i>Euplotes charon</i> (dividing) |
| 13. <i>Trichodina pediculus</i> | 27. <i>Chætonotus larius</i> |



XIII.

The inhabitants of these shells are Amœbæ, mere masses of jelly, and some forms may be found sliding over the weeds in almost every rock-pool. The anchor-mud, already spoken of, always contains some, and many forms

may be found in the sand from sponges, which should be passed through a series of wire sieves, of increasing fineness, and the residuum in each case be examined dry under a one-inch power. The shells may be picked up with a needle which has been slightly greased by being passed over the hair, and they may be mounted by sticking them to the slide with thin starch paste, putting on a cover-glass properly supported, and then running turpentine under the cover-glass, heating to expel the air, and finally filling up with balsam. Or, as opaque objects, they may be mounted in a cell dry. The forms are endless, but all are beautiful, and a few examples are given in Plate IX. Fig. 4 (*Miliolína*), and Plate XII. Fig. 7, which is a portion of the shell to show the holes, Fig. 13 (*Polystomella*), Fig. 14 (*Truncatulína*), Fig. 15 (*Polymorphína*), Fig. 16 (*Miliolína*, partly fossilised), Fig. 18 (*Lagéna*). and Fig. 20 (*Biloculína*).

Allied to these are the lovely Radiolaria, whose shells, constructed on a similar plan, are composed of flint. They are found in remarkable profusion in the deposit from Cambridge, Barbados, but also in a living state at even enormous depths in the ocean. The present writer has obtained many forms from *Challenger* soundings, and the great authority on this subject is Haeckel's report in the official accounts of the expedition of the above-named vessel.

The Hydroid Zoophytes are represented by several examples. These creatures are soft and almost gelatinous, and are furnished with tentacles or lobes by which they can catch and retain their prey. In order to support their tender structure they are endowed with a horny skeleton, sometimes outside and sometimes inside them, which is called the polypidom. They are very common on our coasts, where they may be found thrown on the shore, or may be dredged up from the deeper portions of the sea.

Fig. 13 is a portion of one of the commonest genera, *Sertularia*, showing one of the inhabitants projecting its tentacles from its domicile. Fig. 15 is the same species, given to show the egg-cells. This, as well as other zoophytes, is generally classed among the sea-weeds in the shops that throng all watering-places.

The form just referred to is a near relative of the Hydra, already described, and belongs to the same great family as the sea-anemones. One form, shown in Fig. 26, is the *Hydra Tuba*, long thought to be a distinct animal, but now known to be the young form of a jelly-fish, or Medusa. The Hydra Tuba throws off joints at intervals, each of which becomes a perfect jelly-fish. One of them is shown in Fig. 27. Fig. 28 represents a very small and pretty Medusa, the *Thaumantias*. When this animal is touched or startled, each of the purple globules round the edge flashes into light, producing a most beautiful and singular appearance. Fig. 29 exhibits the so-called compound eye of another species of Medusa, though it would appear that these are really connected with the nervous system of the animal, and have to do with the pulsating contractions of the bell by which it is propelled through the water.

In my *Common Objects of the Sea-Shore* the *Actíniæ*, or Sea-Anemones, are treated of at some length. At Fig. 16 is shown part of a tentacle flinging out the poison-darts by which it secures its prey; and Fig. 17 is a more magnified view of one of these darts and its case.

Much more might be said under this head, but we must pass on to another group, which, whilst possessing a certain general resemblance to the hydroid zoophytes, differs utterly from them in internal organisation. We have already referred to the fresh-water polyzoa. The marine forms are vastly more numerous, and more easily found, since not only pieces of weed upon which they grow are to be found upon every beach, but whole masses of leaf-like colonies, forming what is known as horn-wrack, may be plentifully found. Instead of the tentacles armed with sting-cells, like the anemone's, possessed by the Hydrozoa, the Polyzoa have arms clothed with active cilia, by which the food is swept into the mouth, passing on into the stomach, and then through the intestine to another opening.

Fig. 19 is a very curious zoophyte called *Anguinaria*, or snake-head, on account of its shape, the end of the polypidom resembling the head of the snake, and the tentacles looking like its tongue as they are thrust forward and rapidly withdrawn. Fig. 21 is the same creature on an enlarged scale,

and just below is one of its tentacles still more magnified. Fig. 23 is the ladies'-slipper zoophyte (*Eretea*); and Fig. 24 is called the tobacco-pipe or shepherd's-purse zoophyte (*Notamia*).

Fig. 22 is a portion of the *Bugula*, with one of the curious "birds'-head" processes. These appendages have the most absurd likeness to a bird's head, the beak opening and shutting with a smart snap (so smart, indeed, that the ear instinctively tries to catch the sound), and the head nodding backward and forward just as if the bird were pecking up its food. On Plate XII. Fig. 2, is a pretty zoophyte called *Gemellaria*, on account of the double or twin-like form of the cells; and Fig. 5 represents the *Antennularia*, so called on account of its resemblance to the antennæ of an insect. Fig. 22 is an example of a pretty zoophyte found parasitic on many sea-weeds, and known by the name of *Membranipora*. Two more specimens of zoophytes may be seen on Plate XI. as they appear under polarised light. Fig. 17 is the *Cellularia reptans*; and Fig. 20 is the *Bowerbankia*, one form of which occurs in fresh water.

Among the worms we may refer to the beautiful little *Spirorbis*, whose tiny coiled spiral tube may be found attached to almost every sea-weed, and which, when placed in a trough of sea-water, protrudes its beautiful crown of plumes. In chalk or other soft rocks, again, the tubes of *Spio*, with its two long waving tentacles, may be found by hundreds. Then there are the centipede-like worms, which may be found under nearly every stone, and which belong to the great family of Nereids, provided with formidable jaws and stiff bristles of various forms. The Serpulæ are allied to the *Spirorbis* already mentioned. Parts of the so-called feet of one of these worms are shown in Fig. 36, where the spears or "pushing-poles" are seen gathered into bundles, as during life. One of them, on a larger scale, is shown in Fig. 32. The gorgeous hairs of *Aphrodite* have already been alluded to.

In the sea the few species of Crustacea which fresh water offers to the observer in the shape of Cyclops and its allies become thousands, and their changes during development are numerous and puzzling. Who, for example, would suppose that the young stage of the Cyclops was indistinguishable in

habits, and almost in form, from that of the barnacle which adheres to the rocks? Yet such is the case, and there are other metamorphoses even more startling. Fig. 25 is the larva of the common crab, once thought to be a separate species, and described as such under the name of *Zoæa*.

The Mollusca proper will not afford us many objects, except in the form of their lingual ribbon, which may be extracted from the mouth, gently heated in *liquor potassæ*, and mounted in balsam after well washing, when the rows of teeth form splendid objects by polarised light. The palate of a whelk is shown in Plate XI. Fig. 19.

Again, the gills of the mussel will afford a beautiful illustration of ciliary action. If a portion of the thin plates which lie along the edge of the shell be examined in a little of the liquor, the action may be splendidly seen, and watched for a long time (Fig. 39).

The structure of shell, *e.g.* oyster-shell, is well shown in three examples: Fig. 34 is a group of artificial crystals of carbonate of lime; and on Figs. 38 and 39 may be seen part of an oyster-shell, showing how it is composed of similar crystals aggregated together. Their appearance under polarised light may be seen on Plate XI. Figs. 1 and 6.

We now pass on to the Echinoderms, including the star-fishes and sea-urchins.

The old story of the goose-bearing tree is an example of how truth may be stranger than fiction. For if the fable had said that the mother goose laid eggs which grew into trees, budded and flowered, and then produced new geese, it would not have been one whit a stranger tale than the truth. Plate IX. Fig. 33, shows the young state of one of the common star-fishes (*Comátula*), which in its early days is like a plant with a stalk, but afterwards breaks loose and becomes the wandering sea-star which we all know so well. In this process there is just the reverse of that which characterises the barnacles and sponges, where the young are at first free and then become fixed for the remainder of their lives. Fig. 30 is the young of another kind of star-fish, the long-armed Ophiúris, or snake-star.

Fig. 37 is a portion of the skin of the common sun-star (*Solaster*), showing the single large spine surrounded by a circle of smaller spines, supposed to be organs of touch, together with two or three of the curious appendages called pedicellariæ. These are found on star-fishes and Echini, and bear a close resemblance in many respects to the bird-head appendages of the zoophytes. They are fixed on foot-stalks, some very long and others very short, and have jaws which open and shut regularly. Their use is doubtful, unless it be to act as police, and by their continual movements to prevent the spores of algæ, or the young of various marine animals, from effecting a lodgment on the skin. A group, of pedicellariæ from a star-fish is shown on a large scale on Plate XII. Fig. 6, and Fig. 9 of the same Plate shows the pedicellariæ of the Echinus.

Upon the exterior of the Echini, or sea-urchins, are a vast number of spines having a very beautiful structure, as may be seen by Fig. 35, Plate IX., which is part of a transverse section of one of these spines. An entire spine is shown on Plate XII. Fig. 12, and shows the ball-and-socket joint on which it moves, and the membranous muscle that moves it. Fig. 8 is the disc of the snake-star as seen from below. Fig. 1 is a portion of skin of the sun-star, to show one of the curious madrepor-like tubercles which are found upon this common star-fish. Fig. 3 is a portion of cuttle “bone,” very slightly magnified, in order to show the beautiful pillar-like form of its structure; and Fig. 4 is the same object seen from above. When ground very thin this is a magnificent object for the polariscope.

One or two miscellaneous objects now come before our notice. Fig. 11 is one of those curious marine plants, the Corallines, which are remarkable for depositing a large amount of chalky matter among their tissues, so as to leave a complete cast in white chalk when the coloured living portion of the plant dies. The species of this example is *Jania rubens*.

Fig. 19 is part of the pouch-like inflation of the skin, and the hairs found upon the rat’s tail, which is a curious object as bearing so close a similitude to Fig. 22, the sea-mat zoophyte. Fig. 23 is a portion of the skin taken from the finger, which has been injected with a coloured preparation in order to

show the manner in which the minute blood-vessels or “capillaries” are distributed; and Fig. 26 is a portion of a frog’s lung, also injected.

The process of injection is a rather difficult one, and requires considerable anatomical knowledge. The principle is simple enough, being merely to fill the blood-vessels with a coloured substance, so as to exhibit their form as they appear while distended with blood during the life of the animal. It sometimes happens that when an animal is killed suddenly without effusion of blood, as is often seen in the case of a mouse caught in a spring trap, the minute vessels of the lungs and other organs become so filled with coagulated blood as to form what is called a natural injection, ready for the microscope.

Before leaving the subject I must ask the reader to refer again for a moment to the frog’s foot on Plate X., and to notice the arrangement of the dark pigment spots. It is well known that when frogs live in a clear sandy pond, well exposed to the rays of the sun, their skins are bright yellow, and that when their residence is in a shady locality, especially if sheltered by heavy overhanging banks, they are of a deep blackish-brown colour. Moreover, under the influence of fear they will often change colour instantaneously. The cause of this curious fact is explained by the microscope.

Under the effects of sunlight the pigment granules are gathered together into small rounded spots, as seen on the left hand of the figure, leaving the skin of its own bright yellow hue. In the shade the pigment granules spread themselves so as to cover almost the entire skin and to produce the dark brown colour. In the intermediate state they assume the bold stellate form in which they are shown on the right hand of the round spots. Very remarkable forms of these cells may be found in the skin of the cuttle-fish.

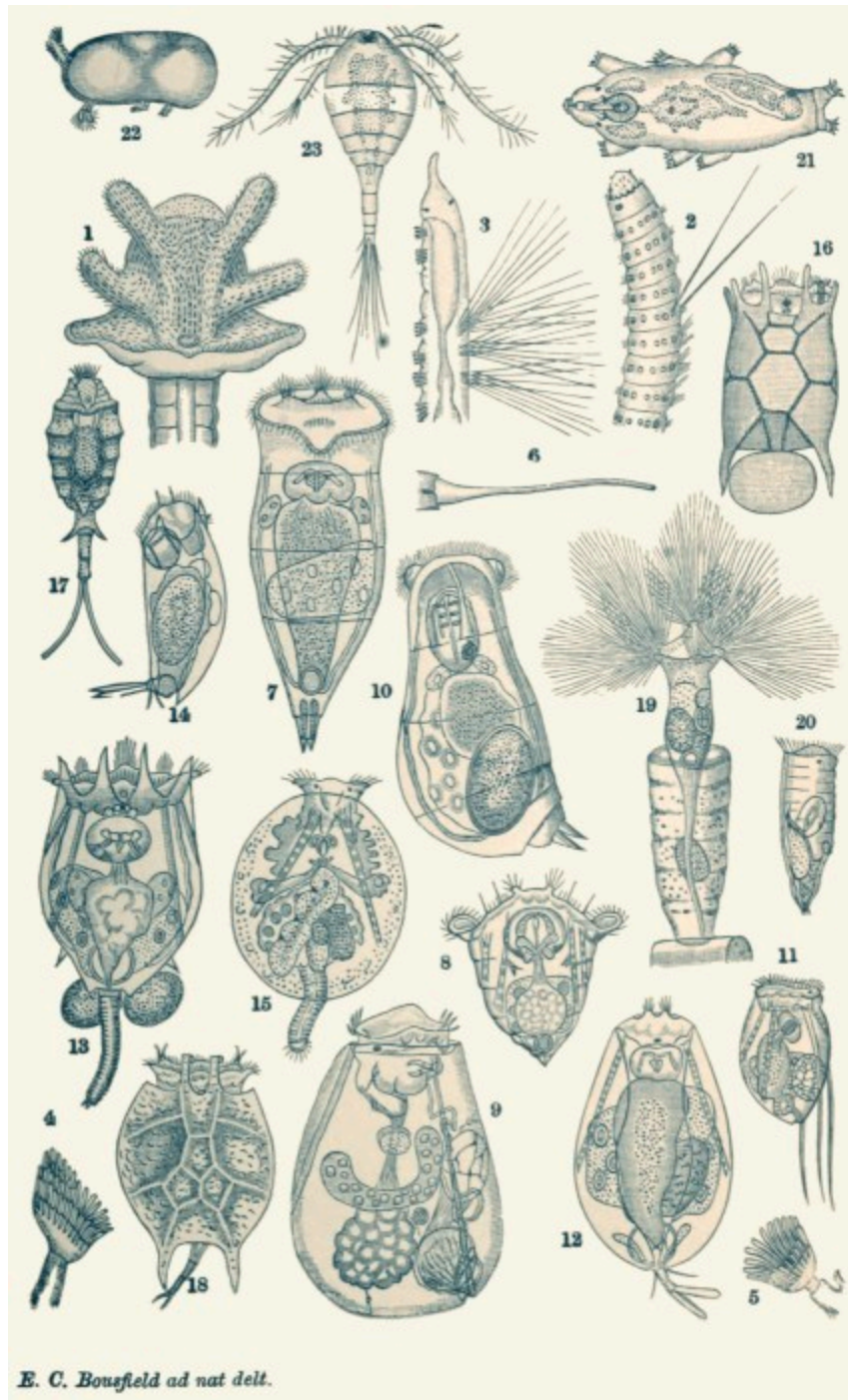
Figs. 24 and 25 are two examples of coal, the former being a longitudinal and the latter a transverse section, given in order to show its woody character. Fig. 17 is a specimen of gold-dust intermixed with crystals of quartz sand, brought from Australia; and Fig. 21 is a small piece of copper-ore.

Every possessor of a microscope should, as soon as he can afford it, add to his instrument the beautiful apparatus for polarising light. The optical explanation of this phenomenon is far too abstruse for these pages, but the practical application of the apparatus is very simple. It consists of two prisms, one of which, called the polariser, is fastened by a catch just below the stage; and the other, called an analyser, is placed above the eye-piece. In order to aid those bodies whose polarising powers are but weak, a thin plate of selenite is generally placed on the stage immediately below the object. The colours exhibited by this instrument are gorgeous in the extreme, as may be seen by Plate [XI](#), which affords a most feeble representation of the glowing tints exhibited by the objects there depicted. The value of the polariser is very great, as it often enables observers to distinguish, by means of their different polarising properties, one class of objects from another.

If the expense of a polarising apparatus be too great for the means of the microscopist, he may manufacture a substitute for it by taking several thin plates of glass, arranging them in a paper tube so that the light may meet the surface of the lowest one at an angle of about 52° , and placing the bundle above the eye-piece to act as an analyser; whilst, by using a plate of glass, and so arranging the lamp that the light falls upon it at the above angle, and is reflected up the tube of the microscope, he will find on rotating the extemporised analyser that the phenomena of polarisation are to a great extent reproduced; whilst by splitting an extremely thin film from the surface of a sheet of mica, such as is employed for making smoke-screens above glass globes, he will have a substitute for the selenite by means of which alone can the gorgeous effects be produced. The extemporised apparatus will not, of course, give such perfect effects, but this is sometimes an advantage, and the present writer has used the same means with considerable success in photographing starch-granules.

FIG.

- | | |
|---------------------------------------|----------------------------------|
| 1. <i>Dero latissima</i> | 13. <i>Brachionus amphiceros</i> |
| 2. <i>Slavina serpentina</i> | 14. <i>Salpina mucronata</i> |
| 3. <i>Pristina longiseta</i> | 15. <i>Pterodina patina</i> |
| 4. <i>Lophopus crystallinus</i> | 16. <i>Anurœa brevispina</i> |
| 5. <i>Fredericella sultana</i> | 17. <i>Dinocharis tetractis</i> |
| 6. <i>Stylaria proboscidea</i> (head) | 18. <i>Noteus quadricornis</i> |
| 7. <i>Hydatina senta</i> | 19. <i>Floscularia ornata</i> |
| 8. <i>Synchoeta mordax</i> | 20. Young <i>Melicerta</i> |
| 9. <i>Asplanchna Brightwellii</i> | 21. <i>Macrobiotus</i> (sp.?) |
| 10. <i>Notommata aurita</i> | 22. <i>Cypris fusca</i> |
| 11. <i>Triarthra longiseta</i> | 23. <i>Cyclops quadricornis</i> |
| 12. <i>Euchlanis triquetra</i> | |



CHAPTER XI

Hints on the Preparation of Objects—Preservative Fluids— Mounting Media—Treatment of Special Objects.

The microscopist who relies altogether on the dealer for his permanent preparations may expend a good deal of money, but the satisfaction which he derives from his hobby will be very inferior to that experienced by the worker who endeavours to secure, for exhibition or for reference, specimens of the objects which he finds most interesting and instructive to himself.

It will be our endeavour in the following pages to give a summary of the elementary principles upon which reliance is to be placed, though it must be clearly understood that the technique of the subject, already occupying a vast amount of literature, is extending day by day, so that it is impossible to deal exhaustively even with one single section of it. Reference must be made, for further information, to such publications as the *Journal of the Royal Microscopical Society*, or that of the Quekett Club, or to the monographs on the various departments. Davies' work on the general subject will also be found useful by the beginner.

Taking first the question of reagents, we may mention five which leave the cells of a tissue as nearly as possible in the natural condition, but fit for permanent preservation. The first of these, in order of importance and of general applicability, is alcohol, represented for most purposes by methylated spirit, which contains about 84 per cent. of absolute alcohol, though, unfortunately for our purpose, there is a certain quantity of mineral naphtha in it in addition. This last has the effect of making it go milky upon dilution with water, which is a considerable disadvantage, though the milkiness disappears to some extent on standing, and it is rarely worth the while of the ordinary microscopist to go through the formalities necessary to obtain permission to purchase unmineralised spirit, which cannot be had

in quantities of less than five gallons (as it is only to be had from the distillers under an Excise permit), and distillers may not supply less.

Four parts of methylated spirit with one of water forms the classical “70 per cent.” alcohol, the most generally useful of all fluids for hardening and preserving purposes. A considerable quantity of this fluid should always be available.

Whatever other fluid may be used to begin with, spirit must almost always be used to finish the process, and fit the tissue for section-cutting and staining.

Of purely preservative, or fixative, fluids, we may mention “formalin,” a 40 per cent. solution of formic aldehyde, which is rapidly coming to the front, as indeed it deserves to do. It is but slightly poisonous, if at all, and leaves in the tissue nothing which requires subsequent removal before proceeding to harden for section-work, whilst it is an admirable preservative of cell-form.

Another admirable but highly poisonous reagent is corrosive sublimate, in saturated solution, with 2 per cent. of acetic acid.

A fourth is osmic acid, used in 1 per cent. solution. This is a highly valuable reagent, but extremely expensive, very poisonous, and giving off fumes which are most irritating to the eyes.

The fifth, a very gentle, but in many respects very satisfactory one, is picric acid in saturated solution. Tissues preserved in this medium must not be washed out with water, as it enters into very feeble combination with protoplasm, and the cells swell and disintegrate as the reagent is dissolved out.

Of mounting media we may mention glycerine, glycerine jelly (made by dissolving starch in glycerine with the aid of heat), and Canada balsam, dissolved in xylol or benzole. The Canada balsam must be dried hard by evaporation over a water-bath, and dissolved as wanted. Under no

circumstances should raw balsam be used, as it takes years to set hard, and turns of a deep yellow colour in the process.

Chloroform is frequently used as a solvent, but it has the disadvantage of attacking and extracting a large number of the aniline dyes used for staining structures, an objection from which the mineral solvents are free.

We will now proceed to go through the objects already referred to, and indicate the method of preservation.

For the study of the cell-structures of plants the portion to be examined is to be placed in spirit of about 30 per cent. strength, which is changed after twenty-four hours for 40 per cent., after a further twenty-four hours for 55 per cent., and finally, as regards our present purpose, in 70 per cent. spirit, in which it may remain until required for section-cutting. The effect of this treatment is to extract the bulk of the water from the tissue, with the minimum of shrinkage of the cells, the latter being preserved in their natural relations to surrounding parts.

In some cases, however, it is desirable to examine and preserve delicate structures, or parts, or dissections, in a medium which allows of the retention of the greater part of the natural moisture, and in such a case the tissue is immersed in glycerine diluted very much in the same way as the alcohol in the last process, but with very much longer intervals between the alterations of strength, until it reaches pure glycerine, in which it remains for a considerable time, as the exchange between the tissue and the dense fluid surrounding it goes on very slowly.

A combination of the two methods is also possible, the spirit-hardening being carried out for a portion of the time, and the tissue being thereafter transferred to glycerine, diluted or pure.

The object of using glycerine at all is merely that it has a much lower refractive index than balsam, so that delicate structures may sometimes be better seen in the former medium, but balsam is to be preferred wherever it is possible to use it, *i.e.* almost always. The writer has not mounted a

preparation in glycerine or a medium containing it for many years, nor, with proper staining, does he think it can ever be necessary to do so, except in the case of dissections in which the glycerine can be slowly run in without disturbing the arrangement, as spirit would be pretty sure to do. The harder portions of plants, woody stems, shells of fruit, or the like, require different treatment, and must, as a rule, be allowed to dry thoroughly before being cut.

Starch granules are somewhat troublesome to mount satisfactorily. The writer has tried many methods, and, on the whole, prefers a glycerin-gelatin medium, which keeps for an almost indefinite time, and may be made as follows: Thirty grains of gelatine (Nelson's "brilliant" or other transparent gelatine is to be preferred) are allowed to soak in water, and the swollen gelatine is drained, and dissolved in the water which it has absorbed, by the aid of a gentle heat. An equal bulk of pure glycerine is then added. In using, a small portion is transferred to a slide with the point of a knife and melted, a small quantity of starch granules added, and stirred into it with a needle. The cover-glass is then laid up on the still-fluid drop, pressed gently down so that the drop is extended to the margin of the cover, and the whole allowed to cool. It is then to be painted round with several layers of Brunswick black, or Hollis's glue, or zinc-white cement, to prevent evaporation,—Hollis's glue being perhaps the best medium for the purpose.

Petals, or other parts of which it is desired to obtain a surface view, must be mounted in cells, which may be made by the use of button-moulds of suitable size, cemented to the glass slide with marine glue. The slide must be free from grease, as the tissue must be fixed in position by the use of gum, and allowed to dry thoroughly before closing the cell, or the cover-glass will be bedewed with moisture when the cell is closed. The best plan is, after air-drying for a couple of days, to place the preparation on a metal plate over a beaker of boiling water for an hour or more, and then to close the cell immediately with Brunswick black, maintaining the heat at first to ensure rapid drying, and then slowly withdrawing it. When cool, another

coat should be given, and rather thick covers should be used, as these preparations are never required to be examined with high powers.

To mount pollen-grains, they should be sprinkled upon the surface of a slide which has been previously moistened with thin gum, and allowed to dry until it has become just “tacky”; the drying is then completed by gentle heat and a drop of balsam placed upon the grains, with a cover-glass over all. Bubbles will probably form, but with Canada balsam this is not of the slightest importance, as they always come out of their own accord, and balsam mounts should never be closed with cement of any kind until thoroughly dry.

Air-bubbles in other media may be eliminated by the use of the air-pump shown in Fig. 16, which may be obtained from Baker at a very reasonable rate, and which is useful not only for that purpose, but for accelerating the drying of moist tissues. To do this, there is placed upon the plate of the pump a porcelain dish containing strong sulphuric acid, and upon this is placed a little triangle of platinum wire, which serves to support the preparation. The air is now exhausted; the tissue parts with moisture to supply its place, and this moisture is in turn greedily absorbed by the sulphuric acid, so that drying is rapid and continuous, as well as very thorough, whilst the process has the great advantage of dispensing entirely with the use of heat.

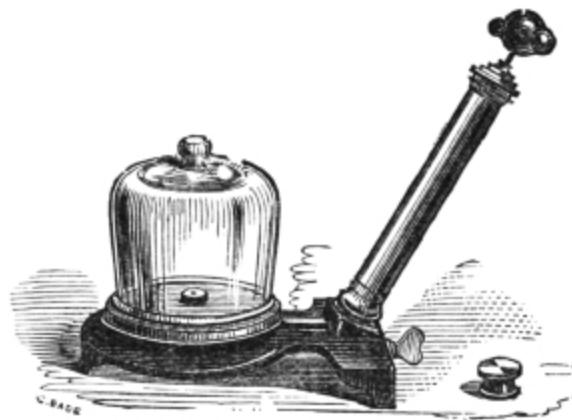


FIG. 16.

Portions of many of the delicate algæ may be mounted in glycerine, having previously been soaked in it as already described; whilst the unicellular forms, such as desmids and diatoms, may be preserved in almost exactly the natural condition by simply mounting them in a saturated solution of picric acid.

Probably formalin, in a solution of 10 per cent. strength, would answer the purpose equally well, but the writer has not tried it. It is hardly necessary to say that, with such extremely fluid media, great care is required in closing the cell. A thin layer of Hollis's glue should be first painted on, to secure the cover in position, and when this is thoroughly dry, several successive layers must be added in the same way.

It may be said here, that it is advisable in all cases to use circular cover-glasses, as far as possible, as they lend themselves with great facility to a mechanically accurate closure. This slide is placed upon a turn-table, carefully adjusted until the cover is seen to be central when rotated, and a brush, preferably a small camel-hair pencil, charged with the desired fluid, but not in large excess, is held against the junction of the slide and cover, whilst the table is rapidly spun. A little experience will teach better than any description what amount of fluid there should be in the brush, and how thick the cement should be. If too thick, it will drag off the cover; if too thin, it will flow over the latter and over the slide.

The preparation of diatom-skeletons as permanent objects is easy. Consisting, as they do, of pure siliceous, or flint,—*i.e.*, practically glass,—they resist long boiling in acids, so that there is little difficulty in isolating them from any organic matter with which they are mingled. It is generally recommended to treat them with strong nitric acid. This is a mistake. The acid acts much more powerfully and less violently when diluted with an equal bulk of water, and it is in an acid so diluted that portions of water-plants, or other diatomaceous material, should be boiled in a glass beaker until all the organic matter is dissolved. The beaker should be covered with a glass plate, to prevent dissipation of the acid fumes. When the process is complete, usually in about half an hour, the contents of the beaker are

thoroughly stirred with a glass rod, poured rapidly off into a larger bulk of cold water, and allowed to settle for another half-hour. The process is then repeated with a smaller bulk of water, several times, to allow the removal of the last traces of acid, and finally with distilled water. The separation of the diatoms into grades is effected by settlement. The final result is poured into a tall glass vessel, and allowed to settle for, at first, a minute, the supernatant fluid again poured off, and allowed to settle for two minutes, and so on, the period being gradually increased, and each sediment preserved apart. The first will probably only be sand, but the proportion of diatoms will increase with each separation, though there will always be a certain proportion of sand of such a size as to settle at the same rate as the diatoms. Marine plants especially will furnish a rich harvest by treatment as described.

Solid diatomaceous deposits, such as kiesel-guhr, mountain-meal, and especially the famous Oamaru deposit from New Zealand, demand different treatment, and perhaps the best way is to disintegrate the mass, either by boiling with Sunlight soap (though the alkali attacks the flint to some extent) or to mix the mass with a super-saturated solution of acetate of soda (made by saturating water with the crystals whilst boiling), and by successive coolings, heatings, and stirrings to cause the process of crystallisation to break up the mass, which it will do very thoroughly. The diatoms are then separated by sedimentation, as above described.

A small portion of the deposit may now be spread thinly on a glass slide, allowed to dry thoroughly, be treated with balsam, and covered.

If it be desired to select individual diatoms, this must be done under the microscope, by means of a bristle fixed in a handle either with glue or sealing-wax. The diatom selected will adhere to the bristle if the latter be slightly greasy, and should then be transferred to a slightly adhesive slide, coated either with thin solution of white shellac, or with thin gum nearly dry. When the forms desired are mounted, the preparation should be covered in balsam. The process is by no means as easily effected as described, however.

The preparation of insects, or parts of insects, as microscopic objects is a tedious and difficult task. The main point is the trouble of softening the integument and eliminating the colour.

The latter can, in any case, be only partially effected. The beginner would do well to begin with a fairly easy form, such as the worker-ant. A good supply of these insects may be placed in a bottle of liquor potassæ, and left there for at least some days until they begin to become clear and limp. From time to time a specimen may be taken, well washed with several waters, then with acetic acid and water of a strength of about 10 per cent., then with weak spirit, about 50 per cent. An attempt may then be made to arrange the insect upon a slide, spreading out the legs so as to exhibit them to the best advantage, and when this has been done a cover-glass may be put on, supported in such a way as to prevent absolute pressure. The spirit is then withdrawn by means of a piece of filtering-paper cut to a point, and strong spirit added. This is again succeeded by absolute alcohol, then by a mixture of turpentine and crystal carbolic acid in equal proportions, and finally the cover-glass is carefully lifted, and some thick balsam solution dropped on, the limbs finally arranged by means of warm needles, and the cover-glass carefully replaced and pressed gently down by means of a clip, which may be obtained for a few pence. The whole is then set aside to harden, the deficiency caused by evaporation made good, the balsam allowed to dry, and the preparation finally painted round.

The contents of the body, in large insects, must be removed, and this is effected during the washing in water by gentle pressure with a camel-hair brush, the process being aided, if necessary, by a small incision made through the integument at the root of the tail. Sections of insects require very special methods, which will hardly fall within the scope of this work.

CHAPTER XII

Section-Cutting—Staining

No method of examination can equal, for general applicability and usefulness, that of section-work. The relations of the parts to each other being preserved, it is possible to draw conclusions as to their actual relations which no other mode allows of, and we shall devote this concluding chapter to some account of the methods to be employed to this end.

The apparatus required is not necessarily complicated. Reduced to its elements, it consists essentially only of a razor to cut the sections and a dish to receive them. It but seldom happens, however, that the relations of the parts in sufficiently thin sections can be preserved by such a rough-and-ready method, and frequently the object to be cut is of such small dimensions as to render it impossible to deal with it in this way. It is therefore necessary to “imbed” it, so as to obtain a handle by which to hold it, in such a way that it shall be equally supported in all directions. Moreover, since the human hand can only in exceptional cases be brought to such a pitch of skill as to cut a series of sections, or even single ones, of the needful delicacy, some mechanical means of raising the object through a definite distance is highly desirable. The writer has cut many thousands of sections with the “free hand,” but the personal equation is a large one, and is not always the same in the same person. For single sections the method will, with practice, succeed very well, but some means of securing a number of sections of more or less the same thickness is usually required.

Let us deal with the imbedding first.

If it be desired to imbed a tissue which has merely been fixed with formalin, the block should be immersed in strong gum (made by saturating water with picked gum arabic, white and clean) for several days. It is then taken out

and, without draining, transferred to the plate of a freezing microtome, and the sections cut from the frozen block, and mounted in glycerine at once.

This plan is of limited usefulness, since it allows of very little differentiation of the tissue elements, and that only optical.

To get the best results, some plan of staining must be adopted. Perhaps the simplest, and certainly a very excellent one, is as follows. After the tissue has been passed from the hardening, or fixing, fluid into the successive alcohols, as described, it is placed in the following solution. Take about forty grains of carmine and eighty grains of borax, dissolve in about an ounce of water, add to the mixture an ounce of methylated spirit, and let it stand for some time with frequent shaking; about a week will be sufficient, and the process of solution may be hastened by gentle warming at intervals. The clear upper portion is then poured off, and into this the block of tissue is dropped, and allowed to remain until thoroughly penetrated. Perhaps the best plan is to substitute the carmine solution for the 50 per cent. alcohol, and thus to make the staining a stage in the hardening process. From the carmine solution the tissue is transferred to 70 per cent. alcohol, to each ounce of which two drops of hydrochloric acid have been added, and after remaining in it for a day (with a piece of the usual size) is placed in 70 per cent. alcohol, in two successive quantities. Sections from this material now only require treatment with the carbolic acid and turpentine above mentioned to be fit for mounting and covering in balsam. We now proceed to indicate how the sections may be cut.

A mixture of wax and almond oil, in proportions varying with the heat of the weather, usually about equal ones, is prepared. The piece of tissue is freed from superfluous spirit by being placed on a bit of blotting-paper for a minute or two, and is then immersed in a quantity of the wax-and-oil mixture contained in a little box of paper or lead-foil. The tissue is held on the point of a needle, and lifted up and down until it is coated with the mixture, and, before solidification of the mass sets in, is lowered into the box and left to cool. The block now furnishes a handle, and this should be wrapped round with paper, the sections cut with the keenest possible razor,

and as thin as possible, and placed in spirit as cut. From the spirit, which must be the strongest obtainable, they are placed in the clearing liquid, carbolic and turpentine, and then slid on to the slide, a drop of balsam placed on the section, and the cover over all. Of late years all sections of ordinary soft tissues, animal or vegetable, have been cut by one of the infiltration methods, in which the interstices of the tissue are filled up by some material which prevents the relations of the cells from being altered during the process of cutting. To enter fully into this matter would occupy too much space, and would serve no useful purpose, for the worker who requires to make use of such means will find it indispensable to obtain Bolles Lee's *Microtomist's Vade Mecum*, in which the whole matter is exhaustively treated.

The simple method above detailed will answer most ordinary purposes, provided that a few precautions be attended to. The chief are as follows. The outside of the block of tissue must be sufficiently dry for the wax-and-oil to adhere to it. The razor must be extremely sharp, and must be kept so by application to a Turkey stone during the section-cutting. The blade must be drawn across the tissue from heel to point, and kept wetted with spirit the whole time, so as to prevent any dragging of the section. The transference of the section to the slide must be effected by means of a section-lifter, which may be made by beating out a piece of stout copper wire to a thin flat blade; or a small palette-knife, or German-silver lifter, may be purchased for a few pence. The carbolic turpentine is best used by placing a little in a watch-glass, and floating the sections on to it by lifting them singly with the lifter, freeing them from superfluous spirit by draining on to blotting-paper, and allowing them to float on to the surface of the liquid in the watch-glass, so that the spirit may evaporate from above, and be replaced by the clearing agent from below. The balsam solution should be thin, and the cover-glass must be allowed to settle down into place without pressure.

The question of staining sections is a very large one, and is becoming of daily increasing complexity.

We cannot go into it here, further than to say that most sections cut from unstained tissue will yield excellent results if stained first with Delafield's logwood solution (to be obtained at Baker's) to a very slight extent, and then with a solution of safranin. The sections should be washed with tap-water after the logwood stain, and should be of a pale violet colour. If overstained, the colour may to a great extent be removed by washing with a very weak solution of hydrochloric acid, about two drops of acid to each ounce of water, and repeated washing in tap-water to remove the acid, and restore the violet. The safranin stain should be weak, and should be allowed to act for some time. From this last the sections are transferred to strong spirit, the latter being renewed until the sections cease to give up the red dye; and they may then be mounted as described. The results with most tissues are superb, every detail of the structure being splendidly brought out. Safranin alone is also an admirable stain for many purposes.

Further information must be sought in the book already mentioned. Let us, in closing, warn the beginner of two things which are of general application in practical microscopy. The first is, not to be discouraged by failures. The manipulations are in many cases very delicate, and premiums must be paid to experience for insurance against failure in every one of the processes.

The second is, that the most scrupulous cleanliness will hardly suffice to prevent contamination of preparation by the all-pervasive dust which, invisible to the eye, assumes colossal proportions under the microscope, and the particles of which have an unpleasant habit of collecting on the most interesting or most beautiful portion of the preparation. This can only be guarded against by careful filtration of all fluids, and constant watchfulness.

A preparation properly made is a thing of beauty, and a joy for ever,—or if not for ever, at any rate for many years; and one such will repay an infinitude of pains taken in its production.

INDEX

	PAGE		PAGE
Air-pump,	174	" mounting of,	177
Algæ,	78	Jelly-fish,	158
" marine,	92	Larva of <i>Chironomus</i> ,	152
Anemones, sea,	159	Light, arrangement of,	29
Antennæ,	96	Live-box,	17
Ants,	97	Logwood solution,	183
Bacillaria,	87	Magnification, to measure,	27
Balancers of Fly,	112	Mare's tail,	91
Bark,	61	Marine life,	155
Blights,	89	Microscope, Baker's,	14
Blood, circulation of,	129	" "	15
" corpuscles of,	128	" "portable",	
Bone,	123	" primitive,	5
Breathing-tubes,	109	" simple,	12, 13
Bull's-eye, use of,	32	Mildew,	89
Camera lucida,	25	Mirror, concave,	29
Canada Balsam,	170	Mollusca,	161
Cartilage,	124	Mounting,	168
Cells, animal,	122	" dry,	173
" circulation in,	40	" foraminifera,	157
" mounting dry in,	173	Mosses,	96
" pigment,	121, 165	Muscle,	127
" spiral,	46	Nails,	124
" vegetable,	37	Needles,	22
Ceramidia,	93	Nerve,	127
Chlorophyll,	40	Net,	133
Compressorium, Beck's,	18	Nucleus,	40
Condenser, bull's-eye,	19		
" substage,	19		

Confervæ,	84	Objectives,	16
Conjugation,	82 , 84	Objects, drawing of,	24
Convex lenses,	7	" photography of,	36
" foci of,	7 , 8	Oil-cells,	58 , 61
" image	10	Oscillatoria,	84
formed by,		Osmic acid,	170
" virtual	11		
image,			
Corallines,	164		
Corrosive sublimate,	170	Parasites,	153
Cover-glasses,	18	Petals,	69
		Picric acid,	170
Desmids,	81	Pigment,	121
Diatoms,	85	Pocket magnifiers,	13
" preparation of,	175	Polariscope,	166
Dipping-tubes,	22	Pollen,	71
Dissection,	20	Polysa,	147
" instruments,	21	Pond-hunting,	132
" under	24	Preservatives,	169
microscope,			
Drawing,	25		
" squares,	26	Radiolaria,	157
		Rotifers,	147
Echinoderms,	162		
Entomostraca,	152	Safranin stain,	183
Epidermis, animal,	122	Sap,	128
" vegetable,	68	Scent-glands,	57
Extemporised apparatus,	5	Sea-weeds,	92
		Section-cutting,	178
Feathers,	119	Seeds,	75
Fish, scales of,	118	Skin,	120
" parasites of,	153	Spiracles,	102
Fixation of cell-forms,	171	Sponge, fresh-water,	135
Focus of mirror,	29	" spicules,	155

Foraminifera,	<u>156</u>	Sporangia,	<u>92</u>
Formalin,	<u>164</u>	Stage-forceps,	<u>116</u>
Frog-plate,	<u>129</u>	Starch,	<u>63</u>
		" mounting,	<u>172</u>
Gills of mussel,	<u>122</u>	Stomata,	<u>49</u>
Gizzard of insects,	<u>109</u>	Suckers,	<u>108</u>
Glycerine-gelatine,	<u>172</u>		
" jelly,	<u>171</u>	Teeth,	<u>125</u>
		Troughs, glass,	<u>18</u>
Hairs, animal,	<u>116</u>		
" vegetable,	<u>53</u>	Water-bears,	<u>152</u>
Heads of Insects,	<u>104</u>	Wings,	<u>110</u>
		Wool,	<u>116</u>
Illumination, correct,	<u>31</u> , <u>32</u>	Worms, fresh-water,	<u>14</u>
" dark-field,	<u>34</u>	" marine,	<u>160</u>
" for opaque	<u>33</u>		
objects,			
Imbedding,	<u>180</u>	Yeast,	<u>89</u>
" by infiltration,	<u>182</u>		
Infusoria,	<u>135</u>	Zoœa,	<u>161</u>
Injection,	<u>164</u>	Zoophytes,	<u>157</u>
Insects,	<u>97</u>	Zygnemaceæ,	<u>85</u>

PRINTED BY MORRISON AND GIBB LIMITED, EDINBURGH

*** END OF THE PROJECT GUTENBERG EBOOK COMMON
OBJECTS OF THE MICROSCOPE ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.